

Kinora Clinical Reasoning & Practice

AI Orientation (Part 4)

Structured clinical decision-making reference for RAG / AI-assisted physiotherapy
consultation

Pain Referral · Differential Diagnosis · Outcome Measures · Red Flags · Sport Biomechanics
· Prevention · Clinical Reasoning · Exercise Prescription

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Disclaimer

This document is an educational orientation resource for Kinora AI clinical consultation support. It is NOT a substitute for professional clinical judgment, direct patient examination, licensed medical or physiotherapy care, or emergency services. Content reflects established musculoskeletal medicine, sports rehabilitation, and clinical reasoning frameworks but must be verified against current peer-reviewed literature, local protocols, and individual patient presentation. RED FLAGS identified in this document require urgent medical referral — including but not limited to fracture, infection, tumor, myelopathy, stroke, acute tendon rupture, septic arthritis, vascular compromise, and progressive neurological deficit. Never delay emergency care for AI-generated guidance. Diagnostic probabilities, sensitivity/specificity, MCID/MDC values, and likelihood rankings are approximate and vary by study population. Clinical reasoning pathways are illustrative templates, not prescriptive protocols for any specific patient.

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1. Pain Referral Patterns

RECORD: Shoulder pain — cervical referral

ID: REF-001

Pain Location: Lateral or posterior shoulder, deltoid region, occasionally upper arm; may skip mid-arm

Possible Sources: Cervical disc herniation (C4-C7), cervical radiculopathy, cervical facet arthropathy, cervical spondylosis, nerve root compression, brachial plexus irritation at scalene/first rib

Probability: Moderate-high in patients with concurrent neck pain, upper extremity paresthesia, or positive Spurling test; ~15-25% of chronic shoulder pain cases have cervical contribution in mixed cohorts

Red Flags: Progressive weakness, bilateral symptoms, bowel/bladder changes, gait disturbance, fever, unexplained weight loss, history of malignancy, night pain unrelieved by position change

Differential Diagnosis: Rotator cuff tendinopathy, adhesive capsulitis, AC joint pathology, glenohumeral OA, labral tear, referred cardiac/pulmonary pain, brachial neuritis

Associated Symptoms: Neck stiffness, ipsilateral scapular pain, dermatomal numbness/tingling (C5 lateral arm, C6 thumb, C7 middle finger, C8 ring/little finger), headache at occiput, pain worsened by neck rotation/extension

Physical Examination: Cervical ROM (rotation, side bend often limited), segmental palpation tenderness, upper limb tension tests, myotome/dermatome screen, shoulder exam may be relatively benign unless concurrent pathology

Clinical Tests: Spurling test (radiculopathy), upper limb tension test (ULTT/A), cervical distraction (may relieve), shoulder impingement tests may be negative or secondary; Hoffman/Babinski if myelopathy concern

Imaging: MRI cervical spine if radiculopathy/myelopathy signs; plain films if trauma/instability; shoulder MRI only if local findings predominate; consider both regions if mixed presentation

Evidence: Cervical radiculopathy commonly refers to shoulder girdle via dermatomal patterns; dual pathology frequent in age >50; treat cervical drivers when shoulder exam incongruent with symptoms

References: Wainner et al. reliability cervical radiculopathy cluster; Bogduk cervical referral patterns; see PubMed [cervical radiculopathy shoulder referral]

RECORD: Shoulder pain — cardiac referral

ID: REF-002

Pain Location: Left shoulder, upper arm, jaw, scapular region; may be bilateral or right-sided in atypical presentations

Possible Sources: Myocardial ischemia, acute coronary syndrome (ACS), pericarditis, aortic dissection (rare, catastrophic)

Probability: Low in young athletic MSK clinic but MUST-NOT-MISS; higher in age >40, cardiovascular risk factors, exertional onset

Red Flags: Crushing chest pressure, diaphoresis, nausea, dyspnea, syncope, pain with exertion relieved by rest/nitroglycerin, new ECG changes, troponin elevation

Differential Diagnosis: Rotator cuff strain, biceps tendinopathy, AC sprain, cervical radiculopathy, pulmonary embolism, GI referral

Associated Symptoms: Chest tightness, shortness of breath, fatigue, anxiety, radiation to jaw/teeth, arm heaviness without clear mechanical pattern

Physical Examination: Shoulder ROM often full or minimally restricted; reproduction not clearly mechanical; vitals may show tachycardia, hypotension; diaphoresis; no focal shoulder tenderness pattern typical of cuff disease

Clinical Tests: No MSK special test confirms cardiac origin; pain not consistently reproduced by shoulder loading; consider cardiopulmonary exam (auscultation); 12-lead ECG in emergency setting

Imaging: Not shoulder MRI first line; emergency department: ECG, troponin, chest X-ray; echocardiography, coronary angiography per cardiology

Evidence: Referred pain via shared spinal segments (C3-C5, T1-T4) and viscerosomatic convergence; ACS may present as isolated arm/shoulder pain especially in women and diabetics

References: AHA/ACC chest pain guidelines; McBane referred cardiac pain patterns; see PubMed [atypical ACS shoulder pain]

RECORD: Shoulder pain — pulmonary referral

ID: REF-003

Pain Location: Shoulder tip (especially right), scapular area, upper chest wall

Possible Sources: Pneumonia, pleuritis, pulmonary embolism, pneumothorax, lung apex tumor (Pancoast), subphrenic abscess

Probability: Low in isolated MSK shoulder but elevated with fever, cough, dyspnea, immobilization history, malignancy, recent surgery

Red Flags: Hemoptysis, severe dyspnea, pleuritic chest pain, unilateral leg swelling, recent long-haul travel, night sweats, weight loss, Horner syndrome (Pancoast)

Differential Diagnosis: Rotator cuff pathology, cervical radiculopathy, diaphragmatic irritation, thoracic outlet syndrome, rib fracture

Associated Symptoms: Cough, fever, sputum production, pleuritic pain with deep breath, tachypnea, reduced exercise tolerance, constitutional symptoms with malignancy

Physical Examination: Shoulder motion may be limited secondarily due to pain guarding; pulmonary auscultation may reveal crackles, diminished breath sounds, pleural rub; no primary impingement signs required

Clinical Tests: Shoulder special tests nonspecific; respiratory expansion assessment; percussion/auscultation; no single orthopaedic test rules in pulmonary cause

Imaging: Chest X-ray, CT pulmonary angiography if PE suspected, CT chest for mass; shoulder imaging low yield unless concurrent MSK findings

Evidence: Diaphragmatic and pleural irritation refers to shoulder via phrenic nerve (C3-C5); Kehr sign historically described diaphragmatic irritation with shoulder tip pain

References: Bockenbauer viscerosomatic referral; Fleischner Society pulmonary nodule guidelines; see PubMed [referred shoulder pain pulmonary]

RECORD: Shoulder pain — diaphragmatic referral

ID: REF-004

Pain Location: Shoulder tip (classically right), scapular angle, upper trapezius region

Possible Sources: Diaphragmatic irritation from subphrenic abscess, perforated viscus, hepatic/splenic pathology, post-laparoscopic shoulder pain (CO₂ insufflation), peritonitis

Probability: Low in outpatient physio unless post-surgical or systemic illness; higher post-abdominal surgery (24-72 hours)

Red Flags: Acute abdomen, fever, rigidity, hypotension, recent abdominal surgery with worsening pain, free air on imaging

Differential Diagnosis: Rotator cuff tendinopathy, cervical referral, gallbladder disease, pulmonary causes, psychogenic pain

Associated Symptoms: Abdominal pain/distension, nausea, shoulder pain worse with lying flat or deep inspiration, post-operative timeline correlation

Physical Examination: Shoulder exam often normal except global guarding; abdominal exam may reveal tenderness, distension; shoulder pain may not change with isolated shoulder movement

Clinical Tests: Shoulder impingement/instability tests typically negative or irrelevant; inspiration-related shoulder pain increase suggests diaphragmatic component

Imaging: Abdominal CT or ultrasound; chest X-ray for free air/elevated hemidiaphragm; shoulder imaging not primary

Evidence: Phrenic nerve (C3-C5) afferents converge with somatic input at spinal cord producing referred shoulder tip pain from diaphragmatic/peritoneal irritation

References: Guyton viscerosomatic referral; surgical literature on laparoscopic shoulder pain mechanism; see PubMed [phrenic referred shoulder pain]

RECORD: Shoulder pain — gallbladder / visceral referral

ID: REF-005

Pain Location: Right shoulder, right scapular region, interscapular area; may accompany epigastric/RUQ pain

Possible Sources: Biliary colic, acute cholecystitis, gallstones, peptic ulcer disease (less common posterior referral), pancreatitis (belt-like with back referral)

Probability: Low overall; moderate in patients with fatty meal association, RUQ tenderness, Murphy sign positive

Red Flags: Fever, jaundice, persistent vomiting, hemodynamic instability, Murphy sign, elevated liver enzymes, peritoneal signs

Differential Diagnosis: Rotator cuff disease, cervical radiculopathy, hepatic flexure colonic pathology, right lower lobe pneumonia, T4 syndrome

Associated Symptoms: Postprandial pain especially fatty foods, nausea, bloating, epigastric/RUQ pain, pain may precede shoulder component

Physical Examination: RUQ palpation tenderness, Murphy sign; shoulder strength/ROM may be preserved; pain not reproduced consistently by resisted shoulder tests

Clinical Tests: Murphy sign (RUQ); Boas sign (hypersensitivity T8-T10, historical); shoulder tests non-diagnostic for visceral source

Imaging: Right upper quadrant ultrasound first line for gallstones; abdominal CT if complicated; shoulder MRI not indicated unless dual pathology

Evidence: Diaphragmatic irritation from gallbladder inflammation refers via phrenic nerve to right shoulder tip — classic viscerosomatic pattern taught in clinical medicine

References: Bile duct/gallbladder clinical guidelines; Bogduk referred pain maps; see PubMed [cholecystitis referred shoulder pain]

RECORD: Shoulder pain — rotator cuff source

ID: REF-006

Pain Location: Anterolateral shoulder over deltoid insertion region, deep glenohumeral joint line, may radiate mid-arm not past elbow typically

Possible Sources: Supraspinatus tendinopathy/tear, infraspinatus/teres minor tendinopathy, subacromial impingement, calcific tendinitis, partial/full thickness RTC tear

Probability: High — most common local source of shoulder pain in adults; prevalence increases with age, overhead activity, degenerative change

Red Flags: Acute traumatic fall with inability to abduct (massive tear), fever with hot swollen joint (septic arthritis), night pain with weight loss (tumor), acute ecchymosis with trauma (fracture)

Differential Diagnosis: Adhesive capsulitis, AC joint OA, biceps tendinopathy, labral tear, cervical radiculopathy, glenohumeral OA

Associated Symptoms: Painful arc 60-120 degrees abduction, night pain lying on affected side, weakness in external rotation/abduction, catching, crepitus, overhead activity intolerance

Physical Examination: Inspect atrophy (supraspinatus fossa), palpation GT tenderness, ROM active may lag passive in tears, painful arc, weakness resisted ER and abduction

Clinical Tests: Neer impingement, Hawkins-Kennedy, empty can (Jobe), external rotation lag sign, drop arm test (full thickness), painful arc test cluster; ultrasound/MRI correlation

Imaging: X-ray for OA, calcification, acromial morphology; MRI or ultrasound for cuff integrity; not required day 1 for typical tendinopathy

Evidence: Rotator cuff disorders account for majority of shoulder pain presentations in primary care and sports medicine; staged rehab effective for many partial tears and tendinopathy

References: AAOS rotator cuff guideline; Michener et al. clinical examination accuracy; see PubMed [rotator cuff tendinopathy management]

RECORD: Shoulder pain — labral source

ID: REF-007

Pain Location: Deep shoulder joint, posterior-superior or anterior shoulder; athletes may localize deep in joint with mechanical symptoms

Possible Sources: SLAP lesion, Bankart lesion (anterior labrum with instability), posterior labral tear, paralabral cyst, superior labrum anterior-posterior variants

Probability: Moderate in overhead/throwing athletes with mechanical symptoms; lower in sedentary older adults unless instability history

Red Flags: Recurrent dislocations, vascular injury after dislocation, fracture, sepsis, acute locking with inability to move after trauma

Differential Diagnosis: Rotator cuff tear, biceps tendinopathy (SLAP overlap), glenohumeral instability, AC joint injury, scapular dyskinesis

Associated Symptoms: Clicking, catching, locking, sense of instability, pain with overhead/throwing, dead arm sensation, decreased velocity in throwers

Physical Examination: Apprehension/relocation, load and shift, sulcus sign (if multidirectional), biceps load II (SLAP), O'Brien test, crank test, Kim test

Clinical Tests: O'Brien (SLAP sensitivity moderate), anterior apprehension, relocation, jerk test (posterior), dynamic labral shear; MRI arthrogram gold standard imaging correlate

Imaging: MRI arthrogram preferred for labral detail; plain films for instability-associated bony lesions (Hill-Sachs, Bankart fracture)

Evidence: Labral pathology common in throwing athletes; clinical tests have variable specificity; surgery vs rehab depends on instability, tear type, sport demands

References: Snyder SLAP classification; Wilk et al. throwing athlete shoulder; see PubMed [SLAP lesion rehabilitation]

RECORD: Neck pain — shoulder / UE referral (reverse pattern)

ID: REF-008

Pain Location: Cervical spine, upper trapezius, periscapular region; may present as neck pain but source is shoulder

Possible Sources: Rotator cuff tendinopathy causing compensatory scapular/cervical overload, scapular dyskinesis, AC joint pathology altering scapulothoracic rhythm, long head biceps tendinopathy, shoulder instability

Probability: Moderate — shoulder dysfunction frequently drives cervicothoracic pain via altered kinematics and muscle overload without primary cervical pathology

Red Flags: Myelopathy signs, radiculopathy with progressive deficit, trauma, infection, malignancy as for any neck pain

Differential Diagnosis: Cervical radiculopathy, cervicogenic headache, TOS, thoracic outlet, facet arthropathy, myofascial pain, whiplash

Associated Symptoms: Shoulder pain with overhead activity, scapular winging, painful arc, night pain on affected shoulder side, neck pain improves when shoulder treated

Physical Examination: Scapular dyskinesis assessment, shoulder impingement signs, cervical exam may show secondary upper trap hypertonicity rather than primary segmental dysfunction

Clinical Tests: Shoulder impingement cluster, scapular assistance test, cervical tests may be negative; treat shoulder and reassess neck symptoms

Imaging: Shoulder imaging if local findings; cervical MRI if neuro signs; often clinical diagnosis of secondary cervical pain from shoulder

Evidence: Kinetic chain models demonstrate shoulder-scapular dysfunction increases cervical and thoracic load; addressing primary shoulder pathology often reduces referred neck symptoms

References: Kibler scapular dyskinesis; Cools et al. scapular muscle balance; see PubMed [shoulder neck referred pain kinetic chain]

RECORD: Neck pain — cervical primary with UE referral

ID: REF-009

Pain Location: Neck, occiput, periscapular, may radiate to shoulder, arm, hand per dermatome

Possible Sources: Disc herniation, foraminal stenosis, facet arthropathy, whiplash-associated disorder, cervical spondylosis, myelopathy (central stenosis)

Probability: Very common primary neck presentation; UE referral in ~30-40% of radiculopathy cases

Red Flags: Myelopathy (Hoffman, gait, hand clumsiness, Lhermitte), progressive motor deficit, bilateral symptoms, saddle anesthesia, fever, history of cancer, trauma with step deformity

Differential Diagnosis: Thoracic outlet syndrome, peripheral nerve entrapment, shoulder pathology, inflammatory arthropathy, meningitis

Associated Symptoms: Dermatomal paresthesia, weakness in myotome distribution, headache, tinnitus (upper cervical), dizziness (controversial association)

Physical Examination: Cervical ROM, Spurling, ULTT, reflexes (biceps C5-C6, brachioradialis C6, triceps C7), sensation mapping, myotome strength

Clinical Tests: Spurling, distraction, ULTT1-3, VBI screening if manual therapy planned (historical), upper cervical stability tests if trauma

Imaging: MRI if persistent radiculopathy, progressive deficit, or red flags; X-ray for alignment/instability post-trauma

Evidence: Clinical prediction rules for cervical radiculopathy include Spurling, rotation limited, ULTT; majority improve with conservative care over 6-12 weeks

References: Wainner cervical radiculopathy CPR; Childs et al. neck pain guidelines; see PubMed [cervical radiculopathy conservative management]

RECORD: Thoracic pain — cardiac referral

ID: REF-010

Pain Location: Mid-thoracic spine, interscapular region, may radiate to arms or jaw

Possible Sources: Myocardial ischemia, aortic dissection (tearing interscapular pain), pericarditis

Probability: Low but critical; higher with cardiovascular risk, exertional pain, known CAD

Red Flags: Chest pressure, syncope, unequal pulses (dissection), tearing pain radiating to back, ECG changes, troponin elevation

Differential Diagnosis: Thoracic facet pain, costovertebral dysfunction, T4 syndrome, scapular pain, MSK strain

Associated Symptoms: Dyspnea, diaphoresis, nausea, pain unrelated to thoracic movement, pain with exertion

Physical Examination: Thoracic ROM may be globally limited; vitals abnormal; no segmental reproducibility; cardiac auscultation

Clinical Tests: Thoracic PA mobilization should NOT reproduce isolated cardiac pain pattern; emergency cardiac workup if suspected

Imaging: ECG, troponin, CT angiography for dissection, chest X-ray; thoracic spine MRI only after cardiac exclusion if indicated

Evidence: Interscapular pain classic in aortic dissection and posterior MI; always screen cardiovascular history in thoracic pain

References: AHA acute coronary syndromes; IRAD aortic dissection registry; see PubMed [interscapular pain cardiac]

RECORD: Elbow pain — cervical referral (C6-C7)

ID: REF-011

Pain Location: Lateral elbow, forearm, may mimic tennis elbow distribution

Possible Sources: C6-C7 radiculopathy, cervical facet referral, peripheral nerve double crush (median/radial overlap)

Probability: Moderate in patients with neck pain and distal symptoms; ~5-10% of lateral elbow pain may have cervical component

Red Flags: Progressive weakness (triceps C7), myelopathy, bilateral symptoms, trauma

Differential Diagnosis: Lateral epicondylalgia, radial tunnel syndrome, posterolateral rotatory instability, olecranon bursitis

Associated Symptoms: Neck pain, triceps weakness, middle finger numbness (C7), pain with cervical extension/rotation

Physical Examination: Cervical screen, resisted wrist extension may be painful locally and via nerve tension; compare cervical vs local provocative tests

Clinical Tests: Spurling, ULTT median/radial bias, Cozen test (local), radial nerve palpation; differentiate by response to cervical vs elbow provocation

Imaging: Cervical MRI if radiculopathy; elbow imaging if local epicondyle tenderness and negative cervical screen

Evidence: Double crush phenomenon documented; cervical treatment may reduce distal symptoms when radiculopathy present

References: Upton and McComas double crush; Cleland cervical vs local elbow differentiation; see PubMed [cervical radiculopathy elbow pain]

RECORD: Wrist pain — cervical referral (C6-C8)

ID: REF-012

Pain Location: Wrist, hand, forearm; may mimic carpal tunnel or de Quervain distribution

Possible Sources: C6 (thumb/radial wrist), C7 (middle finger/wrist extensors), C8 (ulnar side/hand intrinsic), T1 (intrinsic hand)

Probability: Moderate when wrist exam incongruent or bilateral with neck signs

Red Flags: Thenar atrophy, progressive weakness, myelopathy, acute trauma with deformity (fracture)

Differential Diagnosis: Carpal tunnel syndrome, de Quervain tenosynovitis, TFCC tear, wrist OA, inflammatory arthritis

Associated Symptoms: Nocturnal paresthesia, neck pain, dermatomal numbness, symptoms with overhead neck posture

Physical Examination: Phalen/Tinel at wrist vs cervical tension tests; grip strength; sensory mapping; cervical ROM

Clinical Tests: ULTT median, Spurling, Phalen, Tinel, Finkelstein; compare symptom provocation location

Imaging: NCS/EMG for CTS; cervical MRI if radiculopathy; wrist MRI for TFCC if mechanical symptoms

Evidence: Differentiating cervical vs peripheral entrapment requires systematic dual-region examination; both may coexist

References: Wainner upper quarter screening; Bland and Rudolph carpal tunnel vs cervical; see PubMed [double crush wrist]

RECORD: Heart — left arm and shoulder referral (source pattern)

ID: REF-013

Pain Location: Left arm, shoulder, jaw, epigastrium — referred FROM cardiac source

Possible Sources: Myocardial ischemia, MI, angina equivalent presentations

Probability: Must consider in any upper extremity pain with cardiac risk profile regardless of apparent MSK cause

Red Flags: Exertional onset, relief with rest, associated chest symptoms, abnormal vitals, known CAD, diabetes (atypical presentation)

Differential Diagnosis: Any MSK shoulder/arm diagnosis; PE; aortic pathology

Associated Symptoms: Dyspnea, diaphoresis, nausea, sense of doom, fatigue disproportionate to activity

Physical Examination: MSK exam may be misleadingly normal; always take vitals and history before focused shoulder treatment in at-risk patients

Clinical Tests: No orthopaedic test confirms or excludes; low threshold for emergency referral

Imaging: ECG, troponin, cardiology workup — not shoulder MRI as first step when suspicion exists

Evidence: Convergence of cardiac afferents with somatic C3-T4 segments produces classic and atypical referred patterns

References: AHA/ACC guidelines; McCabe cardiac referred pain review; see PubMed [referred pain myocardial ischemia]

RECORD: Lung apex — shoulder referral (Pancoast pattern)

ID: REF-014

Pain Location: Shoulder, scapula, medial arm, forearm along ulnar distribution

Possible Sources: Pancoast tumor (superior sulcus), apical pneumonia, apical pleural disease

Probability: Rare but must-not-miss in smokers, age >50, unilateral shoulder pain without MSK mechanism, systemic symptoms

Red Flags: Horner syndrome (ptosis, miosis, anhidrosis), weight loss, hemoptysis, night pain, tobacco history, brachial plexopathy

Differential Diagnosis: Cervical radiculopathy C8-T1, TOS, rotator cuff disease, thoracic outlet tumor

Associated Symptoms: Gradual progressive pain, intrinsic hand weakness, ulnar paresthesia, cough, chest symptoms

Physical Examination: Possible Horner signs, thenar/interossei weakness, supraclavicular fullness, shoulder exam may be secondary

Clinical Tests: Cervical and shoulder tests may mimic radiculopathy; high index suspicion triggers chest imaging not repeated steroid injections

Imaging: Chest CT with apical views; MRI brachial plexus if tumor suspected; PET-CT oncology pathway

Evidence: Apical lung tumors invade brachial plexus and stellate ganglion producing shoulder/arm pain and autonomic signs

References: Pancoast original description; NCCN lung cancer guidelines; see PubMed [superior sulcus tumor shoulder pain]

RECORD: Diaphragm — shoulder cascade (integrated referral chain)

ID: REF-015

Pain Location: Shoulder tip bilateral or unilateral, neck/trapezius, scapular angle — part of visceral-somatic cascade

Possible Sources: Diaphragm, liver, gallbladder, spleen, peritoneum, post-operative CO₂, pulmonary base disease

Probability: Context-dependent; consider when shoulder pain lacks MSK mechanism or follows abdominal/thoracic events

Red Flags: Acute abdomen, sepsis, post-op complications, free air, hemodynamic instability

Differential Diagnosis: Full MSK shoulder workup diagnoses; cervical; cardiac; pulmonary

Associated Symptoms: Abdominal or chest primary symptoms, pain with inspiration, supine intolerance, surgical timeline

Physical Examination: Shoulder isolated tests poor correlation; integrated cardiopulmonary-abdominal screen required

Clinical Tests: Inspiration aggravation; Murphy sign (hepatobiliary); shoulder impingement tests non-specific

Imaging: Abdominal ultrasound/CT; chest X-ray; shoulder MRI low priority until visceral causes excluded

Evidence: Phrenic nerve referral explains shoulder-neck cascade from diaphragm through shared segmental levels linking to cervical/scapular pain regions

References: Bockenbauer viscerosomatic; anatomy phrenic nerve C3-C5; see PubMed [viscerosomatic shoulder referral]

RECORD: Rotator cuff — downstream neck overload (cascade)

ID: REF-016

Pain Location: Neck, upper trapezius, levator scapulae — secondary to primary rotator cuff dysfunction

Possible Sources: Rotator cuff tendinopathy/tear causing altered scapulohumeral rhythm and cervical overload

Probability: High in chronic shoulder patients with concurrent neck complaints; treat shoulder as primary in many cases

Red Flags: Standard neuro/red flag screen for neck; primary shoulder red flags as REF-006

Differential Diagnosis: Primary cervical radiculopathy, facet pain, TOS, myofascial pain syndrome

Associated Symptoms: Shoulder painful arc, weakness, scapular dyskinesis, neck pain with overhead reaching

Physical Examination: Shoulder primary findings; neck tenderness often trapezius/levator secondary; scapular upward rotation deficits

Clinical Tests: Shoulder impingement cluster, scapular assistance test; cervical Spurling may be negative

Imaging: Shoulder ultrasound/MRI; cervical imaging if neuro symptoms persist after shoulder treatment trial

Evidence: Scapular dyskinesis and cuff weakness increase compensatory cervical muscle activity; integrated rehab addresses both regions with shoulder priority when indicated

References: Kibler et al. scapula role; Ludewig & Reynolds shoulder-neck linkage; see PubMed [scapular dyskinesis neck pain]

RECORD: Labrum — instability cascade to neck and UE

ID: REF-017

Pain Location: Deep shoulder, posterior shoulder, periscapular neck, occasional arm parasthesia with nerve stretch

Possible Sources: Labral tear with instability causing secondary muscular guarding, brachial plexus stretch in anterior instability events

Probability: Moderate in young athletes with mechanical symptoms and neck guarding post-subluxation

Red Flags: Vascular injury post dislocation (axillary artery), recurrent dislocation with bone loss, nerve deficit after event

Differential Diagnosis: Cervical radiculopathy, RTC tear, biceps pathology, multidirectional instability

Associated Symptoms: Apprehension, clicking, dead arm, neck spasm after overhead, decreased throwing confidence

Physical Examination: Apprehension, laxity testing, labral tests; cervical guarding; periscapular hypertonicity

Clinical Tests: Anterior apprehension, relocation, O'Brien, load and shift; rule out axillary nerve (deltoid) post dislocation

Imaging: MRI arthrogram for labrum; plain films for glenoid bone loss (instability severity)

Evidence: Instability events produce reflexive cervical-scapular guarding; labral pathology and instability rehabilitation requires scapular and rotator cuff retraining

References: Wilk throwing athlete guidelines; Dick et al. instability classification; see PubMed [shoulder instability labrum rehabilitation]

RECORD: Neck pain — cardiac referral (atypical)

ID: REF-018

Pain Location: Neck, jaw angle, upper trapezius — referred from myocardial ischemia

Possible Sources: Angina, ACS, pericarditis

Probability: Low in isolation but elevated with exertional pattern and risk factors

Red Flags: Exertional onset, chest pressure, diaphoresis, nausea, known CAD, diabetes

Differential Diagnosis: Cervical facet pain, cervicogenic headache, MSK strain

Associated Symptoms: Dyspnea on exertion, fatigue, arm heaviness concurrent

Physical Examination: Neck ROM may be full; pain not reproduced segmentally; vitals and cardiac history critical

Clinical Tests: No MSK test diagnostic; treadmill history; ECG if suspected

Imaging: Cardiac workup not cervical MRI first if ACS suspected

Evidence: Women and diabetics more atypical neck/jaw referral without chest pain

References: AHA ACS atypical presentation; see PubMed [neck pain cardiac referral]

RECORD: Thoracic pain — pulmonary referral

ID: REF-019

Pain Location: Thoracic back, chest wall, shoulder referral pleuritic

Possible Sources: Pneumonia, pleurisy, PE, pneumothorax

Probability: Moderate when fever, cough, dyspnea present

Red Flags: Hemoptysis, tachycardia, hypoxia, unilateral leg swelling, sudden dyspnea

Differential Diagnosis: Thoracic facet pain, costochondritis, MSK strain

Associated Symptoms: Cough, fever, pleuritic pain with inspiration

Physical Examination: Decreased breath sounds, crackles, pleural rub; thoracic movement may worsen MSK component only

Clinical Tests: Auscultation paramount; thoracic PA may reproduce MSK not pure pleuritic

Imaging: Chest X-ray, CT-PA if PE suspected

Evidence: Pleuritic pain requires pulmonary exclusion before chronic MSK treatment

References: See PubMed [pleuritic pain referral thoracic]

RECORD: Elbow pain — radial tunnel vs local epicondyle

ID: REF-020

Pain Location: Lateral forearm distal to epicondyle — radial nerve vs LE epicondyle

Possible Sources: Radial tunnel syndrome, lateral epicondylalgia, cervical C6

Probability: Radial tunnel ~5% lateral elbow pain; distinguish by epicondyle tenderness vs distal

Red Flags: Progressive motor weakness wrist/finger extension

Differential Diagnosis: LE, cervical radiculopathy, PIN palsy

Associated Symptoms: Pain worse at night forearm; aching without clear epicondyle point tenderness

Physical Examination: Tenderness 3-4 cm distal to epicondyle over radial nerve; resisted supination pain

Clinical Tests: Tinel radial tunnel, Cozen at epicondyle comparison, Spurling cervical

Imaging: MRI if mass; NCS if equivocal; ultrasound local tendons

Evidence: Radial tunnel diagnosis clinical primarily; surgical decompression controversial outcomes

References: See PubMed [radial tunnel vs lateral epicondylalgia]

RECORD: Wrist pain — cervical C6-C7 referral

ID: REF-021

Pain Location: Dorsal wrist, thumb, radial hand — C6; middle finger dorsum C7

Possible Sources: Cervical radiculopathy, CTS overlap C6-C7, de Quervain local

Probability: Moderate when wrist tests inconclusive and neck positive

Red Flags: Progressive weakness, myelopathy signs

Differential Diagnosis: CTS, de Quervain, scaphoid fracture, TFCC

Associated Symptoms: Neck pain rotation, dermatomal numbness, symptoms with overhead neck posture

Physical Examination: Compare Phalen vs Spurling provocation; sensory mapping

Clinical Tests: ULTT median, Spurling, Finkelstein, scaphoid compression

Imaging: Cervical MRI if radiculopathy; wrist X-ray FOOSH

Evidence: Double crush cervical and wrist co-occurrence documented

References: Wainner upper quarter; see PubMed [C6 radiculopathy wrist pain]

RECORD: Finger pain — cervical C8-T1 referral

ID: REF-022

Pain Location: Ring/small finger, medial hand intrinsic — C8-T1 dermatomal

Possible Sources: Cervical radiculopathy C8-T1, ulnar neuropathy cubital tunnel, TOS

Probability: Moderate with neck signs and ulnar distribution without elbow tenderness

Red Flags: Intrinsic wasting, progressive weakness, Pancoast apical tumor

Differential Diagnosis: Cubital tunnel, TOS, jersey finger local, trigger finger
Associated Symptoms: Neck pain, hand clumsiness, dropping objects
Physical Examination: Intrinsic strength, Froment sign, Tinel elbow vs cervical tension
Clinical Tests: Spurling, elbow flexion test, Tinel cubital, ULTT ulnar bias
Imaging: Cervical MRI; NCS ulnar; chest if TOS tumor concern
Evidence: C8-T1 radiculopathy mimics ulnar neuropathy — differentiate by provocation site
References: See PubMed [C8 radiculopathy ulnar hand]

RECORD: Shoulder pain — biceps long head source

ID: REF-023

Pain Location: Anterior shoulder biceps groove, deep anterior joint
Possible Sources: Biceps tendinopathy, SLAP lesion, instability-related biceps load
Probability: Common anterior shoulder pain source especially overhead athletes
Red Flags: Sudden pop proximal biceps deformity Popeye sign
Differential Diagnosis: Subscapularis, labral tear, AC joint, cervical
Associated Symptoms: Speed test pain, clicking anterior, pain resisted supination elbow flexion
Physical Examination: Palpation bicipital groove tenderness; Speed's, Yergason's tests
Clinical Tests: Speed's, Yergason's, O'Brien SLAP overlap, empty can subscap differential
Imaging: MRI biceps sheath; ultrasound dynamic
Evidence: Biceps tendinopathy often coexists cuff pathology; treat both
References: See PubMed [biceps tendinopathy shoulder]

RECORD: Neck pain — thoracic outlet referral pattern

ID: REF-024

Pain Location: Neck, supraclavicular fossa, medial arm, ulnar hand
Possible Sources: Neurogenic TOS, vascular TOS, cervical rib, scalene hypertrophy
Probability: Low overall; higher overhead workers musicians athletes with posture load
Red Flags: Vascular TOS ischemia color change; progressive atrophy; Pancoast tumor
Differential Diagnosis: Cervical radiculopathy, cubital tunnel, MSK neck pain
Associated Symptoms: Arm heaviness overhead, numbness ulnar digits, symptoms worse carrying loads
Physical Examination: Roos test, Adson maneuver (interpret cautiously), supraclavicular tenderness, posture forward head rounded shoulders
Clinical Tests: Roos/EAST, ULTT, cervical Spurling, Tinel supraclavicular
Imaging: MRI brachial plexus; chest X-ray cervical rib; NCS
Evidence: Neurogenic TOS diagnosis clinical primarily; conservative posture and nerve glides first line many cases
References: Povlsen TOS review; see PubMed [neurogenic TOS neck pain]

RECORD: Thoracic pain — gallbladder visceral referral

ID: REF-025

Pain Location: Right interscapular, right shoulder tip, epigastric/RUQ primary
Possible Sources: Gallbladder inflammation, biliary colic, hepatic flexure
Probability: Consider when thoracic pain post fatty meals with RUQ symptoms
Red Flags: Fever, jaundice, Murphy sign, peritoneal signs
Differential Diagnosis: Thoracic facet, rib dysfunction, MSK strain

Associated Symptoms: Nausea, bloating, right shoulder tip pain with abdominal primary

Physical Examination: RUQ tenderness; thoracic PA may not reproduce if pure visceral

Clinical Tests: Murphy sign abdominal; inspiration shoulder pain phrenic pattern

Imaging: RUQ ultrasound gallstones; abdominal CT if complicated

Evidence: Phrenic referral classic right shoulder from hepatobiliary irritation

References: See PubMed [visceral referred thoracic shoulder pain]

RECORD: Shoulder pain — subacromial bursitis pattern (local)

ID: REF-026

Pain Location: Lateral shoulder subacromial region painful arc distribution

Possible Sources: Subacromial bursa inflammation secondary to RTC tendinopathy impingement mechanics calcific tendinitis

Probability: Very common as component subacromial pain syndrome often coexists cuff tendinopathy rather than isolated bursitis

Red Flags: Hot swollen shoulder sepsis; acute trauma fracture; systemic infection

Differential Diagnosis: RTC tear capsulitis AC joint OA cervical referral biceps tendinopathy

Associated Symptoms: Painful arc 60-120 night pain side lying catching crepitus occasionally

Physical Examination: Neer Hawkins painful arc positive; injection test may temporarily relieve if purely bursal component debatable

Clinical Tests: Imp ingement cluster empty can; distinguish instability labral mechanical symptoms

Imaging: X-ray acromion morphology calcification; ultrasound bursal effusion cuff; MRI if persistent

Evidence: Bursitis often secondary; subacromial injection short-term relief adjunct exercise primary long-term

References: See REF-006 rotator cuff; Michener subacromial pain

2. Differential Diagnosis by Chief Complaint

RECORD: DDx — Neck Pain

ID: DDX-001

Chief Complaint: Neck Pain

Possible Diagnoses: 1) Mechanical neck pain / facet dysfunction (most common). 2) Cervical radiculopathy. 3) Whiplash-associated disorder. 4) Cervicogenic headache. 5) Myofascial pain syndrome. 6) Cervical spondylosis. 7) TOS. 8) Inflammatory spondyloarthritis. 9) Malignancy (must-not-miss). 10) Spinal infection.

Likelihood: Common → less common → must-not-miss ranking as listed; mechanical WAD/facet highest in outpatient MSK

Red Flags: Myelopathy (gait, Hoffman, Babinski, Lhermitte), fever, IV drug use, immunosuppression, history of cancer, trauma with neuro deficit, progressive bilateral symptoms, night pain unrelieved, weight loss

Yellow Flags: Prior neck injury, sedentary work, stress, depression, compensation claim (WAD), repeated microtrauma, poor ergonomics, sleeping posture

Key Clinical Tests: Spurling, ULTT, cervical ROM, segmental mobility, neuro screen (reflexes, myotomes, dermatomes), upper cervical stability if trauma, TOS tests if arm symptoms

Imaging: X-ray post-trauma or if instability suspected; MRI if radiculopathy >6 weeks, progressive deficit, or red flags; no routine imaging uncomplicated mechanical neck pain

Referral Criteria: Emergency: myelopathy, cauda equina equivalent cervical cord compression, fracture, infection. Urgent: progressive radiculopathy, vascular dissection post trauma. Routine ortho/spine if failed 6-12 weeks conservative

Evidence: Majority mechanical neck pain improves within 6 weeks; exercise and manual therapy modest benefit; imaging does not improve outcomes uncomplicated cases

References: Childs et al. neck pain guidelines; Blanpied et al. APTA CPG; see PubMed [neck pain differential diagnosis]

RECORD: DDx — Shoulder Pain

ID: DDX-002

Chief Complaint: Shoulder Pain

Possible Diagnoses: 1) Rotator cuff tendinopathy/partial tear. 2) Subacromial pain syndrome. 3) Adhesive capsulitis. 4) AC joint pathology. 5) Biceps tendinopathy. 6) Glenohumeral OA. 7) Instability/labral tear. 8) Cervical radiculopathy referral. 9) Referred visceral/cardiac. 10) Septic arthritis / fracture (must-not-miss).

Likelihood: RTC tendinopathy and subacromial pain most common age 40+; instability/labral in young athletes; capsulitis middle-aged females

Red Flags: Acute trauma with deformity, fever/hot joint, inability to lift arm (massive RTC tear), neurovascular compromise post dislocation, night sweats, cancer history, chest pain/dyspnea (cardiac/pulmonary)

Yellow Flags: Overhead occupation/sport, diabetes (capsulitis), prior shoulder injury, scapular dyskinesis, smoking (cuff healing), psychosocial factors

Key Clinical Tests: Neer, Hawkins, painful arc, empty can, ER lag, apprehension/relocation, Speed's/biceps, cross-body AC test, cervical screen, neurovascular

Imaging: X-ray baseline many chronic cases; ultrasound/MRI cuff/labrum when surgery contemplated or persistent >3-6 months; MRI arthrogram labrum

Referral Criteria: Emergency: dislocation with neurovascular injury, sepsis, fracture. Urgent: acute massive tear young patient. Refer visceral if systemic/cardiac signs

Evidence: Exercise-based rehab first-line RTC tendinopathy; subacromial decompression not superior to conservative care many cases per UK FIT trial literature

References: Michener shoulder exam; AAOS cuff guidelines; see PubMed [shoulder pain differential]

RECORD: DDx — Thoracic Pain

ID: DDX-003

Chief Complaint: Thoracic Pain

Possible Diagnoses: 1) Thoracic facet dysfunction / mechanical pain. 2) Costovertebral/costotransverse strain. 3) Scheuermann / postural pain. 4) Compression fracture (osteoporotic). 5) Disc herniation (rare). 6) T4 syndrome. 7) Referred cardiac/aortic. 8) Pulmonary pathology. 9) Spinal infection/malignancy.

Likelihood: Mechanical thoracic pain common but often underdiagnosed; fracture in elderly osteoporotic; serious pathology low but high impact

Red Flags: Age >50 new unexplained pain, cancer history, fever, IV drug use, progressive neuro deficit, trauma, osteoporosis, corticosteroid use, night pain, weight loss, cardiac symptoms

Yellow Flags: Prolonged sitting, kyphotic posture, heavy lifting, repetitive rotation, stress, inadequate physical activity

Key Clinical Tests: Segmental PA pressure, rib springing, thoracic rotation ROM, repeated movement testing, neuro screen, cardiac screen if atypical

Imaging: X-ray if fracture suspected or trauma; MRI if infection, malignancy, progressive neuro; CT if complex fracture

Referral Criteria: Emergency: aortic dissection, cord compression, unstable fracture. Urgent: compression fracture severe pain, infection suspicion

Evidence: Thoracic spine pain often postural/mechanical; manipulation/mobilization modest short-term benefit; red flag screen essential due to visceral referral overlap

References: Heneghan thoracic manipulation review; Finucane T4 syndrome; see PubMed [thoracic spine pain differential]

RECORD: DDx — Elbow Pain

ID: DDX-004

Chief Complaint: Elbow Pain

Possible Diagnoses: 1) Lateral epicondylalgia (LE). 2) Medial epicondylalgia (thrower's elbow). 3) Olecranon bursitis. 4) UCL sprain/tear. 5) Posterolateral rotatory instability. 6) Radial tunnel syndrome. 7) Cervical radiculopathy C6-C7. 8) Septic bursitis/arthritis. 9) Fracture (must-not-miss).

Likelihood: LE most common in general population; UCL/medial epicondylalgia in throwers; fracture post fall on outstretched hand

Red Flags: Acute trauma deformity, fever, hot swollen joint, inability to move after fall, neurovascular compromise, open wound

Yellow Flags: Repetitive gripping, racquet sports, manual labor, throwing, smoking, prior tendinopathy

Key Clinical Tests: Cozen (LE), Mill (ME), Maudsley, valgus stress (UCL), moving valgus stress, Tinel radial tunnel, cervical screen

Imaging: X-ray if trauma, arthritis, loose bodies; MRI UCL throwers; ultrasound tendons; MRI cervical if radiculopathy

Referral Criteria: Emergency: fracture-dislocation, sepsis. Urgent: complete UCL tear elite thrower, PLRI, progressive neuro deficit cervical

Evidence: Eccentric wrist extensor loading effective LE; majority LE self-limiting 6-24 months; throwers need kinetic chain assessment

References: Coombes LE systematic review; Dines elbow throwing injuries; see PubMed [elbow pain differential]

RECORD: DDx — Wrist Pain

ID: DDX-005

Chief Complaint: Wrist Pain

Possible Diagnoses: 1) De Quervain tenosynovitis. 2) Carpal tunnel syndrome. 3) Wrist sprain/TFCC tear. 4) Scaphoid fracture (occult). 5) Wrist OA. 6) Inflammatory arthritis. 7) Ganglion cyst. 8) C6-C7 radiculopathy. 9) Septic arthritis (must-not-miss).

Likelihood: Overuse tenosynovitis and CTS common; scaphoid fracture must not miss after FOOSH

Red Flags: Snuffbox tenderness post FOOSH, fever, hot swollen wrist, inability to bear weight, open injury, progressive thenar atrophy

Yellow Flags: Repetitive typing, pregnancy, diabetes, racquet sports, new motherhood (De Quervain), manual work

Key Clinical Tests: Finkelstein, Phalen, Tinel, scaphoid compression, TFCC load/shift, grip strength, cervical screen, ULTT

Imaging: Scaphoid series X-ray; MRI if occult fracture suspected; NCS/EMG CTS moderate-severe; MRI TFCC

Referral Criteria: Emergency: open fracture, sepsis, acute carpal dislocation. Urgent: scaphoid fracture (AVN risk), complete CTS with thenar atrophy

Evidence: Scaphoid missed fracture leads to nonunion; CTS splinting and activity modification first-line mild-moderate

References: Cooney scaphoid management; AAOS CTS guideline; see PubMed [wrist pain differential]

RECORD: DDx — Finger Pain

ID: DDX-006

Chief Complaint: Finger Pain

Possible Diagnoses: 1) Trigger finger (stenosing tenosynovitis). 2) MCP/PIP/DIP sprain. 3) Jersey finger (FDP avulsion). 4) Mallet finger (extensor avulsion). 5) Boutonniere/swans neck deformity. 6) Osteoarthritis (DIP Heberden, PIP Bouchard). 7) Inflammatory arthritis. 8) Flexor sheath infection (must-not-miss). 9) Tumor (rare).

Likelihood: Trigger finger and OA common middle age; jersey finger in contact sports; flexor tenosynovitis Kanavel signs emergency

Red Flags: Kanavel signs (flexed posture, fusiform swelling, tenderness sheath, pain passive extension), open wound, neurovascular injury, rapid progression

Yellow Flags: Climbing, gripping sports, manual labor, diabetes (trigger), prior finger injury

Key Clinical Tests: Trigger locking, Bunnell test (flexor sheath infection), Allen test, collateral ligament stress, extensor/flexor tendon function

Imaging: X-ray avulsion fractures mallet/jersey; ultrasound tendon; MRI complex ligament injuries

Referral Criteria: Emergency: flexor sheath infection, open fracture, compartment syndrome. Urgent: jersey finger surgical, mallet with large avulsion

Evidence: Early antibiotics and drainage flexor infection prevents tendon necrosis; mallet splinting 6-8 weeks if closed

References: Doyle mallet finger; Manske jersey finger; see PubMed [finger pain differential]

RECORD: DDx — Numbness

ID: DDX-007

Chief Complaint: Numbness

Possible Diagnoses: 1) Carpal tunnel syndrome. 2) Cervical radiculopathy. 3) Ulnar neuropathy at cubital tunnel. 4) Radial sensory branch irritation. 5) TOS. 6) Peripheral neuropathy (diabetes). 7) Double crush syndrome. 8) Stroke/TIA (must-not-miss). 9) Cervical myelopathy.

Likelihood: Entrapment neuropathies most common localized; cervical radiculopathy if dermatomal; stroke if acute face/arm/leg pattern

Red Flags: Acute onset hemiparesis, facial droop, speech difficulty (stroke); progressive weakness; bilateral symptoms; saddle anesthesia; rapid ascending weakness (GBS); Lhermitte; bowel/bladder

Yellow Flags: Diabetes, pregnancy, repetitive vibration, ergonomic stress, neck trauma history, alcohol use, metabolic syndrome

Key Clinical Tests: Phalen/Tinel, ULTT, Tinel cubital tunnel, elbow flexion test, Spurling, 2-point discrimination, monofilament if systemic neuropathy suspected, FAST stroke screen if acute

Imaging: NCS/EMG localize entrapment; cervical MRI if radiculopathy/myelopathy; brain imaging if stroke suspected

Referral Criteria: Emergency: stroke, cauda equina, cord compression. Urgent: progressive CTS weakness, cervical radiculopathy with motor loss

Evidence: Pattern recognition by distribution critical; double crush common cervical + peripheral coexisting

References: Wainner upper quarter; AAOS CTS; see PubMed [upper extremity numbness differential]

RECORD: DDx — Weakness

ID: DDX-008

Chief Complaint: Weakness

Possible Diagnoses: 1) Rotator cuff tear. 2) Cervical radiculopathy. 3) Brachial neuritis (Parsonage-Turner). 4) CTS (thenar). 5) Ulnar neuropathy (intrinsic). 6) Stroke. 7) Myelopathy. 8) Tendon rupture (biceps/long head, FDP jersey). 9) Neuromuscular disease (rare).

Likelihood: Pain inhibition vs true weakness must differentiate; acute painless weakness brachial neuritis; stroke time-sensitive

Red Flags: Acute facial/arm weakness stroke; fever; painless rapid onset shoulder (brachial neuritis); stepwise myelopathy; trauma with deformity

Yellow Flags: Recent viral illness (brachial neuritis), diabetes, repetitive strain, prior shoulder injury, anticoagulation (compartment/spontaneous)

Key Clinical Tests: Manual muscle testing myotomes, cuff tear lag signs, grip/pinch, intrinsic hand tests, reflexes, Babinski, FAST if acute neuro

Imaging: MRI shoulder cuff; cervical MRI; NCS/EMG; brain MRI if central cause; labs if myopathy suspected

Referral Criteria: Emergency: stroke, cord compression, cauda equina. Urgent: complete tendon rupture, progressive radiculopathy, brachial plexus tumor

Evidence: True weakness indicates neurologic or structural tear — differs from pain inhibition on testing

References: McComas muscle testing; AAOS neuromuscular; see PubMed [upper limb weakness differential]

RECORD: DDx — Instability

ID: DDX-009

Chief Complaint: Instability

Possible Diagnoses: 1) Glenohumeral instability (traumatic anterior). 2) MDI (multidirectional). 3) AC joint separation. 4) SC joint instability (rare). 5) Elbow PLRI. 6) MCP collateral instability. 7) Hypermobility spectrum / EDS. 8) Seizure-related posterior shoulder dislocation.

Likelihood: Traumatic anterior GH instability young males contact sports; MDI hypermobility phenotype; AC sprain fall on shoulder

Red Flags: First-time dislocation with neurovascular compromise; open injury; posterior dislocation (seizure, electric) easily missed; SC posterior dislocation vascular

Yellow Flags: Generalized ligamentous laxity (Beighton), overhead sports, contact sport, seizure disorder, prior dislocations

Key Clinical Tests: Apprehension, relocation, load and shift, sulcus sign, Beighton score, AC joint stress, elbow pivot shift (PLRI), drawer tests digits

Imaging: X-ray pre/post reduction; MRI arthrogram labrum; CT glenoid bone loss if recurrent

Referral Criteria: Emergency: dislocation unable reduce, vascular injury. Urgent: recurrent instability young athlete, significant bone loss

Evidence: Recurrence high under age 25; rehab focuses rotator cuff + scapular control; surgery if recurrent or structural deficit

References: Dick instability guidelines; Wilk rehab phases; see PubMed [shoulder instability differential]

RECORD: DDx — Loss of ROM

ID: DDX-010

Chief Complaint: Loss of ROM

Possible Diagnoses: 1) Adhesive capsulitis. 2) Post-surgical stiffness. 3) Osteoarthritis. 4) Cervical radiculopathy guarding. 5) RTC tear pain-limited ROM. 6) Fracture/ malunion. 7) Complex regional pain syndrome. 8) Septic arthritis. 9) Charcot (rare).

Likelihood: Capsulitis classic progressive loss active and passive; OA older gradual; fracture acute post trauma

Red Flags: Hot swollen joint, fever, trauma with deformity, CRPS signs (color change, allodynia), neuro deficit, night sweats

Yellow Flags: Diabetes, thyroid disease, immobilization, prior shoulder surgery, stroke hemiplegia

Key Clinical Tests: Active vs passive ROM comparison, capsular pattern external rotation loss (capsulitis), end-feel assessment, neuro screen

Imaging: X-ray OA/fracture; MRI if cuff/labrum; labs if inflammatory/septic suspected

Referral Criteria: Emergency: sepsis, dislocation. Urgent: CRPS early referral, frozen shoulder severe disability >6 months failed PT

Evidence: Capsulitis phases: freezing painful, frozen stiff, thawing recovery 1-3 years; early mobilization within pain tolerance

References: Kelley adhesive capsulitis; Neviaser staging; see PubMed [shoulder stiffness differential]

RECORD: DDx — Night Pain

ID: DDX-011

Chief Complaint: Night Pain

Possible Diagnoses: 1) Rotator cuff tendinopathy/tear. 2) Adhesive capsulitis. 3) Shoulder OA. 4) Metastatic bone disease (must-not-miss). 5) Infection. 6) CRPS. 7) Cardiac ischemia (atypical). 8) Cervical radiculopathy. 9) Inflammatory arthritis.

Likelihood: Shoulder night pain lying on side very common cuff/capsulitis; age >50 unremitting night pain consider malignancy

Red Flags: Unremitting night pain, weight loss, fever, history cancer, rest pain chest (cardiac), progressive neuro deficit, pathologic fracture

Yellow Flags: Diabetes, prior cancer, smoking, inflammatory conditions, chronic shoulder disease

Key Clinical Tests: Shoulder exam cuff/capsule; systemic review; palpation for focal bone tenderness; neuro screen; vitals

Imaging: X-ray shoulder + consider labs ESR/CRP; MRI if tumor suspected; bone scan oncology pathway; cardiac workup if atypical

Referral Criteria: Emergency: pathologic fracture, sepsis, cardiac ACS. Urgent: unexplained night pain >4 weeks with systemic signs → oncology

Evidence: Night pain alone rarely indicates cancer but persistent unrelenting pain with red flags mandates imaging

References: British red flags night pain; AAOS oncology bone lesions; see PubMed [night pain musculoskeletal red flags]

3. Outcome Measures

RECORD: DASH (Disabilities of the Arm, Shoulder and Hand)

ID: OM-001

Outcome Measure: DASH (Disabilities of the Arm, Shoulder and Hand)

Region: Full upper extremity

Purpose: Patient-reported function and symptoms for any UE condition

Scoring: 30 items (optional 4 work/sport modules); each 1-5; score = $((\text{sum}/n)-1)*25$; 0=no disability, 100=most disability

Interpretation: Higher = worse; normative ~10 in general population; clinically meaningful improvement when score drops

MCID: MCID ~10-15 points (approximate; literature range 8-14 depending population)

MDC: MDC ~10 points (90% CI; varies by study ~8-12)

Reliability: ICC 0.96 test-retest; internal consistency alpha >0.94

Validity: Construct validity vs SF-36, pain scales; responsive in shoulder, hand, mixed UE

Clinical Use: Use intake, progress, discharge; anchor to PSFS for individualized goals; required many research trials

References: Hudak et al. original DASH; Schoneveld et al. MCID review; see PubMed [DASH MCID]

RECORD: QuickDASH

ID: OM-002

Outcome Measure: QuickDASH

Region: Full upper extremity (short form)

Purpose: Shortened DASH for clinical efficiency

Scoring: 11 items same scoring as DASH; optional modules; 0-100 scale

Interpretation: Correlates highly with full DASH ($r>0.90$); similar interpretation

MCID: MCID ~8-12 points (slightly lower threshold than full DASH in some studies)

MDC: MDC ~6-10 points approximate

Reliability: ICC 0.90-0.96; comparable reliability to DASH

Validity: Strong correlation with DASH; validated multiple languages

Clinical Use: Preferred when time constrained; not for individual digit-only conditions if hand-specific needed

References: Gummesson QuickDASH development; see PubMed [QuickDASH reliability]

RECORD: SPADI (Shoulder Pain and Disability Index)

ID: OM-003

Outcome Measure: SPADI (Shoulder Pain and Disability Index)

Region: Shoulder

Purpose: Shoulder-specific pain and disability

Scoring: 13 items: 5 pain + 8 disability; VAS 0-10 each; total 0-100 or average

Interpretation: 0-100 higher worse; minimal clinically important difference often cited 8-10 points

MCID: MCID ~8-10 points (approximate; 10-point rule of thumb common)

MDC: MDC ~8 points (varies 6-10)

Reliability: ICC 0.86-0.95 shoulder pathology populations

Validity: Valid vs clinician-rated severity; responsive post rehab and surgery

Clinical Use: Excellent shoulder-specific clinic tool; pair with ROM and strength

References: Roach et al. SPADI; Paul et al. psychometrics; see PubMed [SPADI MCID]

RECORD: ASES (American Shoulder and Elbow Surgeons score)

ID: OM-004

Outcome Measure: ASES (American Shoulder and Elbow Surgeons score)

Region: Shoulder

Purpose: Combined patient pain and function for shoulder outcomes (often surgical cohorts)

Scoring: Patient self-report portion: pain VAS + 10 function items; standardized to 0-100 higher better

Interpretation: Higher = better function; used heavily orthopaedic surgery research

MCID: MCID ~12-14 points improvement (approximate surgical literature)

MDC: MDC variable ~10-15 points

Reliability: ICC 0.84-0.95 depending version

Validity: Correlates with SF-36; sensitive post rotator cuff repair

Clinical Use: Common post-op shoulder tracking; clinician portion optional in full ASES

References: Richards et al. ASES; see PubMed [ASES shoulder outcome]

RECORD: PRTEE (Patient-Rated Tennis Elbow Evaluation)

ID: OM-005

Outcome Measure: PRTEE (Patient-Rated Tennis Elbow Evaluation)

Region: Elbow (lateral/medial epicondylalgia)

Purpose: Elbow-specific pain and function

Scoring: 15 items: 5 pain + 10 function; 0-10 each; total 0-150 or 0-100 normalized

Interpretation: Higher = worse; used LE primary outcome many trials

MCID: MCID ~11 points on 0-150 scale (~7% change) approximate

MDC: MDC ~9 points approximate

Reliability: ICC 0.89-0.96 epicondylalgia

Validity: Valid vs pain-free controls; responsive eccentric exercise trials

Clinical Use: Gold standard patient-reported elbow outcome LE research and clinic

References: Rompe PRTEE validation; see PubMed [PRTEE MCID]

RECORD: PRWHE (Patient-Rated Wrist/Hand Evaluation)

ID: OM-006

Outcome Measure: PRWHE (Patient-Rated Wrist/Hand Evaluation)

Region: Wrist and hand

Purpose: Wrist/hand pain and function

Scoring: 15 items pain and function; 0-10; total 0-150 normalized 0-100

Interpretation: Higher worse; used fracture, CTS, TFCC cohorts

MCID: MCID ~11-14 points approximate (0-100 scale ~6-8)

MDC: MDC ~8-12 points approximate

Reliability: ICC 0.91-0.95 wrist/hand conditions

Validity: Construct valid vs DASH subset; responsive post distal radius fracture

Clinical Use: Preferred over DASH when wrist/hand primary complaint

References: MacDermid PRWHE; see PubMed [PRWHE psychometrics]

RECORD: MHQ (Michigan Hand Outcomes Questionnaire)

ID: OM-007

Outcome Measure: MHQ (Michigan Hand Outcomes Questionnaire)

Region: Hand

Purpose: Comprehensive hand function including ADL, work, pain, aesthetics

Scoring: 37 items 6 domains; scaled 0-100 higher better overall function

Interpretation: Domain-specific interpretation; overall hand health score

MCID: MCID domain-specific ~10-15 points approximate varies by domain

MDC: MDC varies by subscale ~8-12

Reliability: ICC 0.81-0.94 subscales

Validity: Extensively validated hand trauma, arthritis, CTS

Clinical Use: Use when aesthetic/satisfaction domains relevant; longer than PRWHE

References: Chung MHQ development; see PubMed [MHQ hand outcome]

RECORD: Neck Disability Index (NDI)

ID: OM-008

Outcome Measure: Neck Disability Index (NDI)

Region: Neck

Purpose: Neck-related disability ADL impact

Scoring: 10 items 0-5 each; total 0-50 or percent disability

Interpretation: 0-4 minimal; 5-14 mild; 15-24 moderate; 25-34 severe; >34 complete — classic bands

MCID: MCID ~3.5-5 points (or 10% relative) approximate across studies

MDC: MDC ~5 points (90% CI ~3-7)

Reliability: ICC 0.82-0.95 neck pain populations

Validity: Correlates with SF-36; responsive manual therapy and exercise trials

Clinical Use: Standard neck disability tracking; combine with NPRS and ROM

References: Vernon NDI; MacDermid NDI MCID review; see PubMed [NDI MCID]

RECORD: Numeric Pain Rating Scale (NPRS)

ID: OM-009

Outcome Measure: Numeric Pain Rating Scale (NPRS)

Region: Global / regional

Purpose: Pain intensity quick measure 0-10

Scoring: 0=no pain, 10=worst imaginable; current, average, or worst often recorded

Interpretation: ≥2-point reduction often considered clinically meaningful in MSK (approximate MCID)

MCID: MCID ~2 points (range 1.1-2.5 in literature depending condition)

MDC: MDC ~1.5-2 points approximate

Reliability: ICC 0.70-0.95 depending recall period

Validity: Strong construct validity with other pain scales

Clinical Use: Universal intake/progress metric; specify which anchor (current vs average)

References: Farrar NPRS MCID anchor-based; see PubMed [NPRS minimum clinically important difference]

RECORD: Patient-Specific Functional Scale (PSFS)

ID: OM-010

Outcome Measure: Patient-Specific Functional Scale (PSFS)

Region: Patient-defined activities

Purpose: Individualized functional goals rated by patient

Scoring: Patient nominates up to 3 activities; rate difficulty 0-10 each; average score

Interpretation: Lower = better; change ≥ 2 points per activity often meaningful

MCID: MCID ~2 points per activity (approximate anchor-based)

MDC: MDC ~1.5-2 points approximate

Reliability: ICC 0.82-0.92 various MSK

Validity: High face validity patient-centered; correlates with regional indices

Clinical Use: Essential for goal-oriented rehab; complements DASH/SPADI generic measures

References: Stratford PSFS development; see PubMed [PSFS MCID]

4. Red Flags

RECORD: Fracture

ID: RF-001

Presentation: Acute trauma with deformity, inability to bear weight/use limb, focal bony tenderness, ecchymosis, crepitus

Possible Diagnosis: Clavicle, humerus, scaphoid, radius/ulna, rib, cervical compression fracture

Urgency: Immediate — same day imaging and immobilization

Immediate Referral: Emergency department if open fracture, neurovascular compromise, shoulder/elbow dislocation; urgent orthopaedic fracture clinic otherwise

Emergency Signs: Open wound, pulseless limb, expanding compartment pain, tenting skin, fat pad sign on X-ray with FOOSH

Clinical Features: FOOSH wrist → scaphoid; fall on shoulder → clavicle/proximal humerus; point tenderness snuffbox; painful arc absent in acute fracture vs cuff

Imaging: X-ray standard; CT complex; MRI occult scaphoid if X-ray negative high suspicion

Evidence: Ottawa rules and scaphoid protocols reduce missed fractures; early immobilization improves outcomes

References: Canadian C-spine rules context; Gelberman scaphoid; see PubMed [occult scaphoid fracture]

RECORD: Infection

ID: RF-002

Presentation: Hot swollen joint or tendon sheath, fever, systemic illness, IV drug use, recent injection or surgery

Possible Diagnosis: Septic arthritis, flexor tenosynovitis, cellulitis, osteomyelitis, discitis

Urgency: Emergency — hours matter for joint preservation

Immediate Referral: Emergency department for aspiration, blood cultures, IV antibiotics

Emergency Signs: Kanavel signs (finger), inability to move joint, sepsis vitals, immunocompromised host

Clinical Features: Shoulder/elbow/wrist monoarticular hot swelling; refusal to use limb; CRP/ESR elevated

Imaging: X-ray baseline; MRI if deep infection; ultrasound joint effusion

Evidence: Delay >7 days septic arthritis increases cartilage destruction risk significantly

References: Mathews septic arthritis; see PubMed [septic arthritis upper extremity]

RECORD: Tumor

ID: RF-003

Presentation: Unrelenting night pain, weight loss, history of cancer, progressive pain without injury, pathologic fracture

Possible Diagnosis: Metastatic bone disease, primary bone tumor, Pancoast tumor, myeloma

Urgency: Urgent — oncology workup within days

Immediate Referral: Primary oncologist or ED if pathologic fracture/spinal cord compression

Emergency Signs: Cord compression, cauda equina, hypercalcemia crisis, fracture instability

Clinical Features: Age >50, night pain unrelieved by position, focal bony tenderness, constitutional symptoms

Imaging: X-ray, MRI, CT chest/abdomen/pelvis per pathway; PET-CT oncology

Evidence: Red flag combinations increase likelihood serious pathology; isolated night pain low specificity but mandates vigilance

References: NICE NG59 red flags; Flynn cancer screening; see PubMed [musculoskeletal malignancy red flags]

RECORD: Myelopathy

ID: RF-004

Presentation: Upper motor neuron signs, gait disturbance, hand clumsiness, bilateral symptoms, bowel/bladder dysfunction

Possible Diagnosis: Cervical spondylotic myelopathy, disc herniation central, tumor, MS (CNS)

Urgency: Urgent — neurosurgical evaluation; emergency if rapid progression

Immediate Referral: Neurosurgery or emergency if acute progression; MRI cervical spine

Emergency Signs: Rapid ascending weakness, bowel/bladder retention, saddle anesthesia

Clinical Features: Hoffman positive, hyperreflexia, Babinski, wide-based gait, Lhermitte, difficulty buttoning

Imaging: MRI cervical spine gold standard; avoid high-velocity manipulation

Evidence: Early surgical decompression improves outcomes progressive cervical myelopathy

References: Bono CSM guidelines; see PubMed [cervical myelopathy diagnosis]

RECORD: Stroke

ID: RF-005

Presentation: Acute face/arm/leg weakness, speech difficulty, visual change, sudden severe headache

Possible Diagnosis: Ischemic stroke, hemorrhagic stroke, TIA

Urgency: Emergency — time is brain (thrombolysis windows)

Immediate Referral: Call emergency services; FAST positive → immediate ED

Emergency Signs: Facial droop, arm drift, speech slurring, sudden worst headache (SAH)

Clinical Features: May mimic peripheral weakness but central pattern with face involvement; sensation cortical pattern

Imaging: Non-contrast CT, MRI diffusion; not MSK clinic imaging

Evidence: Every minute delay reduces salvageable brain tissue in large vessel occlusion

References: AHA stroke guidelines; FAST campaign; see PubMed [acute stroke recognition]

RECORD: Acute tendon rupture

ID: RF-006

Presentation: Sudden pop, immediate weakness, ecchymosis, inability to perform specific action

Possible Diagnosis: Distal biceps rupture, triceps rupture, FDP jersey finger, extensor mallet, rotator cuff massive tear post fall

Urgency: Urgent orthopaedic/hand surgery — especially biceps and jersey finger (timing surgical)

Immediate Referral: Hand surgeon jersey/mallet large avulsion; ortho distal biceps within 2-3 weeks optimal

Emergency Signs: Open tendon injury, vascular injury associated

Clinical Features: Hook test biceps; inability flex DIP jersey; mallet DIP droop; drop arm massive cuff

Imaging: MRI or ultrasound confirms; X-ray avulsion fragments

Evidence: Early surgical repair distal biceps and jersey finger improves strength outcomes

References: Mazzocca distal biceps; see PubMed [acute tendon rupture upper extremity]

RECORD: Septic arthritis

ID: RF-007

Presentation: Monoarticular hot swollen joint, fever, refusal to move, recent bacteremia risk

Possible Diagnosis: Bacterial septic arthritis shoulder/elbow/wrist/finger

Urgency: Emergency — joint aspiration within hours

Immediate Referral: ED aspiration and IV antibiotics; orthopaedic consult

Emergency Signs: Sepsis, immunocompromised, prosthetic joint nearby

Clinical Features: Any joint motion excruciating; joint held splinted; CRP/ESR/WBC elevated

Imaging: X-ray effusion indirect; ultrasound-guided aspiration

Evidence: Synovial WBC >50,000 strongly suggests septic arthritis; gout can mimic

References: Goldenberg septic arthritis review; see PubMed [septic arthritis aspiration]

RECORD: Acute vascular compromise

ID: RF-008

Presentation: Pulseless, pale, painful, paresthetic limb after trauma or dislocation

Possible Diagnosis: Axillary/brachial artery injury, compartment syndrome, acute limb ischemia

Urgency: Emergency — limb-threatening

Immediate Referral: Emergency vascular surgery/orthopaedic reduction; do not delay for imaging if vascular injury suspected post dislocation

Emergency Signs: 6 Ps: pain, pallor, pulselessness, paresthesia, paralysis, poikilothermia; compartment pain out of proportion

Clinical Features: Shoulder dislocation with weak radial pulse; supracondylar fracture child pulseless; crush injury

Imaging: Angiography if stable; clinical diagnosis may precede imaging in acute setting

Evidence: Posterior shoulder dislocation post seizure easily missed on X-ray — check neurovascular

References: AAOS vascular injury shoulder dislocation; see PubMed [upper extremity vascular trauma]

RECORD: Progressive neurological deficit

ID: RF-009

Presentation: Worsening weakness, spreading numbness, new foot drop (cervical), ascending symptoms

Possible Diagnosis: Progressive radiculopathy, cord compression, brachial plexopathy tumor, GBS (ascending)

Urgency: Urgent same-week MRI and neuro/ortho referral; emergency if bilateral or bladder involvement

Immediate Referral: MRI; neurology/neurosurgery; ED if rapid progression or bilateral

Emergency Signs: Cauda equina equivalent, bilateral hand weakness, respiratory weakness (GBS)

Clinical Features: Serial exams show declining MMT; reflex changes; map dermatomal spread

Imaging: MRI cervical or brachial plexus; NCS/EMG if entrapment vs radiculopathy unclear

Evidence: Progressive motor deficit from disc herniation may warrant surgical decompression

References: Wainner radiculopathy; see PubMed [progressive cervical radiculopathy]

5. Sport Biomechanics

RECORD: Tennis

ID: SB-001

Sport: Tennis

Movement: Serve (wind-up, cocking, acceleration, follow-through), groundstrokes, volley

Biomechanics: Serve generates highest shoulder/elbow loads: maximum external rotation 170-180°, scapular upward rotation, trunk rotation, kinetic chain from legs; groundstroke eccentric wrist control

Primary Muscles: Rotator cuff, serratus anterior, lower trapezius, latissimus, wrist extensors/flexors, core rotators

Joint Loads: Superior labrum tension during serve; elbow valgus stress; wrist ulnar deviation loading

Common Injuries: Rotator cuff tendinopathy, labral irritation, tennis elbow, wrist extensor strain, low back

Screening: Glenohumeral IR deficit (GIRD), total arc motion, scapular dyskinesis, hip IR, serve volume

Preventive Exercises: Sleeper stretch, cross-body stretch, scapular strengthening, eccentric wrist extensors, core rotation

Return to Sport: Gradual serve progression 50→75→100% intensity; pain-free groundstrokes before full serve volume

Evidence: GIRD associated with shoulder pathology throwers; workload management reduces overuse

References: Ellenbecker tennis shoulder; see PubMed [tennis serve biomechanics injury]

RECORD: Padel

ID: SB-002

Sport: Padel

Movement: Overhead smash, volley at net, lateral cuts, wall rebound strokes

Biomechanics: Repeated overhead similar tennis but shorter court; rapid direction changes; wrist control for glass/wall shots

Primary Muscles: Shoulder girdle stabilizers, forearm pronators/supinators, gastroc/quad deceleration, obliques

Joint Loads: Shoulder overhead compression; elbow from grip; ankle/knee cutting loads

Common Injuries: Shoulder impingement, epicondylalgia, ankle sprains, calf strains, low back

Screening: Overhead ROM, single-leg stability, prior tennis/padel volume history

Preventive Exercises: RTC endurance, scapular control, proprioceptive ankle drills, eccentric forearm

Return to Sport: Progressive smash volume; match play minutes capped early return

Evidence: Rapid growth sport; overuse patterns mirror racquet sports with added agility demands

References: Emerging padel injury epidemiology; see PubMed [padel sports injuries]

RECORD: Baseball

ID: SB-003

Sport: Baseball (throwing)

Movement: Pitching wind-up through follow-through; batting; base running

Biomechanics: Pitching peak elbow valgus torque; shoulder IR velocity extreme; scapular protraction/retraction timing critical; 6-phase throwing model

Primary Muscles: RTC, serratus, lower trap, lat, triceps, forearm flexors/pronators, hip rotators

Joint Loads: UCL medial elbow; labrum anterior; GIRD adaptive changes

Common Injuries: UCL sprain, medial epicondylalgia, labral tears, RTC impingement, oblique strains

Screening: Total arc motion, GIRD, hip-shoulder separation, pitch count age limits

Preventive Exercises: Throwers ten, plyometric long toss progression controlled, hip mobility, scapular strengthening

Return to Sport: Interval throwing program flat ground → mound → game simulation

Evidence: ASMI interval throwing standard; pitch count guidelines reduce youth injury

References: Wilk throwing rehabilitation; Fleisig pitching biomechanics; see PubMed [baseball UCL injury]

RECORD: Volleyball

ID: SB-004

Sport: Volleyball

Movement: Spike, serve, block, dig, set

Biomechanics: Spike: approach jump, arm cocking, rapid IR/acceleration; landing forces 4-6× body weight; repetitive overhead

Primary Muscles: Shoulder accelerators/decelerators, finger/wrist extensors, quads/calf for jump, core

Joint Loads: Shoulder labrum; finger pulp loading; patellar tendons; ankle landing

Common Injuries: Shoulder impingement, finger sprains/jams, patellar tendinopathy, ankle sprains, low back

Screening: Shoulder IR/ER balance, jump landing mechanics, finger strength

Preventive Exercises: Scapular strengthening, landing neuromuscular training, finger flexor/extensor balance

Return to Sport: Spike progression overhead → game speed; block jump volume managed

Evidence: Landing training reduces ACL/ankle injury in jump sports; shoulder workload key overhead athletes

References: Reeser volleyball injuries; see PubMed [volleyball shoulder overuse]

RECORD: Swimming

ID: SB-005

Sport: Swimming

Movement: Freestyle, backstroke, butterfly, breaststroke pull and kick cycles

Biomechanics: High repetition shoulder arcs; freestyle body roll; butterfly extreme extension; breaststroke whip kick knee valgus

Primary Muscles: Lat, pec major, serratus, subscapularis, triceps, core, hip adductors (breaststroke)

Joint Loads: Shoulder impingement positions repeated; sternoclavicular stress; cervical extension (breath)

Common Injuries: Swimmer shoulder (impingement), breaststroker knee, low back spondylolysis, neck strain

Screening: Shoulder IR, thoracic extension, streamlining pain, weekly yardage

Preventive Exercises: RTC subscapularis emphasis, thoracic mobility, scapular control, hip IR

Return to Sport: Gradual yardage 25% increments; pain-free drill progressions before meet

Evidence: Training volume strongest predictor swimmer shoulder pain

References: Tovin swimmer shoulder; see PubMed [swimming overuse shoulder]

RECORD: Golf

ID: SB-006

Sport: Golf

Movement: Full swing backswing/downswing/follow-through; putting; chipping

Biomechanics: Lead shoulder adduction/across body in follow-through; trail wrist ulnar deviation; trunk rotation 90°+; ground reaction forces

Primary Muscles: Lead shoulder rotators, trail side oblique, hip rotators, forearm pronators, grip muscles

Joint Loads: Lead elbow epicondylar stress; lumbar rotation shear; wrist ulnar deviation

Common Injuries: Medial epicondylalgia lead arm, low back pain, wrist injury, shoulder AC pain

Screening: Thoracic rotation, hip IR bilateral, single leg balance, grip strength asymmetry

Preventive Exercises: Thoracic rotation drills, hip mobility, core anti-rotation, wrist flexibility

Return to Sport: Driving range short iron → driver progression; putt/chip early low load

Evidence: Professional golfers high lumbar loads; amateur lead wrist injuries common

References: McCarroll golf injuries; see PubMed [golf swing biomechanics injury]

RECORD: Basketball

ID: SB-007

Sport: Basketball

Movement: Jumping, landing, shooting, dribbling, contact, cutting

Biomechanics: Repeated jump-landing; overhead shooting smaller arc than volleyball; finger ball control; contact falls

Primary Muscles: Quads/glutes deceleration, calf, shoulder stabilizers for shooting, finger flexors

Joint Loads: Ankle/knee landing; shoulder external rotation shooting; finger axial loads

Common Injuries: Ankle sprains, finger jams, patellar tendinopathy (jumper knee), shoulder strain, concussions

Screening: Landing mechanics, ankle stability, shoulder ROM shooting side

Preventive Exercises: NEUROMUSCULAR landing training, proprioception, finger taping protocol team, eccentric quad

Return to Sport: Jump progression; shooting form before contact drills

Evidence: Balance/landing programs reduce ankle sprain recurrence

References: McKay basketball injury prevention; see PubMed [basketball ankle prevention]

RECORD: Soccer Goalkeeper

ID: SB-008

Sport: Soccer Goalkeeper

Movement: Diving saves, jumping punches, throwing distribution, kicking

Biomechanics: Repeated unilateral landings from dives; shoulder overhead throw; hip abductor loading lateral dives; wrist/finger impact ground

Primary Muscles: Shoulder girdle, core anti-rotation, hip abductors/adductors, wrist/finger grip

Joint Loads: Shoulder labrum throwing; elbow from throws; wrist impact; thumb UCL gamekeeper thumb

Common Injuries: Shoulder impingement, finger sprains, thumb UCL, hip/groin, elbow tendinopathy, concussion

Screening: Throwing volume, landing side dominance pain, thumb stability

Preventive Exercises: Scapular endurance, wrist/finger strengthening, hip stability, dive technique coaching

Return to Sport: Throwing distance progression; dive intensity low → match simulation

Evidence: Goalkeepers unique UE loading vs outfield; impact injuries finger/thumb common

References: FIFA goalkeeper injury reports; see PubMed [goalkeeper upper extremity injury]

RECORD: Rock Climbing

ID: SB-009

Sport: Rock Climbing

Movement: Crimp, sloper, pinch, pocket grips; dynamic dynos; shoulder-heavy pulling

Biomechanics: Finger flexor pulley loads highest; shoulder closed-chain pulling; elbow flexion sustained isometric

Primary Muscles: Finger flexors (FDS/FDP), forearm flexors, lat, biceps, shoulder stabilizers, core tension

Joint Loads: A2/A4 pulleys; elbow medial epicondyle; shoulder labrum chronic pullers

Common Injuries: Pulley rupture/sprain, APE index finger, tennis/golfer elbow, shoulder impingement, lumbrical strains

Screening: Finger strength ratio, shoulder IR deficit, training load (sessions/week)

Preventive Exercises: Progressive hangboard loading, antagonist extensor training, shoulder external rotation endurance

Return to Sport: Grades lowered 2-3 levels; crimp avoidance early; volume 50% first 2 weeks

Evidence: Acute pulley injury ~40% climbing injuries; load management critical

References: Schweizer pulley injury; see PubMed [rock climbing finger injury]

RECORD: CrossFit

ID: SB-010

Sport: CrossFit

Movement: Kipping pull-ups, muscle-ups, Olympic lifts, handstand walks, high-rep push press

Biomechanics: High volume mixed modal; kipping creates shoulder extension peak loads; wrist extension in front rack/clean

Primary Muscles: Full kinetic chain; shoulder girdle high demand; forearm grip; core bracing

Joint Loads: Labrum/RTC kipping; lumbar with lifting; wrist rack position

Common Injuries: Shoulder impingement/labral, low back strain, wrist tendinopathy, rhabdo (systemic red flag)

Screening: Movement quality strict before kipping; load tolerance history

Preventive Exercises: Strict strength prerequisites, scapular control, mobility prerequisites for OH positions

Return to Sport: Strict movements pain-free before kipping reintroduced; scaled WODs

Evidence: Technique and volume progression reduce shoulder injury in high-rep gymnastics movements

References: Weisenthal CrossFit injury survey; see PubMed [CrossFit shoulder injury]

RECORD: Weightlifting

ID: SB-011

Sport: Weightlifting (Olympic)

Movement: Snatch, clean and jerk, front rack, overhead squat

Biomechanics: Triple extension power; rapid elbow/wrist extension catch; shoulder stability overhead maximal loads

Primary Muscles: Quads, glutes, traps, deltoids, triceps, forearm extensors, core

Joint Loads: Wrist extension rack; elbow valgus clean; shoulder impingement overhead squat mobility limited

Common Injuries: Low back, wrist extension pain, shoulder impingement, elbow tendinopathy, patellar

Screening: Ankle/hip/shoulder mobility for overhead squat; wrist extension tolerance

Preventive Exercises: Thoracic/shoulder mobility, wrist prep, positional strength, technique coaching

Return to Sport: Technique bar → percentage 1RM progression; avoid max attempts until mobility restored

Evidence: Mobility restrictions drive compensatory lumbar/wrist stress in overhead positions

References: Calder weightlifting injury profile; see PubMed [Olympic weightlifting injury]

6. Prevention Programs

RECORD: FIFA 11+

ID: PV-001

Program: FIFA 11+ warm-up

Target Population: Soccer players age 14+

Evidence: RCTs show ~30-50% reduction lower extremity injuries when compliance high

Exercises: Running, plyometrics, strength (squats, Nordic, bench), balance, agility — structured 20 min

Frequency: Every training session and pre-match warm-up

Progression: Level 1 beginner → level 2 → level 3 over season; coach-led

Outcome: Reduced ACL, hamstring, ankle injuries; improved neuromuscular control

References: Soligard FIFA 11+ BMJ; see PubMed [FIFA 11 injury prevention]

RECORD: Shoulder injury prevention for overhead athletes

ID: PV-002

Program: Throwing/overhead athlete shoulder program

Target Population: Baseball, volleyball, tennis, swimming athletes

Evidence: GIRD management and scapular strengthening associated reduced shoulder injury rates in cohort studies

Exercises: Sleeper stretch, cross-body, external rotation strengthening, rows, serratus punches, deceleration drills

Frequency: 3-4×/week in-season; daily mobility micro-doses

Progression: Mobility → isometric → isotonic → plyometric throw progression

Outcome: Maintain total arc motion; reduce impingement symptoms; improve scapular control

References: Ellenbecker comprehensive shoulder; Wilk prevention; see PubMed [overhead athlete shoulder prevention]

RECORD: Throwing programs

ID: PV-003

Program: Interval throwing / long toss progression

Target Population: Baseball pitchers and throwers returning or preseason

Evidence: Structured throw counts reduce re-injury vs unstructured return

Exercises: Flat ground throws → long toss → mound → bullpen → game; mandatory rest days

Frequency: 3-4 throwing days/week with rest; pitch counts age-based youth

Progression: ASMI interval throwing program standard distances and steps

Outcome: Safe return post UCL/shoulder injury; workload management

References: ASMI interval throwing; Fleisig pitch counts; see PubMed [throwing program return]

RECORD: Wrist injury prevention

ID: PV-004

Program: Wrist/forearm balance program

Target Population: Gymnasts, CrossFit, racquet sports, manual workers

Evidence: Eccentric and antagonist training reduces tendinopathy recurrence modestly

Exercises: Wrist extensor/flexor eccentric, pronation/supination, grip gradation, rack mobility drills

Frequency: 3×/week 10-15 min adjunct to sport

Progression: Isometric → eccentric → loaded sport-specific

Outcome: Reduced epicondylalgia and wrist tendinopathy incidence

References: Coombes eccentric LE prevention context; see PubMed [wrist tendinopathy prevention]

RECORD: Climbing finger conditioning

ID: PV-005

Program: Progressive finger loading / hangboard protocol

Target Population: Rock climbers recreational to advanced

Evidence: Graduated loading reduces pulley injury vs sudden crimp intensity spikes

Exercises: Open-hand hangs, half-crimp progression, antagonist extensor bands, shoulder pulling balance

Frequency: 2-3 dedicated strength sessions/week; avoid consecutive max crimp days

Progression: 10% load/week rule; open hand before crimp; campus board advanced only

Outcome: Finger strength gains with lower pulley rupture risk

References: Schweizer climbing injuries; López-Rivera hangboard studies; see PubMed [climbing finger training]

RECORD: Tennis elbow prevention

ID: PV-006

Program: Forearm eccentric prevention protocol

Target Population: Racquet sports, manual labor, repetitive grip occupations

Evidence: Eccentric wrist extensor program reduces LE incidence in some workplace athlete studies

Exercises: Tyler twist eccentric, wrist extensor/flexor balance, shoulder scapular prep for kinetic chain

Frequency: Daily 5 min eccentric during high-risk season or job demands

Progression: Low load high rep → sport integration

Outcome: Reduced LE onset and recurrence

References: Tyler eccentric LE; see PubMed [lateral epicondylalgia prevention]

RECORD: Office-worker neck and upper-back programs

ID: PV-007

Program: Ergonomic micro-break strengthening/mobility

Target Population: Desk workers with neck/upper back pain risk

Evidence: Exercise plus ergonomic advice superior to either alone for neck pain prevention

Exercises: Chin tuck, scapular retraction, thoracic extension breaks, upper trap stretch, deep neck flexor endurance

Frequency: Hourly micro-breaks 2-3 min; formal exercise 3×/week 20 min

Progression: Endurance → resistance band rows → overhead tolerance if asymptomatic

Outcome: Reduced neck disability incidence; improved posture tolerance

References: Blangsted neck pain at work; see PubMed [workplace neck pain prevention exercise]

7. Clinical Reasoning Pathways (Core Module)

This core module provides structured clinical reasoning pathways for the most common upper-quarter presentations. Each pathway integrates red and yellow flag screening, ranked differential diagnoses, history and examination priorities, imaging and referral decision points, working diagnosis formulation, treatment and exercise selection, progression criteria, and enriched decision support including sample cases, prognosis timelines, common pitfalls, and interprofessional communication triggers. Use these pathways as decision-oriented templates — always individualize to patient context and escalate when red flags are present.

RECORD: Clinical Pathway — Shoulder Pain

ID: CR-001

Chief Complaint: Patient reports shoulder pain interfering with overhead reaching, dressing, and sleep

Pain Location: Anterolateral shoulder over deltoid region; may radiate mid-arm; distinguish deep joint vs superficial AC

Pain Behavior: Painful arc 60-120° abduction; night pain lying on affected side; improves with rest initially then stiffens if prolonged immobilization

Mechanism of Injury: Gradual overuse overhead work OR acute fall on outstretched hand OR sudden lift — mechanism guides fracture vs cuff vs instability differential

Age: Age stratifies diagnosis: <25 instability/labral; 25-50 cuff/biceps; 50+ cuff/capsulitis/OA; >60 consider referred visceral if atypical

Occupation: Painter, electrician, warehouse overhead — overuse cuff; desk worker may have postural referral from neck

Sport: Swimming, volleyball, tennis — workload and GIRD assessment; collision sports — AC separation/fracture

Red Flags: Fever/hot joint, trauma deformity, chest pain/dyspnea, progressive weakness, cancer history, unremitting night pain with weight loss, pulseless after dislocation

Yellow Flags: Work compensation, fear-avoidance, high pain catastrophizing, diabetes (capsulitis risk), prior failed injections

Differential Diagnoses: Subacromial/RTC tendinopathy, partial/full tear, adhesive capsulitis, AC joint OA, biceps tendinopathy, instability/labral, cervical radiculopathy, referred cardiac/pulmonary/visceral, OA, calcific tendinitis, septic arthritis

Likelihood Ranking: 1) RTC tendinopathy/subacromial pain (most common adult). 2) AC joint pathology. 3) Adhesive capsulitis (diabetic/female 40-60). 4) Cervical referral if neck signs. 5) Instability young athlete. 6) Fracture acute trauma. 7) Referred visceral/cardiac if systemic. 8) Sepsis rare hot joint

History Questions: Onset acute vs gradual? Mechanism? Dominant hand? Night pain side? Neck symptoms? Fever? Chest pain? Prior dislocation? Diabetes/thyroid? What activities reproduce (overhead, reaching behind back)? Work/sport demands? Injection history? Prior imaging/surgery?

Physical Examination: Inspect atrophy supraspinatus/infraspinatus; posture scapular position; palpation GT/AC/biceps groove; AROM/PROM all planes; compare side; strength ER/IR/abduction; neurovascular; cervical screen

Special Tests: Neer, Hawkins, painful arc, empty can, ER lag, lift-off (subscap), Speed's/O'Brien if biceps/labral, cross-body AC, apprehension/relocation if instability, Spurling if neck overlap

Imaging Indications: X-ray if trauma, age >40 chronic baseline, suspected OA/calcification. MRI/ultrasound if persistent >3-6 months, suspected full tear young active, planning injection/surgery, night pain failure rehab. Not routine day 1 uncomplicated tendinopathy

Working Diagnosis: After exam: e.g. 'Subacromial rotator cuff tendinopathy without full-thickness tear, no instability, cervical screen negative' — document confidence and alternatives

Treatment Options: Phase 1: education load management, NSAIDs short course if appropriate, activity modification. Phase 2: supervised exercise cuff/scapular strengthening, manual therapy adjunct. Phase 3: return overhead function. Injection (subacromial/calcific) if plateau. Surgery: full tear active patient failing rehab, instability recurrent, fracture

Exercise Selection: Pain-free isometric ER/IR → band ER rows, scapular retraction, serratus punch, progressive abduction in scapular plane, eccentric lowering, kinetic chain legs/core if overhead athlete

Progression Criteria: NPRS $\leq 3/10$ with ADLs; full passive ROM; MMT ER $\geq 4/5$; painful arc resolving; PSFS goals improving $\geq$ MCID; then advance load velocity sport-specific

Referral Criteria: Emergency: dislocation, sepsis, fracture, cardiac. Urgent: massive tear acute, progressive neuro deficit cervical. Ortho: failed 3-6 months quality rehab, full thickness tear high demand

Clinical Guidelines: AAOS: exercise first-line cuff tendinopathy; avoid routine subacromial decompression without failed conservative care per recent evidence trends

Evidence: Exercise therapy moderate-large effect subacromial pain; manual therapy short-term adjunct; early mobilization post fracture per orthopaedic protocol

References: Michener shoulder CPG; Holmgren exercise vs surgery cuff; see PubMed [shoulder pain clinical reasoning pathway]

Step-by-Step Algorithm Summary: 1 Screen red flags → 2 Mechanism/age stratification → 3 Pain map lateral/anterior/AC/deep → 4 Behavior arc/night/instability → 5 Exam cuff/impingement/instability/cervical → 6 Working dx ranked → 7 Imaging if indicated → 8 Exercise-based rehab → 9 Reassess 2-4 wks → 10 RTS criteria

Sample Case Narrative: 58F office worker presents 6-week insidious right anterolateral shoulder pain, night pain lying on R side, cannot reach high cupboard. No trauma, no neck pain, no systemic symptoms. Exam: painful arc 70-120, Hawkins+, ER 4/5 pain-limited, Spurling negative. Working dx RTC tendinopathy. Plan: education, isometric ER, band rows, SPADI baseline 62, follow-up 3 weeks. No imaging day 1.

Prognosis Timeline: Most subacromial/RTC tendinopathy improves 6-12 weeks with exercise; capsulitis 1-3 years phases; full tear surgical pathway variable

Patient Education Points: Avoid painful arc temporarily; sleep unaffected side; load management not complete rest; home exercise adherence critical; red flags require immediate care

Interprofessional Communication: Refer ortho if failed 3-6 months quality PT, full thickness tear high demand, instability recurrent, or injection/surgery consideration

Medication Considerations: NSAIDs short course if no contraindication adjunct only; not substitute for exercise; corticosteroid injection debate — may provide short relief but tendon health concern repeated doses

Common Clinical Pitfalls: Missing referred cardiac pain age >50 ; operating without rehab trial; ignoring scapular dyskinesis; cervical untreated dual pathology; MRI overuse acute mild cases

Follow-Up Reassessment Protocol: Week 2-4: NPRS SPADI ROM strength; advance exercises if MCID trending; week 6-12: imaging referral decision if plateau; document PSFS patient goals each visit

RECORD: Clinical Pathway — Neck Pain

ID: CR-002

Chief Complaint: Neck pain with stiffness limiting driving and computer work; may include arm symptoms

Pain Location: Cervical paraspinals, upper trapezius, occiput; radicular arm if disc; distinguish shoulder primary referral

Pain Behavior: Mechanical worse with rotation/extension; radicular worse with Spurling; morning stiffness <30 min mechanical vs inflammatory if prolonged

Mechanism of Injury: MVA whiplash, sleeping awkwardly, prolonged desk posture, or insidious degenerative — trauma increases ligament/fracture concern

Age: Young WAD common; middle age disc/facet; elderly myelopathy/fracture osteoporosis risk

Occupation: Desk worker, driver, dental hygienist — sustained posture; overhead worker may have shoulder driving neck pain

Sport: Contact sport stingers; cycling aerodynamic neck extension; swimming breathing rotation

Red Flags: Myelopathy (Hoffman, gait, hand clumsiness), fever, cancer, trauma step deformity, progressive bilateral symptoms, bowel/bladder, arterial dissection post trauma (Hoarseness, Horner, sudden severe)

Yellow Flags: Litigation WAD, high disability NDI, depression, passive coping, prior chronic neck pain

Differential Diagnoses: Mechanical neck pain, facet dysfunction, disc radiculopathy, WAD grades, cervicogenic headache, TOS, inflammatory spondyloarthropathy, meningitis, malignancy, myelopathy

Likelihood Ranking: 1) Mechanical/non-specific neck pain. 2) Facet/myofascial. 3) Radiculopathy if arm symptoms Spurling+. 4) WAD post trauma. 5) Cervicogenic headache if occipital. 6) TOS if ulnar hand weakness atrophy. 7) Myelopathy must rule out. 8) Serious pathology low but high impact

History Questions: Trauma MVA details? Arm numbness/weakness? Headache dizziness? Bladder/bowel? Fever? Cancer? Hand dominance symptoms? Work setup? Prior treatment imaging? Morning stiffness duration? Red flag review systems?

Physical Examination: Posture forward head; cervical AROM/PROM all planes; segmental palpation; neuro myotomes/reflexes/sensation; ULTT; shoulder screen; TOS if indicated; VBI symptoms history before HVLA

Special Tests: Spurling, distraction, ULTT median/radial/ulnar, upper cervical stability if trauma (alar etc.), Tinel supraclavicular if TOS

Imaging Indications: X-ray post trauma, suspected instability. MRI radiculopathy >6 weeks progressive, myelopathy signs, severe trauma. No imaging uncomplicated mechanical <4-6 weeks

Working Diagnosis: Example: 'Mechanical neck pain C5-C6 facet dysfunction with upper trap hypertonicity, no radiculopathy on exam'

Treatment Options: Education reassurance most improve 6 weeks; exercise deep neck flexor endurance, scapular/Thoracic mobility; manual therapy short-term; avoid prolonged collar; radiculopathy epidural if severe; surgery if myelopathy or progressive motor deficit

Exercise Selection: Chin tuck, deep neck flexor training, cervical rotation/snags self-mobilization, upper trap stretch, rows, thoracic extension, postural breaks

Progression Criteria: NDI MCID improvement; full ROM; negative Spurling; return work duties; sport clearance if contact — no radicular symptoms

Referral Criteria: Emergency myelopathy, fracture, dissection suspicion. Urgent MRI progressive radiculopathy motor deficit. Neurosurgery cord compression

Clinical Guidelines: APTA neck pain CPG favors exercise and mobilization over passive modalities alone; imaging not recommended early uncomplicated

Evidence: Exercise and combined interventions superior long-term vs rest; manipulation modest short-term benefit mechanical neck pain

References: Childs neck pain guidelines; Blanpied APTA CPG; see PubMed [neck pain clinical reasoning]

Step-by-Step Algorithm Summary: 1 Myelopathy/red flags → 2 Trauma imaging → 3 Pattern mechanical vs radicular vs headache → 4 Spurling ULTT neuro → 5 Working dx → 6 MRI if progressive → 7 Exercise DNFE scapular → 8 Reassess NDI

Sample Case Narrative: 35M developer 2-week neck stiffness and upper trap ache, no arm symptoms, WFH ergonomic poor. Exam: ROM rotation limited R, segmental tender C5-6, neuro normal, Spurling negative. Mechanical neck pain. Plan: chin tuck hourly, thoracic extension, rows, NDI 18 moderate.

Prognosis Timeline: 80% mechanical neck pain improve 6 weeks; radiculopathy 6-12 weeks conservative many cases; WAD variable weeks to months

Patient Education Points: Movement safe and encouraged; ergonomics monitor height; avoid prolonged static posture; seek urgent care if arm weakness gait changes bladder bowel changes

Interprofessional Communication: Neurosurgery urgent myelopathy progressive deficit; GP pain management if chronic yellow flags; occupational health ergonomics

Medication Considerations: Analgesics adjunct; muscle relaxants short term controversial sedation; avoid long-term opioid

Common Clinical Pitfalls: Manipulation without red flag screen; missing myelopathy subtle; treating neck only when shoulder driver; early MRI low-value uncomplicated

Follow-Up Reassessment Protocol: NDI NPRS every 2-4 weeks; motor reassessment if radicular; MRI trigger progressive deficit 6 weeks persistent

RECORD: Clinical Pathway — Thoracic Pain

ID: CR-003

Chief Complaint: Mid-back pain between scapulae worsened by prolonged sitting and rotation

Pain Location: Thoracic paraspinals, interscapular, costovertebral; cardiac referred if exertional without movement correlation

Pain Behavior: Postural ache end of workday; sharp with rotation/reach; pleuritic if pulmonary; cardiac if exertional not movement-dependent

Mechanism of Injury: Lifting rotation, prolonged kyphotic sitting, or insidious postural deconditioning

Age: Adolescent Scheuermann; adult postural; elderly compression fracture osteoporosis

Occupation: Desk worker, nurse lifting, student backpack

Sport: Rowing, golf rotation, throwing follow-through thoracic rotation demand

Red Flags: Age >50 new pain cancer history, fever, trauma, progressive neuro deficit, exertional chest pain, weight loss, osteoporosis corticosteroids

Yellow Flags: Sedentary lifestyle, fear of movement, repetitive static loading

Differential Diagnoses: Thoracic facet dysfunction, costovertebral strain, myofascial pain, Scheuermann, compression fracture, disc herniation rare, T4 syndrome, referred cardiac/aortic, pulmonary, inflammatory arthritis

Likelihood Ranking: 1) Mechanical thoracic/postural pain. 2) Facet/costovertebral dysfunction. 3) Myofascial trigger points. 4) Scheuermann adolescent. 5) Compression fracture elderly trauma/minor fall. 6) Referred cardiac if atypical. 7) Infection/malignancy red flags

History Questions: Onset? Sitting hours? Trauma? Fever? Cancer? Chest pain exertion? Osteoporosis risk? Breathing pain? Hand numbness T4 syndrome? Response to previous treatment?

Physical Examination: Kyphosis posture; thoracic rotation/extension ROM; segmental PA; rib springing; repeated movement testing; neuro screen; cardiac auscultation if atypical

Special Tests: Segmental mobility PA pressures, rib expansion, Slump if neural, cardiac stress history not orthopaedic test

Imaging Indications: X-ray if fracture suspected deformity trauma osteoporosis. MRI infection tumor neuro. Avoid routine imaging postural pain

Working Diagnosis: Example: 'Mechanical thoracic dysfunction T6-T8 hypomobility with postural kyphosis, no red flags'

Treatment Options: Ergonomic education, thoracic extension mobility, scapular strengthening, manual mobilization/manipulation selected patients, breathing retraining if rib dysfunction

Exercise Selection: Foam roller thoracic extension, cat-camel, open book rotation, rows, serratus, chin tuck postural chain

Progression Criteria: Pain $\leq 2/10$ sitting 1 hour; rotation ROM symmetrical; return sport rotation without pain

Referral Criteria: Emergency cardiac/aortic. Urgent compression fracture severe pain. Oncology if systemic red flags

Clinical Guidelines: Thoracic manipulation may benefit acute mechanical pain selected patients; fracture excluded first elderly

Evidence: Thoracic mobilization improves shoulder overhead indirectly via posture; limited high-quality thoracic-only RCTs

References: Heneghan thoracic manipulation; Finucane T4; see PubMed [thoracic spine clinical reasoning]

Step-by-Step Algorithm Summary: 1 Red flags cardiac/malignancy/fracture → 2 Visceral screen → 3 Thoracic ROM segmental → 4 Mechanical vs serious → 5 Imaging selective → 6 Mobility scapular breathing → 7 Reassess sitting tolerance

Sample Case Narrative: 29F graphic designer interscapular ache end of day 4 weeks. No fever trauma cardiac symptoms. Kyphotic sitting posture thoracic rotation limited bilaterally. Mechanical thoracic dysfunction. Ergonomic plan foam roller extension open book daily.

Prognosis Timeline: Postural mechanical often improves 4-8 weeks with ergonomics and exercise; compression fracture osteoporosis medical management timeline separate

Patient Education Points: Micro-breaks hourly; thoracic movement not just stretching trap; seek care if chest pain exertional fever weight loss

Interprofessional Communication: Cardiology if atypical; ortho/osteoporosis if compression fracture; oncology if red flags

Medication Considerations: Analgesics adjunct postural pain; osteoporosis pharmacology medical not PT scope

Common Clinical Pitfalls: Months manipulation without visceral screen; missing osteoporotic fracture minor fall; ignoring cardiac equivalent

Follow-Up Reassessment Protocol: Sitting tolerance hours; rotation ROM; pain NPRS 2-4 week intervals

RECORD: Clinical Pathway — Elbow Pain

ID: CR-004

Chief Complaint: Lateral elbow pain with gripping and lifting — 'tennis elbow' presentation

Pain Location: Lateral epicondyle origin extensor mass; may radiate forearm; medial if thrower

Pain Behavior: Pain resisted wrist extension/grip; improves with rest early; chronic >6 weeks common

Mechanism of Injury: Repetitive gripping racquet/tools OR acute overload lifting OR cervical contribution

Age: Peak 35-55 lateral epicondylalgia; throwers medial younger

Occupation: Manual labor, typing, carpentry, butcher — grip demands

Sport: Tennis backhand, climbing, CrossFit pull-ups, baseball pitching medial elbow

Red Flags: Trauma deformity fracture, infection hot swelling, acute pop tendon rupture, progressive ulnar nerve weakness

Yellow Flags: Chronic >6 months, litigation, central sensitization features

Differential Diagnoses: Lateral epicondylalgia, medial epicondylalgia, radial tunnel, olecranon bursitis, UCL sprain, cervical C6-C7 radiculopathy, PLRI, arthritis

Likelihood Ranking: 1) Lateral epicondylalgia most common lateral. 2) Radial tunnel if pain distal no epicondyle tenderness. 3) Cervical radiculopathy if neck/dermatomal. 4) Medial epicondylalgia/UCL throwers. 5) Fracture trauma. 6) Septic bursitis olecranon

History Questions: Dominant hand? Racquet/tool use? Neck pain? Numbness? Swelling olecranon? Throwing? Prior cortisone? Morning stiffness?

Physical Examination: Palpation epicondyle extensor mass; resisted wrist extension middle finger (Maudsley); grip dynamometer; cervical screen; ulnar nerve elbow

Special Tests: Cozen, Maudsley, Mill medial, valgus stress UCL, moving valgus stress throwers, Tinel radial tunnel, Spurling cervical

Imaging Indications: X-ray if trauma arthritis; MRI UCL throwers partial tear; ultrasound tendons; cervical MRI if radiculopathy

Working Diagnosis: Example: 'Chronic lateral epicondylalgia ECRB origin tendinopathy, cervical screen negative, no radial tunnel signs'

Treatment Options: Load management, wrist extensor eccentric progression (Tyler twist), forearm strap adjunct debatable, manual therapy, cervical treat if positive, injection if plateau, surgery rare refractory

Exercise Selection: Isometric wrist extension → eccentric slow lowering → grip strengthening → sport-specific plyometric

Progression Criteria: PRTEE MCID; pain-free grip 80% baseline strength; return sport without pain 24h post

Referral Criteria: Urgent UCL complete thrower; fracture; progressive neuro; failed 12 months rehab consider surgery

Clinical Guidelines: Eccentric exercise first-line LE; avoid repeated cortisone long-term tendon health

Evidence: Coombes: eccentric and load management; many self-resolve 6-24 months; cervical treatment if dual pathology

References: Coombes LE review; Dines throwing elbow; see PubMed [elbow pain clinical pathway]

Step-by-Step Algorithm Summary: 1 Red flags fracture infection rupture → 2 Lateral vs medial vs posterior → 3 Cervical screen → 4 Special tests → 5 Working dx → 6 Eccentric loading → 7 PRTEE track

Sample Case Narrative: 48M carpenter 3-month lateral elbow pain gripping tools. Cozen+. Cervical negative. Lateral epicondylalgia. Tyler twist daily putty grip progression ergonomic tool handle modification.

Prognosis Timeline: LE often 6-24 months natural history but exercise accelerates function; many improve 8-12 weeks functional gains

Patient Education Points: Continue modified activity not complete rest; eccentric loading mild pain OK; ergonomics grip size; ulnar nerve elbow padding if medial

Interprofessional Communication: Hand/ortho UCL complete thrower; MRI if thrower partial UCL; surgery LE refractory >12 months rare

Medication Considerations: Topical NSAIDs optional; corticosteroid injection short-term relief debated recurrence

Common Clinical Pitfalls: Repeated steroid without loading program; missing cervical C6; return throw too fast medial pain

Follow-Up Reassessment Protocol: PRTEE every 4 weeks; grip strength; progress eccentric load; throw interval if athlete separate pathway

RECORD: Clinical Pathway — Wrist Pain

ID: CR-005

Chief Complaint: Radial wrist pain with thumb use and lifting — de Quervain or TFCC concern

Pain Location: Radial styloid first dorsal compartment OR ulnar wrist TFCC deep OR diffuse CTS palm

Pain Behavior: De Quervain: thumb/wrist ulnar deviation painful; TFCC: click pain ulnar deviation load; CTS: night numbness thumb-index-middle

Mechanism of Injury: Repetitive thumb lifting newborn care, FOOSH TFCC tear, chronic keyboard/racquet

Age: De Quervain women 30-50; CTS middle age; scaphoid fracture any age FOOSH

Occupation: New parents, hairdressers, gamers, construction

Sport: Tennis serve wrist, gymnastics, weightlifting rack position

Red Flags: FOOSH snuffbox tenderness scaphoid; fever septic; acute deformity; thenar atrophy CTS

Yellow Flags: Pregnancy CTS often resolves postpartum; chronic compensatory patterns

Differential Diagnoses: De Quervain, CTS, TFCC tear, scaphoid fracture, wrist sprain, OA STT, inflammatory arthritis, intersection syndrome

Likelihood Ranking: 1) De Quervain radial thumb pain. 2) CTS numbness predominant. 3) TFCC ulnar wrist mechanical. 4) Scaphoid post FOOSH must rule out. 5) Sprain acute. 6) Inflammatory if bilateral morning stiffness

History Questions: FOOSH? Pregnancy? Night numbness? Clicking? Which fingers numb? Thumb use activities? Prior splinting?

Physical Examination: Finkelstein, Phalen, Tinel, scaphoid tubercle tenderness, TFCC load/shift, grip strength, compare sides

Special Tests: Finkelstein de Quervain, Phalen/Tinel CTS, scaphoid compression, TFCC press test, Allen vascular

Imaging Indications: Scaphoid series FOOSH; MRI TFCC; NCS/EMG CTS moderate-severe; ultrasound tendons

Working Diagnosis: Example: 'De Quervain tenosynovitis first dorsal compartment, negative scaphoid, no CTS signs'

Treatment Options: Splint thumb spica, activity modification, tendon glides, manual therapy, injection compartment, surgery release refractory; CTS splint NCS pathway

Exercise Selection: Tendon glides, thumb abduction strengthening gradual, wrist mobility neutral, nerve glides CTS if appropriate

Progression Criteria: Pain-free Finkelstein negative; return lifting infant/tools; sport grip without flare

Referral Criteria: Scaphoid fracture ortho; CTS thenar atrophy surgery; TFCC tear failing rehab MRI

Clinical Guidelines: Scaphoid missed fracture high morbidity — treat as fracture until proven otherwise high suspicion

Evidence: Splint plus education effective de Quervain; CTS splint first-line mild-moderate

References: AAOS CTS; Cooney scaphoid; see PubMed [wrist pain clinical reasoning]

Step-by-Step Algorithm Summary: 1 FOOSH scaphoid rule → 2 Pattern de Quervain CTS TFCC → 3 Special tests → 4 Cervical if equivocal → 5 Immobilize/splint → 6 Imaging NCS → 7 Progress tendon glides grip

Sample Case Narrative: 34F new mother 6-week radial wrist pain lifting baby. Finkelstein+. de Quervain. Thumb spica splint activity modification tendon glides reassess 2 weeks.

Prognosis Timeline: De Quervain often improves 4-6 weeks splint; CTS variable; scaphoid fracture 8-12 weeks immobilization if confirmed

Patient Education Points: Ergonomic lifting techniques; splint wear consistency; seek urgent care FOOSH snuffbox pain; night numbness CTS pathway

Interprofessional Communication: Hand surgery scaphoid jersey mallet; neurology NCS CTS; ortho TFCC tear failing rehab

Medication Considerations: NSAIDs short course; injection de Quervain compartment if plateau

Common Clinical Pitfalls: Missed scaphoid; CTS surgery without trial conservative; ignoring pregnancy-related fluid retention CTS

Follow-Up Reassessment Protocol: PRWHE or function; Finkelstein symptom; splint compliance check; imaging results action

RECORD: Clinical Pathway — Finger Pain

ID: CR-006

Chief Complaint: Finger pain with locking (trigger) OR sports injury unable to flex/extend DIP/PIP

Pain Location: A1 pulley trigger finger OR DIP mallet OR ring finger jersey OR PIP collateral sprain

Pain Behavior: Trigger: catching locking morning worse; Mallet: DIP droop cannot extend; Jersey: cannot flex DIP; Infection: fusiform swollen finger held flexed

Mechanism of Injury: Repetitive grip trigger; ball strike mallet; tackle jersey finger; crush sprain

Age: Trigger 40-60 diabetes; mallet/jersey sports all ages

Occupation: Climbing, manual labor, musicians

Sport: Football rugby jersey finger, basketball jam, climbing pulley

Red Flags: Kanavel flexor infection — emergency; open fracture; vascular injury

Yellow Flags: Diabetes trigger recurrence; delayed presentation jersey finger >3 weeks surgical outcomes worse

Differential Diagnoses: Trigger finger, mallet finger, jersey finger FDP avulsion, boutonniere, OA nodes, flexor tenosynovitis infection, collateral sprain

Likelihood Ranking: 1) Trigger stenosing tenosynovitis if locking. 2) Mallet if extensor injury DIP. 3) Jersey if athlete cannot flex DIP. 4) Sprain trauma PIP. 5) Infection if Kanavel. 6) OA DIP Heberden chronic

History Questions: Locking? Trauma mechanism? Can you extend/flex DIP? Fever? Which finger sport? Diabetes? Time since injury?

Physical Examination: Trigger palpation A1 nodule; mallet DIP extension lag; jersey no active DIP flexion; collateral stress PIP; Kanavel signs

Special Tests: Locking observation, Bunnell sheath tenderness, extensor/flexor integrity isolated joints

Imaging Indications: X-ray mallet/jersey avulsion fragment size guides surgery; ultrasound trigger

Working Diagnosis: Example: 'Ring finger jersey finger FDP avulsion suspected, no flexion DIP, refer hand surgery'

Treatment Options: Trigger: splint, injection, release; Mallet: splint DIP 6-8 weeks; Jersey: early surgical referral; Infection: emergency IV antibiotics OR

Exercise Selection: Post splint/surgery protocol per hand surgeon; tendon glides when appropriate; avoid loading until integrity confirmed

Progression Criteria: Tendon healing timeline; active flexion/extension milestones; return sport grip when cleared

Referral Criteria: Emergency infection; jersey/mallet large avulsion hand surgery urgent; trigger failed 2 injections

Clinical Guidelines: Jersey finger do not delay surgical consult; mallet splint continuous early

Evidence: Early FDP repair improves flexion strength; trigger injection effective 60-70% single injection

References: Manske jersey finger; Doyle mallet; see PubMed [finger injury clinical reasoning]

Step-by-Step Algorithm Summary: 1 Kanavel infection emergency → 2 Locking mallet jersey sprain → 3 Tendon exam → 4 X-ray avulsion → 5 Splint/surgery/refer → 6 Tendon glides post

Sample Case Narrative: 24M rugby player cannot flex DIP ring finger after tackle. Jersey finger suspected. Refer hand surgery same week splint interim.

Prognosis Timeline: Trigger injection success weeks; mallet splint 6-8 weeks; jersey surgery dependent timing; infection hours-days critical

Patient Education Points: Infection signs seek emergency same day; mallet splint compliance continuous; no return contact until cleared

Interprofessional Communication: Hand surgeon jersey mallet avulsion; ED flexor infection

Medication Considerations: Antibiotics infection medical; analgesics post splint

Common Clinical Pitfalls: Delayed jersey referral; infection as simple sprain; breaking mallet splint early

Follow-Up Reassessment Protocol: Healing milestones DIP extension; infection vitals labs; trigger locking frequency

RECORD: Clinical Pathway — Hand Weakness

ID: CR-007

Chief Complaint: Hand weakness dropping objects, difficulty with buttons, typing, or grip tasks

Pain Location: May localize thenar (CTS), intrinsic ulnar, diffuse brachial plexus, or painless weakness brachial neuritis

Pain Behavior: Weakness with or without pain; acute painless shoulder brachial neuritis; progressive CTS thenar

Mechanism of Injury: Gradual entrapment OR acute nerve injury OR cervical radiculopathy OR stroke

Age: CTS middle age; brachial neuritis 20-60 post viral; stroke any age emergency

Occupation: Vibration tools, keyboard, assembly line

Sport: Climbing finger strength asymmetry; contact brachial plexopathy

Red Flags: Acute facial droop stroke; painless rapid shoulder weakness; progressive bilateral; myelopathy; tumor weight loss

Yellow Flags: Diabetes multiple neuropathies; chronic cervical stenosis

Differential Diagnoses: CTS, ulnar neuropathy, cervical radiculopathy C8-T1, brachial neuritis, RTC tear weakness, stroke, ALS rare, Pancoast

Likelihood Ranking: 1) CTS thenar weakness. 2) Ulnar intrinsic weakness. 3) Cervical C8-T1 radiculopathy. 4) Brachial neuritis acute pain then weakness. 5) Cuff tear if shoulder primary. 6) Stroke if acute central signs

History Questions: Numbness distribution? Neck pain? Acute vs gradual? Viral illness prior? Drop objects which fingers? Morning symptoms? Medical history diabetes?

Physical Examination: Thenar atrophy, intrinsic wasting, grip/pinch, Abductor digiti minimi, finger spread, reflexes, cervical Spurling, FAST if acute

Special Tests: Phalen Tinel, Wartenberg finger spread, Froment sign ulnar, Spurling, manual muscle testing C8-T1 myotomes

Imaging Indications: NCS/EMG localize; cervical MRI if radiculopathy; brain MRI stroke; chest CT if Pancoast suspected

Working Diagnosis: Example: 'CTS moderate with thenar weakness, cervical screen negative, refer NCS and hand surgery discussion'

Treatment Options: CTS splint night, activity mod, nerve glides, surgery if thenar weakness; ulnar nerve elbow padding; cervical PT if radiculopathy; brachial neuritis PT maintain ROM pain management

Exercise Selection: Grip gradation putty, intrinsic strengthening ulnar sparing nerve, tendon glides median/ulnar, postural cervical if indicated

Progression Criteria: Strength MMT improvement; functional pinch grip return; NCS stability post surgery timeline

Referral Criteria: Emergency stroke; urgent CTS thenar atrophy surgery; cervical progressive motor neurosurgery

Clinical Guidelines: Thenar atrophy CTS surgical indication; brachial neuritis usually self-limiting 1-2 years rehab support

Evidence: Early CTS decompression prevents irreversible thenar loss; pattern mapping essential

References: AAOS CTS guideline; Feinberg brachial neuritis; see PubMed [hand weakness pathway]

Step-by-Step Algorithm Summary: 1 Stroke emergency → 2 Map thenar intrinsic wrist extensors → 3 NCS EMG → 4 Cervical MRI if radicular → 5 Treat entrapment neuritis → 6 Grip track

Sample Case Narrative: 52F diabetes gradual thenar weakness dropping cups numbness thumb-index. CTS moderate Phalen+. Refer NCS hand surgery discussion splint started.

Prognosis Timeline: CTS mild improves weeks-months conservative; thenar atrophy needs surgery prompt; brachial neuritis 1-2 years recovery

Patient Education Points: Night splint neutral wrist; activity modification; urgent if acute face arm weakness stroke; report progressive weakness promptly

Interprofessional Communication: Hand surgery CTS; neurology NCS brachial neuritis; emergency stroke

Medication Considerations: Diabetes optimization medical comorbidity; analgesics adjunct

Common Clinical Pitfalls: Confusing pain inhibition with weakness; delaying surgery thenar atrophy; missing brachial neuritis acute painless

Follow-Up Reassessment Protocol: MMT grip pinch monthly; NCS results; thenar bulk observation photo

RECORD: Clinical Pathway — Grip Weakness

ID: CR-008

Chief Complaint: Reduced grip strength affecting sport performance or tool use without clear numbness

Pain Location: Forearm, elbow epicondyle, shoulder if cuff inhibition, cervical if radiculopathy

Pain Behavior: Pain inhibition vs true weakness — retest after pain relief; fatigability vs fixed deficit

Mechanism of Injury: Overuse tendinopathy, partial tendon rupture, nerve entrapment, pain guarding post shoulder injury

Age: Epicondylalgia 35-55; tendon rupture middle age males biceps

Occupation: Manual grip intensive jobs

Sport: Climbing, tennis, weightlifting, Jiu-Jitsu gi grip

Red Flags: Sudden pop distal biceps; infection; acute stroke; compartment syndrome

Yellow Flags: Chronic tendinopathy fear avoidance; bilateral consider systemic

Differential Diagnoses: LE/ME tendinopathy pain inhibition, distal biceps rupture, CTS, cervical radiculopathy, RTC tear, de Quervain, inflammatory arthritis

Likelihood Ranking: 1) Epicondylalgia pain-limited grip. 2) Shoulder cuff pain inhibition. 3) CTS if numbness coexists. 4) Distal biceps rupture acute pop. 5) Cervical if dermatomal

History Questions: Pop sensation? Numbness? Shoulder pain? Which activities fail? One hand dominant symptoms?

Physical Examination: Grip dynamometer compare; hook test biceps; resisted wrist extension; shoulder cuff; Phalen

Special Tests: Cozen, hook test biceps rupture, empty can shoulder, Phalen CTS, Spurling

Imaging Indications: MRI biceps if hook test positive; ultrasound tendons; NCS if CTS

Working Diagnosis: Example: 'Grip weakness secondary lateral epicondylalgia pain inhibition, negative hook test'

Treatment Options: Treat primary source epicondylalgia eccentric; cuff rehab if shoulder; surgical biceps if rupture

Exercise Selection: Progressive grip ladder putty → dynamometer; eccentric wrist; shoulder stability concurrent

Progression Criteria: Grip strength ≥90% unaffected side; sport-specific hold without pain 24h

Referral Criteria: Distal biceps rupture ortho 2-3 week window; CTS weakness surgery pathway

Clinical Guidelines: Differentiate pain inhibition — analgesic window retest strength

Evidence: Grip strength valid functional outcome UE rehab

References: Mazzocca distal biceps; see PubMed [grip weakness clinical reasoning]

Step-by-Step Algorithm Summary: 1 Pop biceps emergency ortho → 2 Pain vs true weakness → 3 Hook test Cozen cuff → 4 Treat primary → 5 Grip ladder

Sample Case Narrative: 44M sudden pop lifting box anterior elbow ecchymosis hook test positive. Distal biceps rupture refer ortho within days.

Prognosis Timeline: Biceps repair 3-6 months; LE grip returns with eccentric 8-12 weeks; cuff tear variable

Patient Education Points: Report pop immediately; do not delay biceps repair; grip training gradual after primary resolved

Interprofessional Communication: Ortho distal biceps urgent; hand CTS if concurrent

Medication Considerations: Post-op pain management surgical

Common Clinical Pitfalls: Missed biceps rupture; strengthening grip through LE without eccentric protocol

Follow-Up Reassessment Protocol: Dynamometer weekly; hook test if acute elbow; LE PRTEE parallel

RECORD: Clinical Pathway — Numbness and Tingling

ID: CR-009

Chief Complaint: Intermittent numbness tingling in hand/arm — positional and nocturnal symptoms common

Pain Location: Median thumb-index-middle (CTS), ulnar ring-small (cubital), dermatomal cervical, diffuse TOS

Pain Behavior: Night waking shaking hand CTS; elbow flexion ulnar; neck rotation radiculopathy; acute stroke unlike peripheral

Mechanism of Injury: Repetitive wrist flexion, elbow leaning desk, cervical posture, trauma brachial plexus

Age: CTS peak 40-60; young TOS posture; elderly cervical stenosis

Occupation: Keyboard, vibration, assembly, cashier scanning

Sport: Cycling ulnar handlebar; climbing finger neural tension

Red Flags: Acute stroke FAST; progressive weakness; bilateral rapid; myelopathy; thenar atrophy

Yellow Flags: Pregnancy CTS; diabetes; double crush cervical plus CTS

Differential Diagnoses: CTS, cubital tunnel, cervical radiculopathy, TOS, peripheral neuropathy diabetes, double crush, stroke TIA

Likelihood Ranking: 1) CTS median distribution. 2) Cubital tunnel ulnar. 3) Cervical radiculopathy if neck Spurling+. 4) TOS medial forearm/hand. 5) Systemic neuropathy bilateral stocking-glove. 6) Stroke acute

History Questions: Which fingers exactly? Night symptoms? Neck pain? Elbow leaning? Diabetes? Pregnancy? Acute onset? Weakness?

Physical Examination: Sensory mapping 2-point; thenar/intrinsic; Phalen Tinel wrist and elbow; Spurling; ULTT; monofilament if systemic

Special Tests: Phalen, Tinel carpal/cubital, elbow flexion test 60 sec, Spurling, ULTT, TOS Roos

Imaging Indications: NCS/EMG gold standard localize and grade; cervical MRI if radiculopathy; avoid MRI wrist alone for CTS diagnosis

Working Diagnosis: Example: 'Clinical CTS moderate Phalen/Tinel positive median nerve distribution, plan NCS confirm'

Treatment Options: CTS splint neutral wrist night, ergonomics, nerve glides, injection, surgery if failed conservative/severe; ulnar padding elbow; cervical PT if dual pathology

Exercise Selection: Median nerve glides, wrist neutral strengthening, cervical posture, ergonomic workstation setup

Progression Criteria: Symptom frequency reduced; NCS stable mild; function PSFS goals; no progression weakness

Referral Criteria: Emergency stroke; urgent thenar atrophy; NCS severe surgery referral

Clinical Guidelines: AAOS CTS: splint and activity modification first-line; NCS guides surgery timing

Evidence: Double crush common — treat both sites if present; NCS correlates symptoms moderate not perfect

References: Wainner upper quarter; AAOS CTS; see PubMed [numbness tingling pathway]

Step-by-Step Algorithm Summary: 1 Stroke FAST → 2 Distribution median ulnar cervical → 3 Phalen Tinel Spurling → 4 NCS EMG → 5 Splint glides → 6 Surgery if severe → 7 Double crush both sites

Sample Case Narrative: 50F night waking shaking hand median digits. Phalen+. CTS clinical. Splint night ergonomics NCS ordered reassess 4 weeks.

Prognosis Timeline: Mild CTS weeks-months conservative; severe thenar surgery; cubital similar; cervical radiculopathy 6-12 weeks

Patient Education Points: Night splint; workstation wrist neutral; seek emergency acute face arm weakness; report thenar wasting

Interprofessional Communication: Hand surgery CTS; neurology NCS; neurosurgery cervical progressive

Medication Considerations: Diabetes control CTS context; no medication replaces ergonomics splint

Common Clinical Pitfalls: Wrist-only focus missing cervical; bilateral without systemic neuropathy workup; surgery without conservative trial mild

Follow-Up Reassessment Protocol: Symptom diary night frequency; 2-point discrimination; NCS follow-up plan

RECORD: Clinical Pathway — Overhead Athlete Shoulder Pain

ID: CR-010

Chief Complaint: Overhead athlete with shoulder pain during serve/spike/throw affecting performance velocity

Pain Location: Deep shoulder anterior/inferior throwers; lateral deltoid impingement pattern; posterior cuff deceleration pain

Pain Behavior: Pain cocking/late cocking throwers; painful arc overhead athletes; dead arm sensation instability

Mechanism of Injury: Overuse volume spike OR instability subluxation OR GIRD adaptive changes OR scapular dyskinesis

Age: Adolescent apophysitis Little League; young adult labral/instability; veteran cuff degenerative

Occupation: Often student/pro athlete — training load key

Sport: Baseball pitch counts, volleyball spike volume, tennis serve, swimming yardage

Red Flags: Recurrent dislocation, vascular post dislocation, sudden velocity drop massive tear, systemic illness

Yellow Flags: Early specialization single sport, year-round play, prior injection masking

Differential Diagnoses: Internal impingement, SLAP, GIRD-related cuff, scapular dyskinesis, biceps tendinopathy, instability, stress fracture proximal humerus rare

Likelihood Ranking: 1) Overuse cuff tendinopathy/GIRD. 2) Internal impingement throwers. 3) SLAP mechanical symptoms. 4) Instability apprehension. 5) Apophysitis adolescent. 6) Stress fracture rare

History Questions: Pitch count/yardage/serve volume change? Apprehension? Which phase hurts throw? Velocity change? Prior dislocation? Skeletal maturity?

Physical Examination: GIRD IR deficit total arc motion, scapular dyskinesis SA test, cuff strength ER/IR, apprehension, O'Brien SLAP, foot position kinetic chain

Special Tests: Sleeper stretch measure GIRD, O'Brien, apprehension/relocation, painful arc, ER strength at 0 and 90 abduction

Imaging Indications: MRI arthrogram labral throwers; X-ray physis adolescent; ultrasound cuff dynamic

Working Diagnosis: Example: 'Internal impingement pattern GIRD 20° deficit, scapular dyskinesis, no instability — overuse cuff primary'

Treatment Options: Load management reduce volume, GIRD stretching cross-body sleeper, scapular strengthening, throw progression interval program, surgery SLAP/instability if structural failure rehab

Exercise Selection: Sleeper stretch, cross-body, ER band at 90 abduction, serratus wall slide, lower trap, kinetic chain hip-shoulder separation drills

Progression Criteria: Pain-free throw flat ground → long toss → mound/meet; total arc within 5° bilateral; ER strength ≥90%

Referral Criteria: Ortho sports: recurrent instability, SLAP failed throwing program, physis fracture

Clinical Guidelines: Wilk/Ellenbecker throwing rehab phases; pitch count limits youth

Evidence: GIRD correction reduces shoulder symptoms overhead athletes; workload strongest injury predictor

References: Wilk throwing athlete; Ellenbecker GIRD; see PubMed [overhead athlete shoulder pathway]

Step-by-Step Algorithm Summary: 1 Instability red flags → 2 GIRD measure → 3 Scapular dyskinesis → 4 Labral tests → 5 Workload → 6 GIRD program → 7 Interval throw

Sample Case Narrative: 19M volleyball hitter deep shoulder pain late season spike volume increased 30%. GIRD 18° IR deficit. Internal impingement pattern. Reduce spike count sleeper stretch scapular program.

Prognosis Timeline: Load reduction often improves 4-6 weeks; labral surgery recovery 6-12 months; GIRD maintenance ongoing

Patient Education Points: Workload management; GIRD stretching not aggressive capsulitis; report instability giving way; cross-training legs core

Interprofessional Communication: Sports ortho MRI labral; coach pitch/spike count; S&C kinetic chain

Medication Considerations: Avoid masking instability with repeated injection without diagnosis

Common Clinical Pitfalls: Year-round play; ignoring hip/core; return spike full volume too fast; missing apophysitis youth

Follow-Up Reassessment Protocol: Total arc motion weekly; pain with spike count log; interval throw stages

RECORD: Clinical Pathway — Throwing Elbow Pain

ID: CR-011

Chief Complaint: Thrower with medial elbow pain during late cocking/acceleration — UCL concern

Pain Location: Medial epicondyle UCL origin; posterior olecranon valgus extension; lateral coexisting

Pain Behavior: Pain valgus stress; decreased velocity/control; may have ulnar nerve symptoms cubital

Mechanism of Injury: Repetitive valgus torque microtrauma UCL; acute pop partial/complete tear

Age: Adolescent apophysitis medial epicondyle; college/pro UCL tears peak

Occupation: N/A typically athlete

Sport: Baseball pitching primary; javelin; tennis serve valgus stress

Red Flags: Acute pop complete UCL; elbow dislocation; fracture; ulnar nerve acute deficit

Yellow Flags: Year-round throwing, pitch type volume curveballs youth debate

Differential Diagnoses: UCL sprain/tear, medial epicondyle apophysitis youth, medial epicondylalgia, ulnar neuritis, olecranon stress, loose bodies

Likelihood Ranking: 1) UCL sprain throwers medial pain valgus+. 2) Apophysitis skeletally immature. 3) Medial epicondylalgia flexor tendinopathy. 4) Ulnar nerve cubital. 5) Posterior impingement extension

History Questions: Pitch count? Pain which throw phase? Pop? Numbness ring small finger? Growth plate age? Prior elbow injury?

Physical Examination: Palpation UCL; valgus stress 30° flexion; moving valgus stress; flexor/pronator tenderness; ulnar nerve Tinel elbow

Special Tests: Moving valgus stress test, valgus stress, milking maneuver, Tinel cubital tunnel

Imaging Indications: MRI UCL throwers; X-ray apophysis youth comparison views; ultrasound dynamic valgus

Working Diagnosis: Example: 'Partial UCL sprain thrower, no complete tear MRI, medial epicondyle tenderness, plan rest and throwing program'

Treatment Options: Rest from throwing, valgus load management, forearm flexor/pronator strengthening, elbow extension rehab posterior impingement if present, TJ surgery complete tear high-level athletes failing rehab

Exercise Selection: Wrist flexor/pronator eccentric, scapular/shoulder kinetic chain, hip rotation, gradual interval throwing ASMI protocol

Progression Criteria: Pain-free valgus stress; moving valgus negative; interval throwing completion without symptoms 24h; velocity return

Referral Criteria: Complete UCL elite thrower ortho; apophysis fracture ortho paediatric; ulnar nerve progressive

Clinical Guidelines: ASMI interval throwing mandatory return; youth pitch counts and rest days

Evidence: UCL reconstruction return to play ~12-18 months MLB; partial tears may rehab successfully lower demand athletes

References: Dines elbow throwing; Wilk interval throwing; see PubMed [throwing elbow UCL pathway]

Step-by-Step Algorithm Summary: 1 Pop UCL complete → 2 Apophysitis youth → 3 Moving valgus → 4 MRI UCL → 5 Rest strengthen → 6 ASMI interval → 7 TJ if complete fail

Sample Case Narrative: 17M pitcher medial pain pitch count 90 weekend tournament. Open physis tenderness moving valgus mild. Apophysitis/UCL stress rest 4 weeks flexor strengthening pitch count limits.

Prognosis Timeline: Apophysitis rest until pain-free growth plate; partial UCL 3-6 months rehab; TJ 12-18 months RTP elite

Patient Education Points: Pitch count limits youth mandatory; report pop immediately; no throwing through medial pain; complete interval program before games

Interprofessional Communication: Sports ortho MRI UCL; coach pitch monitoring; hand if ulnar nerve

Medication Considerations: Analgesics post-throw not to mask continuing throw

Common Clinical Pitfalls: Continuing throw medial pain; adult UCL protocol on open physis; skipping interval program
Follow-Up Reassessment Protocol: Moving valgus pain; pitch count log; interval throwing stage checklist

8. Clinical Decision Algorithms

Step-by-step triage algorithms for rapid clinical decision-making. Follow sequential steps; if any red flag branch triggers, stop routine musculoskeletal progression and initiate appropriate urgent medical referral before continuing localized treatment.

RECORD: Decision Algorithm — Shoulder Pain Triage

ID: ALG-001

Algorithm Name: Shoulder Pain Triage

Step-by-Step Decision Pathway: STEP 1: Screen red flags — fever/hot joint, trauma deformity, chest pain/dyspnea, progressive weakness, cancer history, pulseless limb. If positive → STOP MSK routine → emergency/urgent medical referral. STEP 2: Clarify onset mechanism — traumatic vs insidious; age and sport/occupation. Traumatic → X-ray rule fracture/dislocation first. STEP 3: Map pain location — lateral (cuff/impingement), anterior (biceps/instability), AC top (cross-body painful), deep mechanical click (labral), neck association (cervical screen). STEP 4: Pain behavior — painful arc 60-120 suggests subacromial; night pain side-lying cuff/capsulitis; dead arm instability; exertional non-mechanical → cardiac/pulmonary. STEP 5: Physical exam cluster — inspect atrophy; impingement tests Neer/Hawkins/arc; cuff ER lag/empty can; instability apprehension; cervical Spurling; vitals if atypical. STEP 6: Form working diagnosis with ranked differentials and confidence level document alternatives. STEP 7: Imaging if red flags, trauma, failed 3-6 months rehab, planning injection/surgery, or suspected full tear high-demand patient. STEP 8: Treat — load management + staged exercise scapula/cuff; address cervical if positive; refer visceral/cardiac if suspected. STEP 9: Reassess 2-4 weeks — NPRS, SPADI/QuickDASH, PSFS goals, ROM/strength; progress or refer ortho. STEP 10: Return to sport/work criteria — pain ≤2/10 function, strength ≥90%, sport-specific simulation pain-free 24h post.

Chief Complaint Context: Any adult or adolescent presenting shoulder region pain affecting function

Key Branch Points: Red flag positive → medical; trauma → imaging; young athlete mechanical → instability/labral; age 50+ insidious → cuff/capsulitis; neck signs → dual treatment

Documentation Requirements: Record red flag screen, tests performed, working diagnosis, plan, follow-up timeline, return-to-activity criteria

Common Pitfalls: Missing cardiac referral in atypical left arm pain; injecting cuff without rehab; ignoring cervical contribution; over-imaging acute uncomplicated tendinopathy

Evidence: Structured examination clusters improve cuff diagnosis accuracy; early exercise superior passive modalities alone

References: Michener shoulder exam; Holmgren cuff trial; see PubMed [shoulder clinical algorithm]

RECORD: Decision Algorithm — Neck Pain Triage

ID: ALG-002

Algorithm Name: Neck Pain Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — myelopathy (Hoffman, gait, hand clumsiness, Babinski), fever, cancer, trauma, progressive bilateral deficit, bowel/bladder, arterial dissection history (Hoarseness, Horner, sudden severe headache/neck pain). Positive → urgent MRI/neuro/emergency. STEP 2: Trauma mechanism — MVA whiplash grade WAD; fracture/dislocation imaging if high-energy or neuro deficit. STEP 3: Symptom pattern — central neck mechanical vs radicular arm pain dermatomal vs headache occipital cervicogenic vs bilateral TOS pattern. STEP 4: Exam — ROM, Spurling, ULTT, reflexes, myotomes, dermatomes; shoulder screen secondary neck drivers. STEP 5: Working diagnosis — mechanical neck pain vs radiculopathy vs WAD vs serious pathology excluded. STEP 6: Imaging — MRI if radiculopathy persistent >6 weeks, progressive motor deficit, or red flags; avoid early imaging uncomplicated mechanical. STEP 7: Treatment — reassurance most improve 6 weeks; exercise DNFE + scapular + thoracic mobility; manual therapy short-term adjunct; no prolonged collar. STEP 8: Radiculopathy branch — epidural if severe; surgery if progressive motor deficit or myelopathy. STEP 9: Reassess NDI NPRS 2-4 weeks. STEP 10: Discharge prevention posture program ergonomic review.

Chief Complaint Context: Neck pain with or without arm symptoms in outpatient MSK setting

Key Branch Points: Myelopathy → neurosurgery emergency; radiculopathy → cervical MRI pathway; mechanical → conservative exercise first

Documentation Requirements: Red flags explicitly documented negative or positive action taken; dermatomal map if arm symptoms

Common Pitfalls: HVLA manipulation without red flag screen; missing myelopathy subtle signs; treating only neck when shoulder primary driver

Evidence: APTA CPG exercise and mobilization recommended; imaging low value early mechanical neck pain

References: Childs neck CPG; Wainner radiculopathy cluster; see PubMed [neck pain algorithm]

RECORD: Decision Algorithm — Thoracic Pain Triage

ID: ALG-003

Algorithm Name: Thoracic Pain Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — age >50 cancer, fever, trauma osteoporosis, cardiac exertional symptoms, weight loss, progressive neuro deficit. STEP 2: Visceral screen — RUQ/epigastric meals, cough fever, tearing interscapular pain dissection, pleuritic dyspnea. STEP 3: Mechanical assessment — posture kyphosis, thoracic rotation/extension ROM, segmental PA, rib springing. STEP 4: Working diagnosis mechanical vs compression fracture vs referred visceral/cardiac/pulmonary. STEP 5: Imaging — X-ray fracture; MRI infection/tumor; chest/cardiac workup if visceral. STEP 6: Treat mechanical — ergonomic breaks, thoracic mobility extension/rotation, scapular strengthening, breathing if rib dysfunction. STEP 7: Reassess sitting tolerance and rotation ROM 2-4 weeks.

Chief Complaint Context: Mid-back or interscapular pain often postural or activity-related

Key Branch Points: Cardiac/aortic → emergency; fracture elderly → ortho/medical; mechanical → rehab

Documentation Requirements: Cardiac/vascular symptoms explicitly asked; osteoporosis risk factors

Common Pitfalls: Months of thoracic manipulation without visceral screen; missing compression fracture minor fall elderly

Evidence: Thoracic mobilization modest benefit selected mechanical pain populations

References: Heneghan thoracic MT; see PubMed [thoracic pain algorithm]

RECORD: Decision Algorithm — Elbow Pain Triage

ID: ALG-004

Algorithm Name: Elbow Pain Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — trauma deformity, infection hot swelling, acute pop tendon rupture, progressive ulnar motor loss. STEP 2: Location — lateral epicondyle LE vs medial thrower UCL vs posterior olecranon vs radial tunnel distal. STEP 3: Cervical screen C6-C7 if lateral forearm pain. STEP 4: Tests Cozen/Maudsley/valgus stress/moving valgus/radial Tinel. STEP 5: Working diagnosis ranked. STEP 6: Imaging MRI UCL throwers; X-ray trauma; NCS if nerve. STEP 7: Eccentric tendinopathy protocol LE/ME; thrower rest + interval program; surgery UCL complete elite athlete failing rehab. STEP 8: PRTEE track progress 4-6 weeks.

Chief Complaint Context: Elbow pain with gripping, throwing, or repetitive tool use

Key Branch Points: Thrower medial → UCL pathway; lateral → LE eccentric; trauma → fracture

Documentation Requirements: Sport/occupation grip demands; neck screen result

Common Pitfalls: Repeated steroid injections LE without load management; missing cervical contribution; returning thrower too fast

Evidence: Eccentric exercise first-line LE; load management key tendinopathy

References: Coombes LE; Dines throwing elbow; see PubMed [elbow algorithm]

RECORD: Decision Algorithm — Wrist Pain Triage

ID: ALG-005

Algorithm Name: Wrist Pain Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — FOOSH snuffbox tenderness scaphoid, infection, thenar atrophy CTS, acute deformity. STEP 2: Pattern — radial thumb de Quervain vs ulnar TFCC click vs median numbness CTS vs diffuse sprain. STEP 3: Tests Finkelstein Phalen Tinel scaphoid compression TFCC load. STEP 4: Cervical if equivocal. STEP 5: Imaging scaphoid series FOOSH; NCS CTS; MRI TFCC. STEP 6: Splint activity mod tendon glides CTS/De Quervain. STEP 7: Refer hand surgery scaphoid, complete CTS weakness, TFCC tear failing rehab.

Chief Complaint Context: Wrist or hand pain affecting grip and daily tasks

Key Branch Points: FOOSH → scaphoid until cleared; numbness → CTS pathway

Documentation Requirements: FOOSH mechanism; pregnancy status; night symptoms

Common Pitfalls: Missing scaphoid fracture; treating CTS without ergonomic modification; ignoring double crush cervical

Evidence: Scaphoid missed fracture → nonunion; splint effective mild CTS

References: Cooney scaphoid; AAOS CTS; see PubMed [wrist pain algorithm]

RECORD: Decision Algorithm — Finger Pain Triage

ID: ALG-006

Algorithm Name: Finger Pain Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — Kanavel infection emergency; open wound; vascular injury. STEP 2: Locking trigger vs DIP mallet vs jersey no flexion vs PIP sprain. STEP 3: Exam tendon integrity each joint; X-ray avulsion mallet/jersey. STEP 4: Trigger splint/injection; mallet splint 6-8 weeks; jersey urgent hand surgery; infection ED. STEP 5: Tendon glides post immobilization per protocol.

Chief Complaint Context: Finger pain locking deformity or sport injury

Key Branch Points: Kanavel → emergency; jersey → surgery timeline critical

Documentation Requirements: Time since injury jersey; diabetes trigger; sport mechanism

Common Pitfalls: Delayed jersey finger referral; flexor infection misdiagnosed as sprain

Evidence: Early FDP repair improves outcomes; trigger injection effective many cases

References: Manske jersey; Doyle mallet; see PubMed [finger injury algorithm]

RECORD: Decision Algorithm — Hand Weakness Triage

ID: ALG-007

Algorithm Name: Hand Weakness Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — acute stroke FAST; painless rapid shoulder weakness brachial neuritis; progressive bilateral myelopathy; thenar atrophy. STEP 2: Map weakness — thenar CTS vs intrinsic ulnar vs wrist extensors C7 vs deltoid axillary post dislocation. STEP 3: NCS/EMG plan; cervical MRI if radiculopathy; brain if stroke. STEP 4: Treat entrapment conservative/surgical; brachial neuritis support PT; stroke emergency. STEP 5: Grip strength track weekly initially.

Chief Complaint Context: Objective or subjective hand weakness dropping objects

Key Branch Points: Acute painless → brachial neuritis vs stroke; thenar → CTS surgery pathway

Documentation Requirements: Acute vs gradual; numbness distribution; FAST if acute

Common Pitfalls: Assuming weakness is pain inhibition without retest; delaying CTS surgery with thenar atrophy

Evidence: Pattern mapping essential; NCS localizes entrapment

References: AAOS CTS; Feinberg brachial neuritis; see PubMed [hand weakness algorithm]

RECORD: Decision Algorithm — Grip Weakness Triage

ID: ALG-008

Algorithm Name: Grip Weakness Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — pop distal biceps; stroke; infection. STEP 2: Pain vs true weakness — retest after analgesic or differentiate pain inhibition. STEP 3: Hook test biceps; Cozen LE; cuff empty can; Phalen CTS.

STEP 4: Treat primary diagnosis. STEP 5: Progressive grip ladder putty dynamometer.

Chief Complaint Context: Reduced grip sport or occupational performance

Key Branch Points: Pop → biceps rupture ortho; LE pain-limited → eccentric first

Documentation Requirements: Dynamometer baseline both hands

Common Pitfalls: Missing distal biceps rupture; not treating shoulder when cuff inhibits grip

Evidence: Grip valid functional outcome measure UE rehab

References: Mazzocca biceps; see PubMed [grip weakness algorithm]

RECORD: Decision Algorithm — Numbness and Tingling Triage

ID: ALG-009

Algorithm Name: Numbness and Tingling Triage

Step-by-Step Decision Pathway: STEP 1: Red flags — stroke acute; progressive weakness; myelopathy; thenar atrophy. STEP 2: Distribution — median CTS vs ulnar cubital vs dermatomal cervical vs diffuse neuropathy. STEP 3: Phalen Tinel elbow Spurling ULTT monofilament if systemic. STEP 4: NCS/EMG; cervical MRI if radiculopathy. STEP 5: Splint ergonomics nerve glides; surgery if failed conservative/severe CTS. STEP 6: Double crush treat both sites.

Chief Complaint Context: Paresthesia upper extremity without or with pain

Key Branch Points: Acute hemiparesis → stroke; night median → CTS; ulnar elbow flexion → cubital

Documentation Requirements: Exact fingers affected; night vs day; neck association

Common Pitfalls: Wrist surgery without cervical assessment; missing diabetes peripheral neuropathy

Evidence: NCS gold standard entrapment grading; double crush common

References: Wainner upper quarter; AAOS CTS; see PubMed [numbness algorithm]

RECORD: Decision Algorithm — Overhead Athlete Shoulder

ID: ALG-010

Algorithm Name: Overhead Athlete Shoulder

Step-by-Step Decision Pathway: STEP 1: Red flags dislocation neurovascular, systemic, stress fracture severe pain. STEP 2: Measure GIRD IR deficit total arc motion bilateral. STEP 3: Scapular dyskinesis SA test. STEP 4: Instability apprehension labral tests O'Brien. STEP 5: Workload pitch count yardage spike history. STEP 6: Diagnosis internal impingement GIRD vs SLAP vs instability vs cuff. STEP 7: Load reduction GIRD program scapular strengthening interval return throw. STEP 8: MRI arthrogram if mechanical failure rehab 6-12 weeks elite thrower.

Chief Complaint Context: Overhead sport athlete shoulder pain performance decline

Key Branch Points: Instability → apprehension pathway; GIRD → sleeper cross-body; workload → reduce volume

Documentation Requirements: Training volume change; velocity; phase of throw painful

Common Pitfalls: Ignoring kinetic chain hip/core; cortisone mask labral; year-round play adolescent

Evidence: GIRD correction reduces symptoms; workload predictor injury

References: Ellenbecker GIRD; Wilk throwing; see PubMed [overhead athlete algorithm]

RECORD: Decision Algorithm — Throwing Elbow Pain

ID: ALG-011

Algorithm Name: Throwing Elbow Pain

Step-by-Step Decision Pathway: STEP 1: Red flags pop complete UCL acute fracture dislocation ulnar nerve deficit. STEP 2: Skeletally mature vs apophysitis youth. STEP 3: Moving valgus stress medial pain. STEP 4: MRI UCL partial vs complete. STEP 5: Rest from throwing flexor-pronator strengthening. STEP 6: ASMI interval throwing flat ground to mound. STEP 7: TJ referral complete tear elite failing rehab.

Chief Complaint Context: Thrower medial or posterior elbow pain velocity loss

Key Branch Points: Youth → apophysitis rest; partial UCL → rehab; complete → surgery discussion

Documentation Requirements: Pitch counts; pain phase cocking vs release

Common Pitfalls: Continuing throw through medial pain; missing apophysitis youth open physis

Evidence: Interval throwing standard return; partial UCL may rehab lower demand

References: Dines UCL; ASMI interval; see PubMed [throwing elbow algorithm]

9. Case Vignettes

RECORD: Vignette — Shoulder pain age 55 warehouse worker

ID: VIG-001

Presentation: 55M gradual 8-week lateral shoulder pain lifting overhead at work, night pain lying on R side, no trauma

Initial Concerns: Rule out RTC tear vs capsulitis vs cervical; cardiac screen negative sedentary no chest symptoms

Exam Findings: Painful arc positive; Hawkins positive; ER strength 4/5 pain; Spurling negative; no fever

Reasoning Chain: Insidious + painful arc + night side pain → subacromial RTC tendinopathy most likely; AC less likely no cross-body focal; capsulitis possible but ROM not globally restricted yet

Working Diagnosis: Rotator cuff tendinopathy subacromial pain syndrome

Plan: Education load modification; isometric ER → band exercises scapular; NPRS/SPADI baseline; reassess 3 weeks; X-ray if no improvement 6 weeks

Alternative Diagnosis Considered: Cervical radiculopathy excluded clinically; referred pain excluded systemic screen

Follow-Up Triggers: Progressive weakness → MRI; fever → emergency; chest symptoms → cardiac

References: See CR-001 shoulder pathway

RECORD: Vignette — Neck pain with arm numbness secretary

ID: VIG-002

Presentation: 42F neck pain 3 weeks desk job with thumb/index numbness intermittent, no trauma

Initial Concerns: Cervical radiculopathy vs CTS vs double crush; myelopathy screen negative

Exam Findings: Spurling positive R; ULTT median positive; Phalen mild; reduced C6 dermatome light touch; reflexes normal

Reasoning Chain: Spurling + dermatomal + ULTT → cervical radiculopathy C6 primary; CTS may coexist — treat cervical first 2 weeks reassess hand

Working Diagnosis: Cervical radiculopathy C6 with possible concurrent CTS

Plan: Cervical mobilization exercise DNFE; ergonomic workstation; avoid HVLA until improved; MRI if no improvement 6 weeks or motor deficit

Alternative Diagnosis Considered: Pure CTS less likely without night symptoms dominant; TOS less likely distribution

Follow-Up Triggers: Motor weakness → urgent MRI; bilateral symptoms → expand differential

References: See CR-002 CR-009 pathways

RECORD: Vignette — Thrower adolescent medial elbow

ID: VIG-003

Presentation: 16M baseball pitcher medial elbow pain during late cocking, no pop, 30 pitches/game weekend tournament

Initial Concerns: UCL sprain vs medial apophysitis vs medial epicondylalgia; complete tear excluded no pop velocity only mildly down

Exam Findings: Medial epicondyle tenderness open physis; moving valgus mild pain; valgus stress mild laxity age-appropriate; no ulnar Tinel

Reasoning Chain: Skeletally immature + medial pain + high pitch volume → medial epicondyle apophysitis/overuse UCL stress; MRI if rest 2 weeks fail

Working Diagnosis: Medial elbow overuse apophysitis/UCL stress injury

Plan: Complete rest throwing 4 weeks; flexor-pronator strengthening; pitch count education; return ASMI interval when pain-free

Alternative Diagnosis Considered: Complete UCL tear unlikely no pop severe instability

Follow-Up Triggers: Pop → MRI urgent; ulnar numbness → cubital assessment

References: See CR-011 throwing elbow pathway

RECORD: Vignette — FOOSH wrist young adult

ID: VIG-004

Presentation: 28F fell on outstretched hand 2 days ago radial wrist pain snuffbox tender mild swelling

Initial Concerns: Scaphoid fracture must not miss; sprain vs fracture

Exam Findings: Snuffbox tenderness; scaphoid compression pain; Finkelstein negative; ROM limited pain; Phalen negative

Reasoning Chain: FOOSH + snuffbox tenderness → treat as scaphoid fracture until imaging excludes; thumb spica immobilize refer ortho/hand

Working Diagnosis: Suspected scaphoid fracture high probability

Plan: Immobilize thumb spica; X-ray scaphoid series; if negative high suspicion MRI 10-14 days or CT; no loading

Alternative Diagnosis Considered: Simple sprain cannot exclude without imaging given snuffbox

Follow-Up Triggers: AVN nonunion if missed → strict immobilization protocol

References: See CR-005 wrist pathway

RECORD: Vignette — Climber finger pulley

ID: VIG-005

Presentation: 32M rock climber sudden pop ring finger crimp grip localized volar pain swelling

Initial Concerns: A2 pulley rupture vs sprain vs jersey differential no tackle mechanism

Exam Findings: Bowstringing mild; tenderness A2 pulley; flexion intact; no Kanavel; X-ray no avulsion

Reasoning Chain: Crimp pop → pulley injury grade I-II likely; ultrasound confirm; relative rest 6 weeks tape ring; gradual hangboard

Working Diagnosis: Ring finger A2 pulley sprain partial tear

Plan: Immobilize flexion 2 weeks; pulley taping return climb 50% grades; antagonist extensor training; no crimp 6 weeks

Alternative Diagnosis Considered: Flexor tendon rupture excluded active flexion intact

Follow-Up Triggers: Bowstringing severe → grade IV surgical consult

References: Schweizer pulley; see EX-018 climbing

RECORD: Vignette — Thoracic pain desk worker cardiac screen

ID: VIG-006

Presentation: 48M interscapular ache 2 weeks desk job, worse end of day, no trauma

Initial Concerns: Mechanical thoracic vs cardiac — age borderline cardiovascular risk factor hyperlipidemia

Exam Findings: Kyphotic posture; thoracic rotation limited; PA tender T6-T7; vitals normal; pain reproducible rotation not exertional; no cough fever

Reasoning Chain: Mechanical reproduction + no systemic + no exertional component → mechanical thoracic dysfunction; documented negative cardiac red flag questions

Working Diagnosis: Mechanical thoracic postural pain

Plan: Ergonomic assessment; thoracic extension rotation program; reassess 2 weeks; ED if chest pain exertion

Alternative Diagnosis Considered: Cardiac referred pain less likely history; would reassess if risk changes

Follow-Up Triggers: Exertional pain → cardiac workup; fever → infection imaging

References: See CR-003 thoracic pathway

RECORD: Vignette — Tennis elbow recreational player

ID: VIG-007

Presentation: 52M recreational tennis 2-month lateral elbow pain backhand, gripping tools at work aggravates

Initial Concerns: LE vs radial tunnel vs cervical C6

Exam Findings: Cozen positive epicondyle tender; Maudsley positive; radial tunnel non-tender; Spurling negative

Reasoning Chain: Epicondyle focal tenderness + Cozen → lateral epicondylalgia; cervical excluded

Working Diagnosis: Chronic lateral epicondylalgia

Plan: Tyler twist eccentric daily; forearm strap optional; PRTEE baseline; modify grip tasks; reassess 4 weeks

Alternative Diagnosis Considered: Radial tunnel less likely no distal tenderness

Follow-Up Triggers: Progressive neuro → NCS; no improvement 12 weeks → injection discuss

References: See CR-004 elbow pathway

RECORD: Vignette — Hand weakness CTS warehouse

ID: VIG-008

Presentation: 56F dropping light boxes 4 weeks thumb-index numbness nights

Initial Concerns: CTS vs cervical vs stroke — gradual not acute

Exam Findings: Thenar mild weakness 4/5; Phalen positive; Tinel positive; Spurling negative; 2-point discrimination reduced median

Reasoning Chain: Median night symptoms + test positive + thenar weakness → moderate CTS refer NCS hand surgery consult parallel splint

Working Diagnosis: Carpal tunnel syndrome moderate with early thenar weakness

Plan: Night splint neutral wrist; ergonomics; NCS order; surgeon if NCS severe; tendon glides

Alternative Diagnosis Considered: Cervical less likely Spurling negative

Follow-Up Triggers: Thenar atrophy progression → expedite surgery

References: See CR-007 CR-009 pathways

RECORD: Vignette — Volleyball hitter overhead shoulder

ID: VIG-009

Presentation: 21F collegiate volleyball OH hitter deep shoulder pain spiking, velocity down, no instability

Initial Concerns: Internal impingement GIRD vs SLAP vs cuff

Exam Findings: GIRD IR deficit 15°; O'Brien mild discomfort not definitive; apprehension negative; painful arc mild; scapular dyskinesis noted

Reasoning Chain: Overhead athlete + GIRD + dyskinesis + no instability → internal impingement/overuse cuff; reduce spike volume sleeper stretch program

Working Diagnosis: Overhead athlete shoulder pain GIRD-related impingement pattern

Plan: Reduce spike count 50%; sleeper cross-body; scapular strengthening; reassess 3 weeks; MRI if mechanical symptoms persist

Alternative Diagnosis Considered: SLAP possible if clicking worsens → MRI

Follow-Up Triggers: Instability → MRI arthrogram; failed 8 weeks → ortho

References: See CR-010 overhead athlete pathway

RECORD: Vignette — Numbness double crush office worker

ID: VIG-010

Presentation: 45M computer worker ulnar ring-small numbness elbow flexion desk plus neck stiffness

Initial Concerns: Cubital tunnel vs cervical C8 vs TOS vs double crush

Exam Findings: Tinel elbow positive; elbow flexion test positive; Spurling mild positive; Froment mild; no TOS vascular

Reasoning Chain: Both elbow and cervical provoke → double crush treat both cervical PT + elbow padding night extension splint; NCS to grade

Working Diagnosis: Ulnar neuropathy at elbow with cervical C8 contribution double crush pattern

Plan: Elbow pad; avoid prolonged flexion; cervical exercise; NCS; reassess 4 weeks

Alternative Diagnosis Considered: Pure TOS less likely exam

Follow-Up Triggers: Intrinsic wasting → surgical referral

References: See CR-009 numbness pathway

RECORD: Vignette — Grip weakness post fall biceps

ID: VIG-011

Presentation: 38M lifting furniture sudden pop anterior elbow ecchymosis weakened supination grip

Initial Concerns: Distal biceps rupture vs partial tear vs hematoma

Exam Findings: Hook test negative/supinated possibly partial; ecchymosis antecubital; pain supination resisted; no shoulder injury

Reasoning Chain: Pop + ecchymosis + functional supination loss → suspected distal biceps rupture urgent ortho MRI confirm

Working Diagnosis: Suspected distal biceps tendon rupture

Plan: Refer ortho within days; immobilize comfortable; no heavy lifting; MRI confirm

Alternative Diagnosis Considered: Partial tear still may need surgery young active

Follow-Up Triggers: Delayed repair >3 weeks outcomes worse

References: See CR-008 grip weakness pathway

10. Screening & Return-to-Activity Integration

RECORD: Universal MSK Intake Screen — Upper Quarter

ID: SCR-001

Purpose: Standardized first-visit screening before region-specific exam

Vital Signs: Note HR BP if cardiac/vascular concern especially age >40 shoulder/thoracic pain atypical

Red Flag Questions: Fever weight loss cancer history night pain unremitting trauma bowel bladder gait disturbance progressive weakness bilateral symptoms IV drug use immunosuppression

Yellow Flag Questions: Work compensation litigation fear avoidance catastrophizing depression prior chronic pain

Pain Mapping: Primary location radiation pattern diagram dermatomal vs nerve vs joint line

Functional Baseline: NPRS QuickDASH or regional index PSFS 3 activities

System Review: Cardiac respiratory GI symptoms when shoulder/thoracic/neck referral overlap possible

Outcome Tracking Plan: Reassessment interval 2-4 weeks same measures MCID thresholds documented

Documentation Template: Red flags Y/N each item action if positive; working diagnosis; plan; follow-up date

Evidence: Standardized screening reduces missed serious pathology in MSK primary contact care

References: NICE NG59; Childs neck CPG red flags; see PubMed [MSK red flag screening]

RECORD: Return-to-Activity Clearance Checklist

ID: SCR-002

Purpose: Structured clearance before sport/work return upper quarter injuries

Pain Criteria: NPRS $\leq 2/10$ during and 24h after simulated activity

ROM Criteria: Within 5-10° unaffected side or sport-required minimum documented

Strength Criteria: Manual muscle test $\geq 4+/5$ or dynamometer $\geq 90\%$ bilateral grip/limb appropriate

Functional Criteria: PSFS goals rated $\leq 2/10$ difficulty on patient-specific tasks

Sport-Specific Criteria: Completed interval program throwing/hitting/swimming if applicable; no apprehension instability

Patient Education: Flare management load modification ongoing home program

Referral Back Triggers: Pain resurgence $>4/10$ $>48h$; instability giving way; neuro symptoms

Evidence: Criteria-based RTS reduces reinjury vs time-based alone in throwers and overhead athletes

References: Wilk throwing RTS; see PubMed [return to sport criteria upper extremity]

11. Imaging Referral Decision Guide

Imaging and electrodiagnostic referral should follow clinical reasoning — not replace examination. Avoid low-value early imaging for uncomplicated mechanical presentations without red flags or progressive neurological deficit.

RECORD: Shoulder — when to image

ID: IMG-001

Region: Shoulder

Indication: Trauma with deformity, suspected fracture/dislocation, inability to raise arm acute

First-Line Modality: X-ray AP + axillary or Y-view; CT if complex fracture

Second-Line: MRI or ultrasound for cuff/labrum if persistent >3-6 months, suspected full tear active patient, pre-surgical planning

Avoid: Routine MRI day 1 uncomplicated tendinopathy without red flags

Urgent Pathway: Dislocation neurovascular injury → ED imaging before outpatient MRI

Evidence: Imaging does not improve outcomes uncomplicated subacromial pain early conservative care

References: AAOS appropriate use criteria shoulder

RECORD: Cervical spine — when to image

ID: IMG-002

Region: Cervical spine

Indication: Myelopathy signs, progressive motor deficit, severe trauma, red flags infection malignancy

First-Line Modality: MRI cervical spine; X-ray trauma instability screening

Second-Line: CT myelography if MRI contraindicated

Avoid: Early MRI uncomplicated mechanical neck pain <4-6 weeks without neuro deficit

Urgent Pathway: Myelopathy, progressive deficit → urgent MRI neurosurgery consult

Evidence: Clinical prediction rules guide radiculopathy; imaging correlates poorly mild cases

References: Childs neck CPG imaging recommendations

RECORD: Elbow — when to image

ID: IMG-003

Region: Elbow

Indication: Trauma effusion fracture suspicion; thrower medial pain UCL; acute tendon rupture

First-Line Modality: X-ray AP/lateral; MRI UCL partial/complete throwers

Second-Line: Ultrasound dynamic UCL tendon

Avoid: MRI all lateral epicondylalgia without failure conservative 6+ months unless considering surgery

Urgent Pathway: Fracture-dislocation, sepsis → ED

Evidence: Moving valgus clinical correlates UCL; MRI confirms grade

References: Dines UCL imaging

RECORD: Wrist/hand — when to image

ID: IMG-004

Region: Wrist/hand

Indication: FOOSH snuffbox tenderness; mallet/jersey avulsion; TFCC mechanical symptoms; CTS thenar atrophy NCS

First-Line Modality: Scaphoid series X-ray; NCS/EMG CTS; X-ray finger avulsion

Second-Line: MRI scaphoid occult; MRI TFCC; ultrasound De Quervain

Avoid: MRI all wrist pain without mechanical signs or failure rehab

Urgent Pathway: Scaphoid treat as fracture until cleared; flexor infection no delay

Evidence: Scaphoid sensitivity X-ray ~70% early — high suspicion immobilize MRI

References: Cooney scaphoid; AAOS CTS

RECORD: Thoracic spine — when to image

ID: IMG-005

Region: Thoracic spine

Indication: Trauma osteoporosis suspicion, progressive neuro deficit, infection malignancy red flags, severe compression deformity

First-Line Modality: X-ray thoracic spine AP/lateral; MRI if neuro/infection/tumor

Second-Line: CT fracture characterization; bone scan oncology pathway

Avoid: Routine imaging postural mechanical pain without red flags

Urgent Pathway: Cord compression, unstable fracture → emergency

Evidence: Compression fractures common minor fall elderly — screen osteoporosis risk

References: NICE NG59 imaging guidance

RECORD: Peripheral nerve — when to refer NCS/EMG

ID: IMG-006

Region: Peripheral nerve upper limb

Indication: CTS moderate-severe, equivocal diagnosis, pre-surgical planning, progressive weakness, failed conservative 6-12 weeks

First-Line Modality: NCS/EMG electrodiagnostic studies

Second-Line: Ultrasound nerve cross-section CTS cubital tunnel

Avoid: NCS before treatable conservative trial mild CTS unless diagnosis unclear

Urgent Pathway: Thenar atrophy rapid progression → expedited NCS and surgical consult

Evidence: NCS correlates symptom severity; guides surgical timing

References: AAOS CTS; AAEM electrodiagnostic guidelines

12. Evidence & Clinical Guidelines Summary

RECORD: Rotator cuff tendinopathy — conservative first

ID: EVD-001

Topic: Rotator cuff tendinopathy

Recommendation: Exercise-based physiotherapy first-line; surgery for full thickness failed rehab high-demand

Evidence Level: Level B systematic reviews

Key Trials: UK FIT subacromial decompression vs arthroscopy placebo trends

Clinical Application: Educate patient timeline 3-6 months; avoid passive-only treatment

References: Holmgren; see PubMed [rotator cuff exercise]

RECORD: Neck pain — exercise and mobilization

ID: EVD-002

Topic: Mechanical neck pain

Recommendation: Exercise plus manual therapy short-term; no routine imaging early

Evidence Level: Level A/B APTA CPG

Key Trials: Multiple RCTs exercise vs wait list superior

Clinical Application: DNFE scapular thoracic combined programs

References: Blanpied APTA CPG

RECORD: Lateral epicondylalgia — eccentric loading

ID: EVD-003

Topic: Lateral epicondylalgia

Recommendation: Eccentric wrist extensor program; load management; avoid repeated steroid

Evidence Level: Level A/B Cochrane trends

Key Trials: Tyler twist RCT positive

Clinical Application: Daily home program 12 weeks minimum trial

References: Coombes Cochrane LE

RECORD: Carpal tunnel — splint and ergonomics

ID: EVD-004

Topic: Carpal tunnel syndrome

Recommendation: Night splint neutral wrist first-line mild-moderate; surgery if thenar atrophy severe NCS

Evidence Level: Level B AAOS guideline

Key Trials: Splint vs surgery long-term surgery better severe

Clinical Application: Screen cervical double crush; ergonomic workstation

References: AAOS CTS guideline

RECORD: Adhesive capsulitis — phased rehab

ID: EVD-005

Topic: Adhesive capsulitis

Recommendation: Education phased mobilization within pain tolerance; diabetes thyroid workup

Evidence Level: Level B

Key Trials: Hydrodilatation vs PT mixed; most resolve 1-3 years

Clinical Application: Match intensity to phase freezing vs thawing

References: Kelley capsulitis review

RECORD: Throwing athlete — workload management

ID: EVD-006

Topic: Throwing injury prevention

Recommendation: Pitch count limits youth; interval throwing return; GIRD management

Evidence Level: Level B cohort + consensus

Key Trials: ASMI interval throwing standard care

Clinical Application: Never skip stages flat ground to mound

References: Wilk ASMI

RECORD: FIFA 11+ — injury prevention

ID: EVD-007

Topic: Lower extremity prevention soccer

Recommendation: Neuromuscular warm-up reduces LE injuries with compliance

Evidence Level: Level A/B RCT

Key Trials: Soligard BMJ cluster RCT

Clinical Application: Coach-led every session

References: FIFA 11+ program

RECORD: Red flags — screening importance

ID: EVD-008

Topic: Serious pathology screening

Recommendation: Systematic red flag questioning mandatory all MSK intake

Evidence Level: Level B diagnostic accuracy varies

Key Trials: Individual flags low specificity combinations increase yield

Clinical Application: Document positive/negative explicitly; action if positive

References: NICE NG59

RECORD: Outcome measures — MCID tracking

ID: EVD-009

Topic: Patient-reported outcomes

Recommendation: Use region-specific index + NPRS + PSFS goal tracking

Evidence Level: Level B measurement properties

Key Trials: DASH SPADI NDI MCID anchors established approximate ranges

Clinical Application: Reassess same interval 2-4 weeks interpret MCID not only statistical significance

References: Schoneveld MCID review

RECORD: Thoracic manipulation — selected patients

ID: EVD-010

Topic: Thoracic spine manual therapy

Recommendation: Thoracic mobilization/manipulation may benefit acute mechanical pain after fracture excluded

Evidence Level: Level B short-term

Key Trials: Heneghan Cochrane thoracic MT

Clinical Application: Combine with exercise postural program not standalone

References: Heneghan thoracic MT

RECORD: Scapular-focused rehab — shoulder outcomes

ID: EVD-011

Topic: Scapular dyskinesis

Recommendation: Scapular strengthening part of shoulder impingement rehab

Evidence Level: Level B

Key Trials: Cools scapular exercise EMG studies

Clinical Application: Serratus lower trap rhomboid balance

References: Kibler scapular dyskinesis

RECORD: Pain referral — viscerosomatic awareness

ID: EVD-012

Topic: Referred shoulder pain visceral

Recommendation: Screen cardiac pulmonary hepatobiliary when shoulder pain atypical

Evidence Level: Level C mechanism well established

Key Trials: Clinical medicine teaching phrenic referral

Clinical Application: Left shoulder + exertion + risk factors → cardiac before steroid injection

References: AHA atypical ACS

RECORD: Multimodal rehab — education plus exercise

ID: EVD-013

Topic: Multimodal MSK care

Recommendation: Combine patient education, exercise, optional manual therapy; address psychosocial yellow flags

Evidence Level: Level A/B multiple regions

Key Trials: Better outcomes vs single modality passive care

Clinical Application: Set PSFS goals; address fear avoidance; graded exposure

References: See PubMed [multimodal physiotherapy MSK]

RECORD: Imaging appropriateness — avoid low-value care

ID: EVD-014

Topic: Imaging stewardship

Recommendation: Delay imaging uncomplicated MSK unless red flags or failure appropriate conservative trial

Evidence Level: Level B Choosing Wisely

Key Trials: Early MRI neck/shoulder not improving outcomes many cohorts

Clinical Application: Document clinical indication when ordering; patient expectation management

References: Choosing Wisely MSK imaging

RECORD: Return to sport — criteria based progression

ID: EVD-015

Topic: Return to sport upper limb

Recommendation: Pain strength ROM function sport-specific simulation criteria not time alone

Evidence Level: Level B expert consensus throwers

Key Trials: Interval programs reduce reinjury vs unstructured return

Clinical Application: Use checklists SCR-002; 24h symptom response rule

References: Wilk RTS throwing

RECORD: Clinical reasoning documentation — medicolegal quality

ID: EVD-016

Topic: Clinical documentation quality

Recommendation: Document red flag screen results working diagnosis differential ranked plan follow-up criteria referral triggers

Evidence Level: Level C professional standard

Key Trials: Clear documentation supports continuity and safety especially AI-assisted consultation workflows

Clinical Application: Use FIELD:value structured records; record negative red flags explicitly not omitted

References: Professional practice standards MSK physiotherapy

13. Diagnosis-Specific Management Snapshots

RECORD: Subacromial rotator cuff tendinopathy

ID: DM-001

Diagnosis: Subacromial rotator cuff tendinopathy / subacromial pain syndrome

Typical Presentation: Age 40-65 insidious lateral shoulder pain painful arc night pain side-lying overhead aggravation

Key Exam Findings: Painful arc 60-120 Neer Hawkins positive ER weakness pain not true tear lag

Confirmatory Tests: Empty can painful not necessarily weak lag negative; ultrasound/MRI if persistent

Initial Management: Education load management activity modification NSAIDs short course if appropriate

Rehab Phase 1 (0-2 weeks): Isometric ER/IR pain-free; scapular retraction; avoid painful arc range temporarily

Rehab Phase 2 (2-6 weeks): Band ER side-lying ER scapular exercises serratus wall slide

Rehab Phase 3 (6-12 weeks): Scaption progressive loading return overhead functional tasks

Outcome Measures: SPADI QuickDASH NPRS PSFS overhead task

Referral Triggers: Full thickness tear young active; failed 6 months; injection consideration ortho

Prognosis: Good majority improve 3-6 months exercise; some chronic up to 12-24 months

Evidence: Exercise moderate-large effect; decompression not superior conservative many trials

References: Holmgren; Michener

RECORD: Adhesive capsulitis (frozen shoulder)

ID: DM-002

Diagnosis: Adhesive capsulitis

Typical Presentation: Age 40-60 often diabetic female insidious global ROM loss painful then stiff phases

Key Exam Findings: Loss active AND passive ER especially capsular pattern; global restriction all planes

Confirmatory Tests: Clinical diagnosis; X-ray exclude OA calcification; red flags exclude sepsis

Initial Management: Education natural history 1-3 years phases; pain management phase appropriate

Rehab Phase 1 (freezing): Gentle pendulum pain-limited PROM; avoid aggressive stretch severe irritability

Rehab Phase 2 (frozen): Wand exercises joint mobilization grade III-IV selected; capsular stretch progressive

Rehab Phase 3 (thawing): Strengthening scapular cuff return function sport

Outcome Measures: SPADI ROM ER measurement QuickDASH

Referral Triggers: Hydrodilatation if severe disability; ortho if refractory >12 months

Prognosis: Most thaw spontaneously; diabetes prolonged; 10-20% residual stiffness

Evidence: Multimodal PT hydrodilatation steroid mixed evidence

References: Kelley capsulitis

RECORD: Cervical radiculopathy

ID: DM-003

Diagnosis: Cervical radiculopathy

Typical Presentation: Neck pain arm dermatomal paresthesia myotome weakness Spurling positive

Key Exam Findings: Spurling ULTT positive dermatomal sensory motor reflex changes segmental

Confirmatory Tests: MRI cervical if persistent >6 weeks progressive motor or severe pain

Initial Management: Reassurance majority improve 6-12 weeks; avoid cervical collar prolonged

Rehab Phase 1: Cervical mobilization manual traction selected; DNFE; no HVLA if severe

Rehab Phase 2: Scapular strengthening thoracic mobility nerve glides if appropriate

Rehab Phase 3: Return work sport posture maintenance program

Outcome Measures: NDI NPRS arm pain PSFS

Referral Triggers: Progressive motor deficit myelopathy urgent neurosurgery; epidural if severe pain

Prognosis: 80% improve conservative 6-12 weeks; surgery if progressive

Evidence: Wainner cluster; manual therapy + exercise

References: Childs neck CPG

RECORD: Lateral epicondylalgia

ID: DM-004

Diagnosis: Lateral epicondylalgia

Typical Presentation: Lateral elbow pain gripping resisted wrist extension occupational sport

Key Exam Findings: Cozen Maudsley epicondyle tenderness

Confirmatory Tests: Clinical; NCS if ulnar/radial overlap; cervical screen

Initial Management: Load management ergonomic grip modify activity not complete rest

Rehab Phase 1: Isometric wrist extension pain-free

Rehab Phase 2: Eccentric Tyler twist daily 3x15

Rehab Phase 3: Grip strengthening return sport work tasks

Outcome Measures: PRTEE grip strength NPRS

Referral Triggers: 12 months failed rehab injection surgery rare

Prognosis: Self-limiting many 6-24 months; exercise improves function faster

Evidence: Coombes eccentric loading

References: Tyler twist trial

RECORD: Carpal tunnel syndrome

ID: DM-005

Diagnosis: Carpal tunnel syndrome

Typical Presentation: Median thumb-index-middle numbness night shaking hand thenar weakness late

Key Exam Findings: Phalen Tinel positive sensory thenar assessment

Confirmatory Tests: NCS/EMG grade severity

Initial Management: Night splint neutral wrist ergonomics activity modification

Rehab Phase 1: Splint education tendon median nerve glides

Rehab Phase 2: Tendon glides strengthening if appropriate

Rehab Phase 3: Return keyboard tool use ergonomic sustain

Outcome Measures: BCTQ PRWHE grip NPRS

Referral Triggers: Thenar atrophy severe NCS surgery; treat cervical if double crush

Prognosis: Mild-moderate good conservative; severe surgery effective

Evidence: AAOS splint first-line

References: AAOS CTS

RECORD: Glenohumeral instability / labral tear

ID: DM-006

Diagnosis: Shoulder instability / labral pathology

Typical Presentation: Young athlete apprehension dead arm clicking instability history

Key Exam Findings: Apprehension relocation O'Brien load shift

Confirmatory Tests: MRI arthrogram; X-ray glenoid bone loss
Initial Management: Education rotator cuff scapular neuromuscular control
Rehab Phase 1: Pain control ROM within stable range
Rehab Phase 2: RTC scapular closed chain progression
Rehab Phase 3: Return throw overhead sport-specific
Outcome Measures: Instability episodes WOSI SPADI
Referral Triggers: Recurrent dislocation bone loss surgery; SLAP failed rehab
Prognosis: Recurrence high under 25 non-operative; surgery effective selected
Evidence: Wilk instability rehab
References: Dick instability

RECORD: UCL sprain thrower

ID: DM-007
Diagnosis: Ulnar collateral ligament sprain thrower
Typical Presentation: Medial elbow pain throwers late cocking valgus stress positive
Key Exam Findings: Moving valgus stress medial tenderness flexor-pronator
Confirmatory Tests: MRI UCL partial vs complete
Initial Management: Rest throwing 4-6 weeks flexor-pronator strengthening
Rehab Phase 1: Isometric wrist flexion pronation scapular hip core
Rehab Phase 2: Eccentric flexor-pronator ASMI interval throwing
Rehab Phase 3: Return mound game simulation pitch count limits
Outcome Measures: Pain valgus stress velocity PRTEE
Referral Triggers: Complete tear elite thrower TJ; apophysitis youth ortho
Prognosis: Partial often rehab; complete TJ 12-18 months RTP
Evidence: ASMI interval; Dines UCL
References: Wilk throwing

RECORD: De Quervain tenosynovitis

ID: DM-008
Diagnosis: De Quervain tenosynovitis
Typical Presentation: Radial wrist thumb pain Finkelstein positive new mothers
Key Exam Findings: First dorsal compartment tenderness Finkelstein
Confirmatory Tests: Clinical ultrasound; exclude scaphoid if trauma
Initial Management: Thumb spica splint activity modification
Rehab Phase 1: Splint tendon glides
Rehab Phase 2: Gradual thumb wrist strengthening
Rehab Phase 3: Return lifting ergonomic
Outcome Measures: PRWHE Finkelstein symptom NPRS
Referral Triggers: Failed 6 weeks injection or release
Prognosis: Good splint majority
Evidence: Splint effective
References: See PubMed [de Quervain]

RECORD: Trigger finger

ID: DM-009

Diagnosis: Trigger finger stenosing tenosynovitis

Typical Presentation: Locking catching A1 pulley diabetes manual work

Key Exam Findings: Palpable nodule A1 locking

Confirmatory Tests: Clinical

Initial Management: Splint MCP extension

Rehab Phase 1: Splint tendon glides

Rehab Phase 2: Grip if resolved

Rehab Phase 3: Return full grip

Outcome Measures: Locking frequency QuickDASH

Referral Triggers: Injection or A1 release

Prognosis: Good injection surgery

Evidence: Steroid injection effective

References: See PubMed [trigger finger]

RECORD: Thoracic mechanical postural pain

ID: DM-010

Diagnosis: Mechanical thoracic postural pain

Typical Presentation: Interscapular ache end sitting kyphosis desk worker

Key Exam Findings: Thoracic hypomobility rotation extension PA tender

Confirmatory Tests: Exclude fracture cardiac visceral red flags

Initial Management: Ergonomic micro-breaks

Rehab Phase 1: Thoracic extension rotation open book

Rehab Phase 2: Scapular rows lower trap chin tuck

Rehab Phase 3: Sustained sitting tolerance

Outcome Measures: NPRS sitting hours

Referral Triggers: Red flags medical; compression fracture ortho

Prognosis: Good 4-8 weeks

Evidence: Thoracic MT + exercise

References: Heneghan

RECORD: Cubital tunnel syndrome

ID: DM-011

Diagnosis: Cubital tunnel syndrome

Typical Presentation: Ulnar ring-small numbness elbow flexion desk sleep

Key Exam Findings: Tinel elbow flexion test Froment

Confirmatory Tests: NCS/EMG; cervical C8 screen

Initial Management: Elbow pad night extension splint avoid prolonged flexion

Rehab Phase 1: Ergonomic ulnar nerve glides

Rehab Phase 2: Intrinsic strengthening if motor involved

Rehab Phase 3: Return keyboard modify armrest

Outcome Measures: Symptom diary QuickDASH

Referral Triggers: Intrinsic atrophy surgery

Prognosis: Mild good conservative

Evidence: Night splint padding

References: See PubMed [cubital tunnel]

RECORD: Mallet finger

ID: DM-012

Diagnosis: Mallet finger extensor avulsion

Typical Presentation: DIP droop cannot extend after ball strike

Key Exam Findings: DIP extension lag

Confirmatory Tests: X-ray avulsion size

Initial Management: Continuous DIP extension splint 6-8 weeks

Rehab Phase 1: Splint 24/7 no flexion DIP

Rehab Phase 2: Wean splint if healed

Rehab Phase 3: Return sport extension resolved

Outcome Measures: DIP extension function

Referral Triggers: Large avulsion open hand surgery

Prognosis: Good closed splint compliance

Evidence: Continuous splint standard

References: Doyle mallet

RECORD: Jersey finger FDP avulsion

ID: DM-013

Diagnosis: Jersey finger

Typical Presentation: Athlete cannot flex DIP after tackle

Key Exam Findings: No active DIP flexion PIP flexed

Confirmatory Tests: X-ray ultrasound MRI retraction

Initial Management: Hand surgery URGENT refer

Rehab Phase 1: Post-op surgeon protocol

Rehab Phase 2: Protected flexion 8-12 weeks

Rehab Phase 3: Return sport 4-6 months

Outcome Measures: DIP flexion strength

Referral Triggers: Always surgical consult

Prognosis: Early repair best

Evidence: Manske jersey outcomes

References: Manske

RECORD: Scaphoid fracture

ID: DM-014

Diagnosis: Scaphoid fracture

Typical Presentation: FOOSH snuffbox tenderness

Key Exam Findings: Snuffbox scaphoid compression pain

Confirmatory Tests: X-ray MRI if occult

Initial Management: Thumb spica immobilize until excluded

Rehab Phase 1: Immobilization 6-12 weeks if confirmed

Rehab Phase 2: ROM strengthening post union

Rehab Phase 3: Return sport 3-6 months
Outcome Measures: Union imaging grip
Referral Triggers: Orthopaedic all confirmed displaced surgery
Prognosis: Nonunion risk if missed
Evidence: Early immobilization critical
References: Cooney scaphoid

RECORD: Multidirectional shoulder instability

ID: DM-015
Diagnosis: MDI
Typical Presentation: Young female hypermobility bilateral sulcus
Key Exam Findings: Sulcus Beighton load shift multidirectional
Confirmatory Tests: MRI exclude labral; EDS screen
Initial Management: RTC scapular endurance 6 months trial
Rehab Phase 1: Closed chain scapular control
Rehab Phase 2: RTC endurance proprioception
Rehab Phase 3: Return sport if stable
Outcome Measures: Instability episodes WOSI
Referral Triggers: Failed 6 months capsular plication
Prognosis: Good rehab many MDI
Evidence: Kibler MDI program
References: See PubMed [MDI rehab]

RECORD: Finger pulley injury climber

ID: DM-016
Diagnosis: Pulley injury climber
Typical Presentation: Pop crimp swelling bowstringing A2
Key Exam Findings: Pulley tenderness flexor intact
Confirmatory Tests: Ultrasound MRI grade
Initial Management: Rest taping grade dependent
Rehab Phase 1: Avoid crimp open hand
Rehab Phase 2: Gradual hangboard 50% load
Rehab Phase 3: Return climbing lower grades
Outcome Measures: Pain grade climbed
Referral Triggers: Grade IV surgical
Prognosis: Grade I-II 6 weeks typical
Evidence: Schweizer grading
References: Schweizer climbing

RECORD: AC joint sprain / osteoarthritis

ID: DM-017
Diagnosis: Acromioclavicular joint pathology
Typical Presentation: Superior shoulder pain cross-body adduction painful fall on shoulder AC sprain or chronic OA lifters
Key Exam Findings: AC joint tenderness step deformity if severe sprain cross-body test positive

Confirmatory Tests: X-ray weighted stress views if acute sprain; MRI rarely needed

Initial Management: Sling comfort acute; activity modification; avoid heavy bench fly cross-body loading

Rehab Phase 1: Pain-free ROM scapular setting

Rehab Phase 2: Progressive loading pressing horizontal adduction controlled

Rehab Phase 3: Return contact sport if sprain healed per Rockwood grade

Outcome Measures: Cross-body pain NPRS SPADI

Referral Triggers: Grade IV-V AC dislocation surgery; chronic OA refractory distal clavicle excision

Prognosis: Grade I-II 2-6 weeks; OA chronic manage load

Evidence: Rockwood classification guides management

References: Rockwood AC joint injuries

RECORD: Radial tunnel syndrome

ID: DM-018

Diagnosis: Radial tunnel syndrome

Typical Presentation: Lateral forearm pain distal to epicondyle aching night worse extensor mass use

Key Exam Findings: Tenderness 3-4 cm distal to epicondyle over radial nerve resisted supination middle finger extension pain

Confirmatory Tests: NCS often normal; clinical diagnosis exclusion LE; cervical C6 screen

Initial Management: Activity modification ergonomics splinting forearm neutral optional

Rehab Phase 1: Nerve glides radial bias; avoid repetitive supination pronation

Rehab Phase 2: Gradual extensor strengthening low irritability

Rehab Phase 3: Return keyboard tool use modified technique

Outcome Measures: Forearm pain NPRS PRTEE

Referral Triggers: Failed 6-12 months surgical decompression controversial discuss specialist

Prognosis: Variable conservative success; may coexist LE treat both

Evidence: Diagnosis exclusion; surgery outcomes mixed

References: See REF-020 radial tunnel

RECORD: Brachial neuritis (Parsonage-Turner)

ID: DM-019

Diagnosis: Brachial neuritis

Typical Presentation: Acute severe shoulder pain then weakness patchy multiple nerves days later often post viral

Key Exam Findings: Painless weakness after initial pain phase patchy nerve distribution not single root

Confirmatory Tests: NCS/EMG confirms axonal loss brachial plexus; MRI exclude compressive mass

Initial Management: Pain management medical; reassurance long recovery; maintain ROM

Rehab Phase 1: PROM pain-free prevent contracture

Rehab Phase 2: Gradual strengthening as reinnervation occurs months

Rehab Phase 3: Return function 1-2 years timeline realistic

Outcome Measures: MMT grip function QuickDASH

Referral Triggers: Neurology confirm diagnosis; rule out neoplastic plexopathy if progressive

Prognosis: Slow recovery 1-2 years many good functional recovery; some residual weakness

Evidence: Self-limiting inflammatory condition supportive rehab

References: Feinberg brachial neuritis

RECORD: TFCC tear wrist

ID: DM-020

Diagnosis: Triangular fibrocartilage complex tear

Typical Presentation: Ulnar wrist pain click loading rotation grip fall on extended wrist

Key Exam Findings: Ulnar fovea tenderness TFCC load test positive click

Confirmatory Tests: MRI arthrogram wrist; X-ray ulnar variance

Initial Management: Immobilization short course 4-6 weeks ulnar wrist; activity modification

Rehab Phase 1: Pain-free ROM tendon glides

Rehab Phase 2: Grip strengthening progressive ulnar neutral loading

Rehab Phase 3: Return sport modification taping if needed

Outcome Measures: PRWHE grip pain ulnar loading

Referral Triggers: Hand surgery if instability DRUJ or failed 3 months conservative

Prognosis: Peripheral tears better central poor vascularity; conservative trial reasonable

Evidence: Immobilization then rehab standard first line

References: See CR-005 wrist pathway

RECORD: Internal impingement thrower

ID: DM-021

Diagnosis: Internal impingement (posterolateral glenoid labrum contact) thrower

Typical Presentation: Thrower volleyball tennis swimmer deep posterior shoulder pain late cocking maximum ER

Key Exam Findings: GIRD IR deficit often positive O'Brien mild posterior pain apprehension usually negative

Confirmatory Tests: MRI arthrogram posterolateral labrum partial thickness cuff articular side

Initial Management: Reduce throw volume GIRD correction sleeper cross-body scapular strengthening

Rehab Phase 1: Restore total arc motion within 5° bilateral

Rehab Phase 2: RTC scapular endurance closed chain

Rehab Phase 3: Interval throw return sport-specific

Outcome Measures: Total arc GIRD degrees SPADI throw tolerance

Referral Triggers: Failed 3-6 months rehab MRI structural lesion surgery discussion

Prognosis: Good many respond GIRD program without surgery; structural tears variable

Evidence: Burkhart internal impingement theory GIRD management

References: Ellenbecker GIRD; Wilk thrower

14. Exercise Prescription Engine

Exercise prescription entries map interventions to diagnosis, rehabilitation phase, pain tolerance, equipment, and setting (home, gym, clinic). Match exercises to the working diagnosis and phase; use progressions and regressions when symptoms flare or plateau.

RECORD: Cervical retraction (chin tuck)

ID: EX-001

Exercise: Cervical retraction (chin tuck)

Body Region: Neck

Target Structure: Deep cervical flexors, postural alignment

Diagnosis: Mechanical neck pain, cervicogenic headache, postural dysfunction, WAD

Rehabilitation Phase: Acute through maintenance

Pain Level Required: 0-3/10 acceptable

Equipment: None; optional towel roll

Difficulty: Beginner

Starting Position: Seated or supine neutral spine

Execution: Gently glide chin straight back creating double chin without flexing neck down; hold 5-10 sec; repeat 10-15

Coaching Cues: Head slides back on shelf; lengthen back of neck; breathe normally

Common Errors: Flexing chin to chest; shrugging shoulders; holding breath

Contraindications: Acute cervical instability post trauma, myelopathy, vertebral artery symptoms with rotation (use retraction only)

Progressions: Add resistance band anterior; perform quadruped; add arm movements

Regressions: Supine with towel support; smaller range

Home Version: Hourly desk micro-sets 5 reps

Gym Version: Band resisted retractions 3×10

Clinic Version: Biofeedback pressure cuff deep neck flexor training

Evidence Level: Level B — supported neck pain programs

Clinical Guidelines: APTA neck CPG recommends motor control exercises

References: Jull deep neck flexor training; see PubMed [chin tuck neck pain]

RECORD: Deep neck flexor endurance training

ID: EX-002

Exercise: Deep neck flexor endurance (craniocervical flexion)

Body Region: Neck

Target Structure: Longus colli/capitis

Diagnosis: Neck pain, WAD, cervicogenic headache

Rehabilitation Phase: Subacute onward

Pain Level Required: ≤3/10

Equipment: Pressure biofeedback cuff optional

Difficulty: Intermediate

Starting Position: Supine crook lying head neutral

Execution: Nod chin performing upper cervical flexion only lifting head slightly if progressed; hold 10 sec stages 20-30 mmHg cuff pressure

Coaching Cues: Small nod like saying yes minimally; keep sternocleidomastoid relaxed

Common Errors: Lifting head with SCM dominance; excessive range

Contraindications: Unstable cervical fracture, acute radiculopathy severe

Progressions: Increase hold time; seated; add arm elevation

Regressions: Isometric nod no lift; lower cuff pressure target

Home Version: Daily 10×10 sec holds

Gym Version: Superset with rows postural chain

Clinic Version: Pressure cuff staged protocol Jull

Evidence Level: Level B

Clinical Guidelines: Combined with cervical retraction standard care

References: Jull craniocervical flexion test protocol

RECORD: Upper trapezius stretch

ID: EX-003

Exercise: Upper trapezius stretch

Body Region: Neck/shoulder

Target Structure: Upper trapezius, levator scapulae

Diagnosis: Neck pain, TOS posture, desk-related tension

Rehabilitation Phase: All phases

Pain Level Required: Stretch mild discomfort only $\leq 4/10$

Equipment: Chair

Difficulty: Beginner

Starting Position: Seated upright

Execution: Side bend head away while depressing opposite shoulder with hand; hold 30 sec ×3 each side

Coaching Cues: Ear toward shoulder not forward; gentle overpressure

Common Errors: Rotating forward; bouncing

Contraindications: Vertebral artery insufficiency with combined rotation extension — use side bend only

Progressions: Add scapular depression isometric during stretch

Regressions: Smaller range; no hand overpressure

Home Version: Desk breaks hourly

Gym Version: Post-workout cool-down

Clinic Version: Manual stretch adjunct same position

Evidence Level: Level C adjunct

Clinical Guidelines: Pair with strengthening lower/mid trap

References: See PubMed [upper trapezius stretch neck pain]

RECORD: Thoracic extension over foam roller

ID: EX-004

Exercise: Thoracic extension over foam roller

Body Region: Thoracic spine

Target Structure: Thoracic extensors, facet mobility

Diagnosis: Thoracic hypomobility, postural kyphosis, overhead athlete restriction

Rehabilitation Phase: Subacute onward

Pain Level Required: $\leq 4/10$

Equipment: Foam roller

Difficulty: Beginner-intermediate

Starting Position: Supine roller perpendicular mid-thoracic, hands support head

Execution: Extend over roller segmentally 2-3 levels; hold 2-3 sec; move roller cephalad/caudal

Coaching Cues: Exhale into extension; avoid lumbar hyperextension — keep ribs down

Common Errors: Extending lumbar not thoracic; holding breath

Contraindications: Osteoporosis compression fracture risk aggressive extension; acute rib fracture

Progressions: Arms overhead; smaller diameter roller

Regressions: Seated cat-camel without roller

Home Version: Daily 5 min

Gym Version: Pre overhead workout mobility

Clinic Version: Manual PA mobilization alternative

Evidence Level: Level C — widely used clinical practice

Clinical Guidelines: Improve thoracic extension before overhead loading

References: Heneghan thoracic manual therapy context

RECORD: Open book thoracic rotation

ID: EX-005

Exercise: Open book thoracic rotation

Body Region: Thoracic

Target Structure: Thoracic rotators, rib cage mobility

Diagnosis: Thoracic stiffness, rotational sport restriction golf/baseball

Rehabilitation Phase: All phases

Pain Level Required: $\leq 3/10$

Equipment: Mat

Difficulty: Beginner

Starting Position: Side lying hips/knees flexed 90° arms extended together

Execution: Rotate top arm open toward floor behind keeping hips stacked; hold 5 sec $\times 10$

Coaching Cues: Follow hand with eyes; breathe into rib cage

Common Errors: Rolling hips back; forcing range

Contraindications: Acute rib fracture, recent spinal surgery

Progressions: Add band resistance rotation

Regressions: Smaller arc

Home Version: Morning mobility routine

Gym Version: Warm-up rotation prep

Clinic Version: Segmental rotation mobilization combo

Evidence Level: Level C

Clinical Guidelines: Thoracic rotation prerequisite golf/baseball

References: See PubMed [thoracic rotation mobility exercise]

RECORD: Scapular retraction (band row)

ID: EX-006

Exercise: Scapular retraction — band row

Body Region: Scapula/shoulder

Target Structure: Middle/lower trapezius, rhomboids
Diagnosis: Shoulder impingement, scapular dyskinesis, neck/postural pain
Rehabilitation Phase: Early rehab through performance
Pain Level Required: $\leq 3/10$
Equipment: Resistance band or cable
Difficulty: Beginner
Starting Position: Standing band anchored chest height elbows extended
Execution: Pull band squeezing scapulae together without shrugging; hold 2 sec; 3×12-15
Coaching Cues: Chest tall; pinch shoulder blades down and back
Common Errors: Shrugging upper trap; excessive lumbar extension
Contraindications: Acute shoulder dislocation until stable
Progressions: Increase band tension; single arm; face pull height variation
Regressions: Isometric retraction no band
Home Version: Door anchor band rows
Gym Version: Cable row scapular emphasis light load
Clinic Version: Manual scapular squeeze feedback
Evidence Level: Level B shoulder rehab
Clinical Guidelines: Foundation scapular stabilization programs
References: Cools scapular muscle balance

RECORD: Serratus anterior wall slide

ID: EX-007
Exercise: Serratus wall slide
Body Region: Scapula
Target Structure: Serratus anterior
Diagnosis: Scapular winging, impingement, instability rehab
Rehabilitation Phase: Early-mid rehab
Pain Level Required: $\leq 3/10$
Equipment: Wall
Difficulty: Beginner
Starting Position: Forearms on wall elbows bent W position
Execution: Slide forearms upward protracting scapulae maintaining contact; 3×10 slow
Coaching Cues: Push wall away; feel scapula wrap around rib cage
Common Errors: Upper trap dominance; losing wall contact
Contraindications: Painful arc severe without modification range
Progressions: Foam roller wall slide; add mini band
Regressions: Supine serratus punch
Home Version: Wall slides daily
Gym Version: Scapular push-up plus progression
Clinic Version: Kinetic control serratus activation
Evidence Level: Level B
Clinical Guidelines: Kibler scapular dyskinesis protocol component
References: Kibler serratus function

RECORD: Rotator cuff isometric external rotation

ID: EX-008

Exercise: Rotator cuff isometric external rotation

Body Region: Shoulder

Target Structure: Infraspinatus/teres minor

Diagnosis: Rotator cuff tendinopathy, post-op early, impingement acute

Rehabilitation Phase: Acute/early rehab

Pain Level Required: $\leq 4/10$

Equipment: Towel roll optional doorframe

Difficulty: Beginner

Starting Position: Standing elbow 90° flexed at side towel between elbow and ribs

Execution: Press outward into doorframe or hand resisting without movement; hold 5-10 sec $\times 10$

Coaching Cues: Keep elbow glued to side; gentle submaximal effort initially

Common Errors: Abducting elbow; compensating trunk rotation

Contraindications: Acute full thickness tear without ortho protocol

Progressions: Band isotonic ER; side-lying ER

Regressions: Lower effort shorter hold

Home Version: Doorframe isometrics daily

Gym Version: Cable ER 0° abduction light

Clinic Version: Blood flow restriction optional advanced

Evidence Level: Level B early RTC rehab

Clinical Guidelines: Isometrics safe early tendinopathy loading

References: See PubMed [isometric rotator cuff tendinopathy]

RECORD: Side-lying external rotation

ID: EX-009

Exercise: Side-lying external rotation

Body Region: Shoulder

Target Structure: Infraspinatus, teres minor

Diagnosis: RTC tendinopathy, impingement, post-op protocol

Rehabilitation Phase: Mid rehab

Pain Level Required: $\leq 3/10$

Equipment: Light dumbbell or band

Difficulty: Beginner-intermediate

Starting Position: Side lying affected shoulder up elbow 90° flexed

Execution: Rotate hand upward keeping elbow at side; lower slowly 3 $\times$ 12-15

Coaching Cues: Slow eccentric; don't roll backward

Common Errors: Using momentum; excessive weight

Contraindications: Acute impingement painful arc — modify range

Progressions: Increase load; standing ER at 90 abduction

Regressions: No weight isometric

Home Version: Water bottle light weight

Gym Version: Dumbbell side-lying ER

Clinic Version: Manual resisted ER

Evidence Level: Level B

Clinical Guidelines: Standard RTC strengthening

References: Michener RTC exercise review

RECORD: Sleeper stretch (posterior capsule)

ID: EX-010

Exercise: Sleeper stretch

Body Region: Shoulder

Target Structure: Posterior capsule, GIRD management

Diagnosis: GIRD overhead athletes, internal impingement, adhesive capsulitis caution

Rehabilitation Phase: Maintenance/prevention; capsulitis only if indicated

Pain Level Required: Stretch discomfort $\leq 5/10$; avoid aggressive capsulitis freezing

Equipment: Mat

Difficulty: Intermediate

Starting Position: Side lying affected shoulder down arm 90° forward flexion elbow bent

Execution: Gently press hand toward table into IR stretch; hold 30 sec $\times 3$

Coaching Cues: Scapula stabilized; gentle sustained pressure not bounce

Common Errors: Scapula protraction; excessive force

Contraindications: Unstable shoulder; acute capsulitis severe irritability

Progressions: Cross-body stretch combo

Regressions: Less IR range

Home Version: Post throwing cool-down

Gym Version: Partner assisted IR stretch

Clinic Version: Manual IR glides

Evidence Level: Level B overhead athletes

Clinical Guidelines: Ellenbecker GIRD protocol cornerstone

References: Ellenbecker sleeper stretch GIRD

RECORD: Scapular plane elevation (scaption)

ID: EX-011

Exercise: Scapular plane elevation (scaption)

Body Region: Shoulder

Target Structure: Supraspinatus, deltoid, serratus

Diagnosis: RTC tendinopathy, impingement rehab mid phase

Rehabilitation Phase: Mid-late rehab

Pain Level Required: $\leq 3/10$ avoid painful arc if possible

Equipment: Light dumbbells

Difficulty: Intermediate

Starting Position: Standing arms $\sim 30^\circ$ forward from frontal plane thumb up

Execution: Elevate arms to shoulder height; lower slow 3 $\times$ 10-12

Coaching Cues: Lead with thumbs up; stop before pain arc

Common Errors: Shrugging; going into painful range

Contraindications: Massive RTC tear no active elevation protocol per surgeon

Progressions: Increase load; full elevation if tolerated

Regressions: Assisted elevation; reduced range below arc

Home Version: Light cans

Gym Version: Dumbbell scaption

Clinic Version: Cable scaption

Evidence Level: Level B

Clinical Guidelines: Safer elevation plane impingement

References: See PubMed [scaption rotator cuff]

RECORD: Tennis elbow eccentric wrist extension (Tyler twist)

ID: EX-012

Exercise: Eccentric wrist extension — Tyler twist

Body Region: Elbow/forearm

Target Structure: ECRB/Wrist extensors

Diagnosis: Lateral epicondylalgia

Rehabilitation Phase: Subacute-chronic

Pain Level Required: ≤5/10 during exercise acceptable tendinopathy loading

Equipment: FlexBar or dumbbell

Difficulty: Intermediate

Starting Position: Affected arm extended wrist extended holding FlexBar

Execution: Eccentric lowering into flexion controlling 3 sec; 3×15 daily

Coaching Cues: Control the lowering; mild pain OK; stay in plane

Common Errors: Concentric overload too early; using heavy load

Contraindications: Acute tendon rupture

Progressions: Stiffer FlexBar; increase reps

Regressions: Two-hand concentric assist eccentric only

Home Version: Daily FlexBar protocol

Gym Version: Dumbbell eccentric wrist extension off bench

Clinic Version: Isometric first then eccentric progression

Evidence Level: Level A/B LE rehab

Clinical Guidelines: First-line tendinopathy loading

References: Tyler FlexBar study; Coombes LE review

RECORD: Wrist flexor eccentric (medial epicondylalgia)

ID: EX-013

Exercise: Eccentric wrist flexion

Body Region: Elbow

Target Structure: Flexor-pronator mass

Diagnosis: Medial epicondylalgia, thrower's elbow tendinopathy

Rehabilitation Phase: Subacute onward

Pain Level Required: ≤5/10 loading

Equipment: Dumbbell

Difficulty: Intermediate

Starting Position: Forearm supported palm up wrist extended

Execution: Lower weight slowly into flexion 3 sec; assist concentric other hand 3×15

Coaching Cues: Slow eccentric focus; elbow stable on table
Common Errors: Too heavy too soon
Contraindications: Acute UCL tear — avoid valgus load until cleared
Progressions: Increase weight; pronation-supination combo
Regressions: Isometric wrist flexion
Home Version: Light hammer eccentric
Gym Version: Cable wrist curl eccentric emphasis
Clinic Version: Manual resisted flexion
Evidence Level: Level B
Clinical Guidelines: Mirror LE protocol medial side
References: See PubMed [medial epicondylalgia eccentric]

RECORD: Median nerve glide

ID: EX-014
Exercise: Median nerve glide
Body Region: Wrist/hand/neural
Target Structure: Median nerve mobility
Diagnosis: Carpal tunnel syndrome, double crush, post-op nerve
Rehabilitation Phase: All phases when irritability allows
Pain Level Required: Tension $\leq 3/10$ no sustained increase symptoms
Equipment: None
Difficulty: Beginner
Starting Position: Arm extended shoulder abducted 90°
Execution: Sequence wrist extension/flexion with neck side bend opposite/same per chosen technique; 10 slow reps
Coaching Cues: Gentle flossing not stretching into numbness increase
Common Errors: Aggressive stretching; reproducing night symptoms severely
Contraindications: Acute nerve injury; severe CTS with thenar atrophy — surgery pathway parallel
Progressions: Load tolerance sport after glides tolerated
Regressions: Smaller amplitude components
Home Version: Daily morning glides
Gym Version: Pre-grip warmup
Clinic Version: ULTT mobilization combo
Evidence Level: Level C/B adjunct CTS
Clinical Guidelines: Adjunct not replacement splint/ergonomics
References: Rozmaryn nerve gliding CTS

RECORD: TOS neural mobility and posture protocol

ID: EX-015
Exercise: TOS posture and nerve mobility combo
Body Region: Neck/shoulder/neural
Target Structure: Brachial plexus mobility, scalene length, posture
Diagnosis: Thoracic outlet syndrome neurogenic, postural TOS
Rehabilitation Phase: Subacute onward
Pain Level Required: $\leq 3/10$ tension

Equipment: None; doorway stretch

Difficulty: Intermediate

Starting Position: Seated neutral

Execution: Chin tuck + scapular retraction + gentle ULTT median bias; pectoralis doorway stretch; 2× daily

Coaching Cues: Open chest; reduce forward head; never force paresthesia

Common Errors: Overstretching scalenes aggressively

Contraindications: Vascular TOS acute ischemia — medical emergency not exercise

Progressions: Scapular strengthening rows; ergonomic workstation

Regressions: Posture only no nerve tension

Home Version: Ergonomic setup + stretches

Gym Version: Rows face pulls

Clinic Version: First rib mobilization qualified clinicians

Evidence Level: Level C neurogenic TOS

Clinical Guidelines: Conservative first-line many neurogenic TOS

References: Povlsen TOS exercise review

RECORD: De Quervain tendon glides

ID: EX-016

Exercise: Tendon glides thumb/wrist

Body Region: Wrist/hand

Target Structure: APL/EPB tendons first dorsal compartment

Diagnosis: De Quervain tenosynovitis

Rehabilitation Phase: Acute-subacute with splint

Pain Level Required: ≤3/10

Equipment: Thumb spica splint optional

Difficulty: Beginner

Starting Position: Neutral wrist

Execution: Thumb extension/abduction combinations through gentle range 10 reps hourly

Coaching Cues: Smooth gliding within splint if worn

Common Errors: Forceful Finkelstein as exercise

Contraindications: Acute severe irritation — rest first

Progressions: Grip strengthening gradual

Regressions: Active ROM only one plane

Home Version: Hourly glides

Gym Version: Avoid heavy grip until cleared

Clinic Version: Manual soft tissue mobilization

Evidence Level: Level C

Clinical Guidelines: Splint plus activity modification primary

References: See PubMed [de Quervain rehabilitation]

RECORD: Finger flexor tendon glides

ID: EX-017

Exercise: Finger flexor tendon glides (seven positions)

Body Region: Hand

Target Structure: Flexor digitorum tendons pulleys
Diagnosis: Trigger finger post-op, flexor rehab, post-splint stiffness
Rehabilitation Phase: Early rehab post injury
Pain Level Required: $\leq 4/10$
Equipment: None
Difficulty: Beginner
Starting Position: Hand relaxed
Execution: Progress through hook, fist, straight fist, table top positions sequentially 10 reps
Coaching Cues: Hold each position 3 sec; no forcing into lock
Common Errors: Rushing; painful triggering through
Contraindications: Acute flexor sheath infection; unrepaired jersey finger
Progressions: Putty grip after integrity confirmed
Regressions: Partial positions only
Home Version: 5× daily post hand surgery protocol
Gym Version: Climbing return gradual
Clinic Version: Place and hold splint protocol
Evidence Level: Level B hand therapy standard
Clinical Guidelines: Standard post flexor repair protocol component
References: Mayo flexor tendon gliding protocol

RECORD: Climbing antagonist wrist extensor band

ID: EX-018
Exercise: Rubber band finger extension (antagonist)
Body Region: Hand/forearm
Target Structure: Finger extensors, ECU synergy
Diagnosis: Climbing overuse, elbow tendinopathy prevention
Rehabilitation Phase: Prevention and rehab
Pain Level Required: $\leq 3/10$
Equipment: Rubber band around fingers
Difficulty: Beginner
Starting Position: Forearm supported fingers flexed
Execution: Spread fingers against band 3×20
Coaching Cues: Open all fingers evenly; control return
Common Errors: Too stiff band; neglecting extensors
Contraindications: Acute pulley rupture — follow surgeon protocol
Progressions: Thicker band; rice bucket extensors
Regressions: Single finger extension
Home Version: Daily climber routine
Gym Version: Superset with hangs
Clinic Version: Balance flexor/extensor ratio assessment
Evidence Level: Level C prevention
Clinical Guidelines: Climbing injury prevention standard recommendation
References: Schweizer climbing injury prevention

RECORD: Diaphragmatic breathing

ID: EX-019

Exercise: Diaphragmatic breathing

Body Region: Breathing/thoracic

Target Structure: Diaphragm, rib excursion, parasympathetic activation

Diagnosis: Neck pain accessory breathing, rib dysfunction, stress-related tension, TOS

Rehabilitation Phase: All phases

Pain Level Required: 0-2/10

Equipment: None optional belt

Difficulty: Beginner

Starting Position: Supine crook or seated

Execution: Inhale expand lower ribs/belly; exhale long slow; 5 min 6 breaths/min pace

Coaching Cues: Minimal upper chest lift; relax shoulders

Common Errors: Upper trap breathing; hyperventilation

Contraindications: None typical MSK

Progressions: Supine 90-90 wall breathing; integrate with bracing

Regressions: Shorter duration

Home Version: Morning/evening 5 min

Gym Version: Pre-lift bracing reset

Clinic Version: Biofeedback respiratory training

Evidence Level: Level B neck pain adjunct

Clinical Guidelines: Breathing retraining chronic neck pain programs

References: See PubMed [diaphragmatic breathing neck pain]

RECORD: Prone Y-T-W scapular series

ID: EX-020

Exercise: Prone Y-T-W raises

Body Region: Scapula/shoulder

Target Structure: Lower trapezius, middle trapezius, rotator cuff

Diagnosis: Scapular dyskinesis, impingement, postural syndrome

Rehabilitation Phase: Mid rehab

Pain Level Required: $\leq 3/10$

Equipment: Mat light weights optional

Difficulty: Intermediate

Starting Position: Prone forehead supported arms hanging

Execution: Raise arms into Y, T, W positions thumbs up 2×10 each pattern

Coaching Cues: Squeeze scapulae; small range quality

Common Errors: Lifting too high into pain; lumbar extension

Contraindications: Prone intolerance pregnancy etc — use standing variant

Progressions: Light dumbbells; stability ball

Regressions: Smaller arc; one position at a time

Home Version: Floor daily

Gym Version: Incline bench Y raises

Clinic Version: EMG biofeedback lower trap

Evidence Level: Level B

Clinical Guidelines: Lower trap activation protocols

References: Cools scapular exercises

RECORD: Closed chain wall push-up plus

ID: EX-021

Exercise: Wall push-up plus (scapular protraction)

Body Region: Scapula

Target Structure: Serratus anterior

Diagnosis: Serratus weakness, scapular dyskinesis

Rehabilitation Phase: Mid-late rehab

Pain Level Required: ≤3/10

Equipment: Wall

Difficulty: Beginner-intermediate

Starting Position: Wall push-up position arms extended

Execution: At top protract scapulae rounding upper back slightly; 3×15

Coaching Cues: Push wall away at end range; don't bend elbows much

Common Errors: Bending elbows turning into push-up not plus

Contraindications: Wrist pain — modify surface

Progressions: Floor push-up plus; band resisted plus

Regressions: Seated serratus punch

Home Version: Wall daily

Gym Version: Push-up plus progression

Clinic Version: Manual facilitation serratus

Evidence Level: Level B

Clinical Guidelines: Closed chain scapular control progression

References: Kibler push-up plus

RECORD: Pendulum exercises (Codman)

ID: EX-022

Exercise: Pendulum/Codman exercises

Body Region: Shoulder

Target Structure: Glenohumeral mobility gentle

Diagnosis: Adhesive capsulitis early, post-op RTC, acute pain guarded shoulder

Rehabilitation Phase: Acute/early

Pain Level Required: ≤4/10 passive motion

Equipment: Table support

Difficulty: Beginner

Starting Position: Bent forward hand dangling supported

Execution: Small circles clockwise/counter 30 sec; forward-back gentle

Coaching Cues: Relax shoulder let gravity move; trunk still

Common Errors: Active muscle substitution

Contraindications: Unstable fracture non-surgical

Progressions: Add light weight cup; wand exercises

Regressions: Smaller circles

Home Version: Kitchen counter lean

Gym Version: Pre-workout mobility acute phase

Clinic Version: PROM therapist assisted after

Evidence Level: Level C historical standard

Clinical Guidelines: Early capsulitis and post-op gentle motion

References: Kelley adhesive capsulitis rehab

RECORD: Wand flexion and external rotation

ID: EX-023

Exercise: Upper extremity wand exercises

Body Region: Shoulder

Target Structure: Glenohumeral ROM flexion ER

Diagnosis: Capsulitis, post-op ROM, impingement stiff phase

Rehabilitation Phase: Early-mid

Pain Level Required: $\leq 4/10$

Equipment: Wand/broomstick

Difficulty: Beginner

Starting Position: Supine or seated holding wand both hands

Execution: Unaffected arm pushes affected into flexion/ER/elevation to tolerance 10 reps

Coaching Cues: Passive-assisted not forcing pain

Common Errors: Forcing end range sharp pain

Contraindications: Post repair restrictions per protocol — respect limits

Progressions: Increase range; active-assisted

Regressions: Gravity assisted pendulum first

Home Version: Daily ROM program

Gym Version: PVC pipe warmup

Clinic Version: Joint mobilization after wand

Evidence Level: Level B capsulitis

Clinical Guidelines: Standard ROM restoration

References: See PubMed [shoulder wand exercises capsulitis]

RECORD: Rhomboid strengthening — prone horizontal abduction

ID: EX-024

Exercise: Prone horizontal abduction (reverse fly)

Body Region: Scapula

Target Structure: Rhomboids, middle trapezius, infraspinatus

Diagnosis: Rounded shoulder posture, scapular dyskinesis

Rehabilitation Phase: Mid-late

Pain Level Required: $\leq 3/10$

Equipment: Light dumbbells

Difficulty: Intermediate

Starting Position: Prone or bent-over torso $\sim 45^\circ$

Execution: Raise arms to side squeezing scapulae 3×12

Coaching Cues: Thumbs up or neutral; neck relaxed
Common Errors: Using heavy traps; shrugging
Contraindications: Acute impingement painful arc — limit height
Progressions: Cable reverse fly; increase load
Regressions: No weight
Home Version: Band reverse fly
Gym Version: Machine reverse pec fly
Clinic Version: Manual resisted horizontal extension
Evidence Level: Level B
Clinical Guidelines: Postural strengthening programs
References: Cools scapular rehab

RECORD: Levator scapulae stretch with rotation

ID: EX-025
Exercise: Levator scapulae stretch
Body Region: Neck
Target Structure: Levator scapulae
Diagnosis: Neck pain, scapular elevation habit, desk posture
Rehabilitation Phase: All phases
Pain Level Required: Stretch $\leq 4/10$
Equipment: Chair
Difficulty: Beginner
Starting Position: Seated
Execution: Rotate head 45° toward armpit then nod down; gentle overpressure hand 30 sec
Coaching Cues: Nose toward armpit; opposite shoulder down
Common Errors: Forward flexion only without rotation
Contraindications: VBI symptoms with rotation — omit or medical clearance
Progressions: Add scapular depression isometric
Regressions: Less rotation
Home Version: Desk breaks
Gym Version: Cooldown stretch
Clinic Version: Manual levator release
Evidence Level: Level C
Clinical Guidelines: Adjunct to strengthening lower trap
References: See PubMed [levator scapulae stretch]

RECORD: Cubital tunnel nerve glide (ulnar)

ID: EX-026
Exercise: Ulnar nerve glide
Body Region: Elbow/hand neural
Target Structure: Ulnar nerve
Diagnosis: Cubital tunnel syndrome, ulnar neuropathy
Rehabilitation Phase: Subacute
Pain Level Required: Tension $\leq 3/10$

Equipment: None

Difficulty: Beginner-intermediate

Starting Position: Arm extended shoulder abducted

Execution: Elbow flex/extend combined wrist positions per flossing sequence 10 reps

Coaching Cues: Gentle do not hold sustained stretch at end

Common Errors: Aggressive elbow flexion stretch sleeping position not addressed

Contraindications: Severe motor deficit — surgical consult parallel

Progressions: Ergonomic elbow pad night splint 30° flexion

Regressions: Smaller amplitude

Home Version: Avoid prolonged elbow flexion desk

Gym Version: Modify curl position

Clinic Version: NCS correlation

Evidence Level: Level C

Clinical Guidelines: Night elbow extension splint common adjunct

References: See PubMed [cubital tunnel conservative]

RECORD: Grip strengthening progressive (putty)

ID: EX-027

Exercise: Therapeutic putty grip strengthening

Body Region: Hand

Target Structure: Flexor digitorum, intrinsics, grip

Diagnosis: Post immobilization weakness, return sport grip, CTS post-op late

Rehabilitation Phase: Mid-late rehab

Pain Level Required: ≤3/10

Equipment: Therapy putty

Difficulty: Beginner-intermediate

Starting Position: Seated hand supported

Execution: Squeeze putty full grip hold 5 sec; pinch variants 3×15

Coaching Cues: All fingers participate; neutral wrist

Common Errors: Wrist flexion compensation; too firm putty too soon

Contraindications: Acute tendon repair before clearance

Progressions: Firmer putty; dynamometer tracking

Regressions: Softer putty; ball squeeze

Home Version: Daily grip routine

Gym Version: Farmer carry progression

Clinic Version: Grip strength metrics weekly

Evidence Level: Level B functional outcome

Clinical Guidelines: Standard hand rehab progression

References: See PubMed [hand grip strengthening rehabilitation]

RECORD: Cross-body shoulder stretch (posterior capsule)

ID: EX-028

Exercise: Cross-body adduction stretch

Body Region: Shoulder

Target Structure: Posterior capsule, posterior deltoid
Diagnosis: GIRD, internal impingement prevention
Rehabilitation Phase: Maintenance
Pain Level Required: $\leq 5/10$ stretch
Equipment: None
Difficulty: Beginner
Starting Position: Standing
Execution: Pull affected arm across chest with other hand hold 30 sec $\times 3$
Coaching Cues: Scapula neutral; don't torque neck
Common Errors: Shrugging; bouncing
Contraindications: Acute AC joint pain may irritate
Progressions: Combine sleeper stretch protocol
Regressions: Less adduction range
Home Version: Post overhead sport
Gym Version: Pre-workout
Clinic Version: Manual cross-body stretch
Evidence Level: Level B GIRD
Clinical Guidelines: Ellenbecker overhead athlete protocol
References: Ellenbecker cross-body stretch

RECORD: Lower trap wall slide at 135°

ID: EX-029
Exercise: Lower trapezius wall slide Y
Body Region: Scapula
Target Structure: Lower trapezius
Diagnosis: Scapular dyskinesis, impingement
Rehabilitation Phase: Mid rehab
Pain Level Required: $\leq 3/10$
Equipment: Wall
Difficulty: Intermediate
Starting Position: Back to wall arms overhead Y
Execution: Slide arms down wall squeezing lower trap maintaining contact 3×10
Coaching Cues: Elbows straight-ish; feel shoulder blades depress
Common Errors: Rib flare; upper trap takeover
Contraindications: Painful overhead — reduce angle
Progressions: Foam roller between back and wall
Regressions: Smaller range
Home Version: Wall drill daily
Gym Version: Cable Y pull
Clinic Version: Lower trap EMG verification
Evidence Level: Level B
Clinical Guidelines: Scapular retraining standard
References: Cools lower trapezius activation

RECORD: Radial nerve glide

ID: EX-030

Exercise: Radial nerve glide

Body Region: Forearm/neural

Target Structure: Radial nerve

Diagnosis: Radial tunnel syndrome adjunct, lateral arm pain neural component

Rehabilitation Phase: Subacute

Pain Level Required: Tension $\leq 3/10$

Equipment: None

Difficulty: Intermediate

Starting Position: Arm at side elbow extended

Execution: Wrist flexion/pronation with shoulder abduction/extension sequence 10 reps

Coaching Cues: Smooth sequence not static stretch

Common Errors: Confusing with aggressive wrist extension stretch LE

Contraindications: Acute nerve injury

Progressions: Address cervical double crush if present

Regressions: Component isolation

Home Version: Daily flossing

Gym Version: Modify pressing grip if symptomatic

Clinic Version: ULTT radial bias test correlate

Evidence Level: Level C

Clinical Guidelines: Adjunct radial tunnel conservative trial

References: See PubMed [radial nerve mobilization]

RECORD: Standing thoracic rotation with band

ID: EX-031

Exercise: Standing thoracic rotation with band

Body Region: Thoracic

Target Structure: Thoracic rotators, obliques, rib cage

Diagnosis: Thoracic stiffness, golf/baseball rotation restriction, postural pain

Rehabilitation Phase: Mid-late rehab

Pain Level Required: $\leq 3/10$

Equipment: Resistance band anchored

Difficulty: Intermediate

Starting Position: Athletic stance band at chest height arms extended

Execution: Rotate trunk pulling band keeping hips relatively stable 3×10 each direction

Coaching Cues: Rotate from mid-back; exhale into rotation

Common Errors: Rotating hips only; arm pull without trunk

Contraindications: Acute rib fracture; severe osteoporosis twist

Progressions: Increase band tension; sport stance

Regressions: Open book floor version

Home Version: Door anchor band twist

Gym Version: Cable wood chop pattern

Clinic Version: Manual segmental rotation

Evidence Level: Level C sport prep

Clinical Guidelines: Golf/baseball return prerequisite rotation

References: See sport biomechanics SB-006 SB-003

RECORD: Empty can scaption progression

ID: EX-032

Exercise: Empty can to full can scaption progression

Body Region: Shoulder

Target Structure: Supraspinatus, deltoid

Diagnosis: RTC tendinopathy mid-phase strengthening

Rehabilitation Phase: Mid-late

Pain Level Required: ≤3/10

Equipment: Light dumbbell

Difficulty: Intermediate

Starting Position: Scapular plane thumb down empty can or thumb up full can

Execution: Elevate to shoulder height lower slow 3×10-12

Coaching Cues: Small range below painful arc if needed; scapula set

Common Errors: Shrugging; excessive weight

Contraindications: Empty can controversial pain — use full can thumb up alternative

Progressions: Increase load; slow eccentric lowering

Regressions: Isometric hold at low angle

Home Version: Water bottle

Gym Version: Dumbbell scaption

Clinic Version: Cable scaption

Evidence Level: Level B RTC rehab

Clinical Guidelines: Full can preferred some protocols over empty can irritation

References: Michener RTC exercises

RECORD: Quadruped rock back breathing

ID: EX-033

Exercise: Quadruped rock back with diaphragmatic breathing

Body Region: Breathing/thoracic

Target Structure: Diaphragm, lumbar-thoracic mobility, core

Diagnosis: Breathing dysfunction, thoracic pain, postural syndrome, TOS adjunct

Rehabilitation Phase: All phases

Pain Level Required: 0-3/10

Equipment: Mat

Difficulty: Beginner

Starting Position: Quadruped neutral spine

Execution: Rock hips toward heels while maintaining neutral spine breathe into posterior rib cage 10 reps

Coaching Cues: Slow exhalation; feel expansion low ribs

Common Errors: Flexing lumbar excessively; holding breath

Contraindications: Knee weight bearing intolerance modify standing

Progressions: Add band around ribs feedback

Regressions: Child pose static breathing
Home Version: Daily morning routine
Gym Version: Warm-up breathing prep
Clinic Version: Combine with manual rib mobilization
Evidence Level: Level C/B breathing retraining
Clinical Guidelines: Part of TOS and neck programs
References: See EX-019 diaphragmatic breathing

RECORD: Isometric hip abduction wall press (kinetic chain)

ID: EX-034
Exercise: Isometric hip abduction wall press
Body Region: Hip/kinetic chain
Target Structure: Gluteus medius, core stability for throwing/overhead
Diagnosis: Overhead athlete kinetic chain deficit, thrower hip weakness, shoulder secondary to hip
Rehabilitation Phase: Mid rehab return sport
Pain Level Required: $\leq 3/10$
Equipment: Wall
Difficulty: Beginner
Starting Position: Side lying or standing side to wall knee/hip flexed
Execution: Press knee or foot into wall hold 10 sec $\times 10$ activate glute
Coaching Cues: Feel glute medius not TFL; pelvis stable
Common Errors: TFL dominance; holding breath Valsalva
Contraindications: Hip acute pathology
Progressions: Band monster walks; single leg RDL
Regressions: Shorter holds
Home Version: Wall press daily throwers
Gym Version: Cable hip abduction
Clinic Version: Part Wilk throwing phase 2
Evidence Level: Level B kinetic chain
Clinical Guidelines: Throwing programs include hip-shoulder separation
References: Wilk kinetic chain throwing

RECORD: Wrist flexor stretch forearm

ID: EX-035
Exercise: Wrist flexor stretch
Body Region: Forearm/elbow
Target Structure: Wrist flexors pronators
Diagnosis: Medial epicondylalgia, wrist flexor tightness, climbing forearm
Rehabilitation Phase: Maintenance
Pain Level Required: Stretch $\leq 4/10$
Equipment: None
Difficulty: Beginner
Starting Position: Arm extended palm up elbow straight
Execution: Extend wrist and fingers with other hand hold 30 sec $\times 3$

Coaching Cues: Elbow straight; gentle stretch not painful
Common Errors: Flexing elbow; bouncing
Contraindications: Acute UCL tear valgus stretch caution
Progressions: Combine nerve glides if ulnar
Regressions: Smaller wrist extension range
Home Version: Post climbing cool-down
Gym Version: Between sets forearm work
Clinic Version: Before flexor eccentric ME protocol
Evidence Level: Level C adjunct
Clinical Guidelines: Adjunct to eccentric strengthening not replacement
References: See EX-013 eccentric flexor

RECORD: Pec minor doorway stretch

ID: EX-036
Exercise: Pec minor / anterior chest doorway stretch
Body Region: Shoulder/chest
Target Structure: Pectoralis minor/major, anterior shoulder
Diagnosis: Rounded shoulder posture, TOS posture, scapular dyskinesis
Rehabilitation Phase: All phases
Pain Level Required: ≤4/10 stretch
Equipment: Doorway
Difficulty: Beginner
Starting Position: Forearm on doorframe elbow shoulder height lunge forward
Execution: Lean forward gentle chest stretch hold 30 sec ×3
Coaching Cues: Scapula retract slightly; don't crank shoulder into extreme ER pain
Common Errors: Hyperextending lumbar; forcing painful range
Contraindications: Anterior instability apprehension — avoid aggressive
Progressions: Combine chin tuck retraction after stretch
Regressions: Higher or lower arm angle
Home Version: Hourly desk break
Gym Version: Pre rows face pulls
Clinic Version: TOS program EX-015 combo
Evidence Level: Level C posture programs
Clinical Guidelines: Pair with scapular strengthening not stretch alone
References: Povlsen TOS posture

RECORD: Intrinsic hand strengthening (finger abduction)

ID: EX-037
Exercise: Finger abduction rubber band
Body Region: Hand
Target Structure: Interossei, lumbricals
Diagnosis: Ulnar nerve recovery, grip weakness, hand post immobilization
Rehabilitation Phase: Mid-late rehab
Pain Level Required: ≤3/10

Equipment: Rubber band around fingers
Difficulty: Beginner
Starting Position: Fingers together hand relaxed
Execution: Spread fingers against band 3×15
Coaching Cues: Spread evenly; wrist neutral
Common Errors: Using only index finger
Contraindications: Acute fracture tendon repair until cleared
Progressions: Thicker band; small finger bias
Regressions: Single finger abduction
Home Version: Daily hand routine
Gym Version: Farmers carry follow-up
Clinic Version: Post ulnar nerve glide progression
Evidence Level: Level B hand therapy
Clinical Guidelines: Standard intrinsic strengthening
References: See EX-027 grip putty progression

RECORD: Seated nerve floss ULTT median bias

ID: EX-038
Exercise: Seated median nerve floss simplified
Body Region: Neural
Target Structure: Median nerve C6-C7 distribution
Diagnosis: CTS, cervical radiculopathy adjunct, post-op nerve
Rehabilitation Phase: Subacute when irritability low
Pain Level Required: Tension ≤3/10
Equipment: Chair
Difficulty: Beginner
Starting Position: Seated upright shoulder abducted 90° elbow extended
Execution: Extend wrist and neck away side bend opposite then reverse sequence 10 slow reps
Coaching Cues: Gentle oscillation not sustained aggressive stretch
Common Errors: Provoking severe numbness increase
Contraindications: Acute severe CTS thenar atrophy — parallel surgical pathway
Progressions: Standing ULTT full sequence
Regressions: Wrist motion only no neck
Home Version: Morning evening CTS program
Gym Version: Pre grip work warmup
Clinic Version: After Spurling negative cervical cleared
Evidence Level: Level C/B adjunct
Clinical Guidelines: With splint ergonomics CTS
References: Rozmaryn nerve gliding

RECORD: Single leg balance reach (Y balance pattern)

ID: EX-039
Exercise: Single leg balance reach anterior/lateral
Body Region: Lower limb/kinetic chain

Target Structure: Proprioception, glute, core for UE athlete stability

Diagnosis: Overhead athlete landing instability, kinetic chain screening, ankle-knee-shoulder chain

Rehabilitation Phase: Return sport

Pain Level Required: $\leq 3/10$ lower limb

Equipment: None tape lines optional

Difficulty: Intermediate

Starting Position: Single leg stance knee soft

Execution: Reach non-stance foot anterior and lateral tap floor return balance 3×8 each leg

Coaching Cues: Hips level; knee tracks toe; trunk stable

Common Errors: Trunk collapse; hip drop

Contraindications: Acute ankle knee injury

Progressions: Catch medicine ball overhead during balance

Regressions: Two-leg staggered stance

Home Version: Thrower return program

Gym Version: Warm-up overhead athletes

Clinic Version: Y-balance test training correlation

Evidence Level: Level B neuromuscular

Clinical Guidelines: Lower quarter affects throwing landing

References: Wilk kinetic chain

RECORD: Suboccipital release self-massage

ID: EX-040

Exercise: Suboccipital release with lacrosse ball

Body Region: Neck/headache

Target Structure: Suboccipital muscles, upper cervical

Diagnosis: Cervicogenic headache adjunct, upper cervical tension, WAD

Rehabilitation Phase: Subacute maintenance

Pain Level Required: $\leq 4/10$ pressure

Equipment: Lacrosse ball or rolled towel

Difficulty: Beginner

Starting Position: Supine ball at suboccipital base skull

Execution: Gentle pressure 30-60 sec perform slow chin tuck 5 reps

Coaching Cues: Mild pressure not aggressive; breathe

Common Errors: Aggressive ball high cervical fracture risk elderly — caution

Contraindications: Upper cervical instability trauma; osteoporosis aggressive pressure

Progressions: Combine DNFE after release

Regressions: Towel roll only no ball

Home Version: Headache routine

Gym Version: Post overhead sport cool-down

Clinic Version: Manual suboccipital release alternative

Evidence Level: Level C adjunct headache

Clinical Guidelines: Adjunct exercise primary cervicogenic

References: Jull headache management